Note: The following is a redacted version of the original report published March 20, 2026 [26 pgs].

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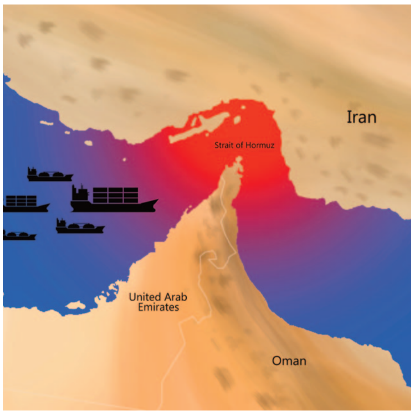
TOP of MIND

IRAN CONFLICT: HOW LONG, AND HOW BAD?



The unprecedented US and Israeli coordinated attack on Iran has resulted in the largest energy supply disruption in history. With seemingly no end to the conflict in sight, its potential duration, impact on global energy supplies, and economic and market implications are Top of Mind. Chatham House's Sanam Vakil and long-time US Middle East advisor Dennis Ross agree that an end to the conflict doesn't look imminent owing to (the lack of) incentives on Iran's side and its chokehold on the Strait of Hormuz, which stands in the way of the US declaring victory. To that end, Vice Admiral Kevin Donegan, USN (ret) explains that military convoys would be capable of ensuring safe transit through the Strait, but not of restoring oil flows to anywhere near normal levels. So, we explore the growing implications of these disruptions for energy prices, the global economy, and markets, finding that assets are pricing an inflationary shock, but not a growth shock, which could be the next shoe to drop.

"For now, all evidence suggests that the conflict will persist—Iran isn't yet signaling that it wants out, and the US has yet to achieve an outcome it can plausibly claim as a victory.

- Sanam Vakil

The key to the US being able to walk away is the Strait of Hormuz… [Trump] likely can rightly claim today that Iran won't be able to pose a conventional threat to its neighbors for at least five years. But he can't say he's won and stop as long as Iran controls who can and can't export oil and sail through the Strait.

- Dennis Ross

I'm confident the US and its allies have the capability to escort commercial ships through the Strait... [but] it is important to distinguish between capability and capacity. Convoys would never be able to restore oil flows through the Strait to normal levels.

- Kevin Donegan"

WHAT'S INSIDE

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Macro news and views

We provide a brief snapshot on the most important economies for the global markets

US

Latest GS proprietary datapoints/major changes in views

- We lowered our 2026 US growth forecast and raised our 2026 US inflation forecast to reflect higher oil prices amid the Iran war, with growth risks skewed further to the downside and inflation risks to the upside.
- We pushed back our forecast for the next Fed cut from Jun to Sep, with a second cut in Dec to the same terminal rate of 3-3.25% given the higher inflation path we now expect.

Datapoints/trends we're focused on

- Labor market, which we expect to soften further.

Japan

Latest GS proprietary datapoints/major changes in views

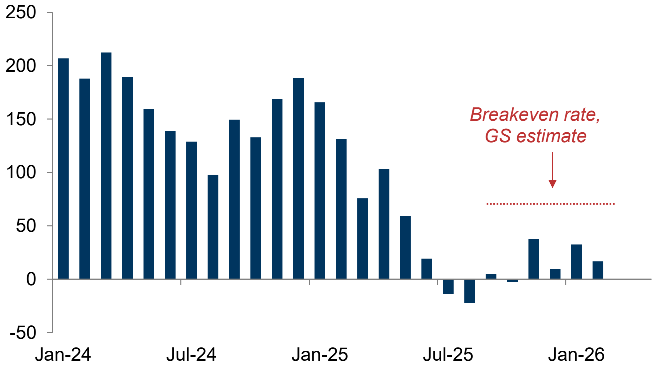
- We lowered our CY26/27 real GDP growth forecasts and raised our 2026 core CPI inflation forecast to reflect higher energy prices, though the sensitivity of growth to rising oil prices is smaller than in the past.

Datapoints/trends we're focused on

- BoJ policy; we continue to expect the next hike in July, but the uncertainty around timing remains high as inflationary pressures from higher energy prices could prompt an earlier hike while rising growth concerns could delay further hikes.
- 2026 *shunto* negotiations; the first round of collected data will likely confirm high wage hike momentum this year.

Labor market softening concerns: alive and well

Underlying trend job growth, GS estimate, thousands per month

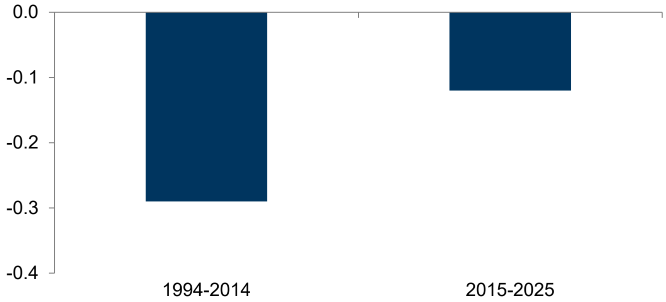


*Note: We estimate underlying trend job growth as 0.75*3-month average payroll growth + 0.25*9-month average household employment growth.*

Source: Goldman Sachs GIR.

Higher oil prices: less shocking than in the past

Impact of a 1% deterioration in terms of trade on GDP growth, pp



Note: Impulse response from a VAR model with terms of trade (logarithmic difference) and GDP (logarithmic difference) as endogenous variables, and the real effective exchange rate (logarithmic difference) and overseas GDP (logarithmic difference) as exogenous variables.

Source: Goldman Sachs GIR.

Europe

Latest GS proprietary datapoints/major changes in views

- We further lowered our 2026 Euro area real GDP growth forecast and raised our 2026 headline inflation forecast to reflect more prolonged oil disruptions amid the Iran war.
- We now expect the BoE to remain on hold for longer and expect Bank Rate to remain at 3.75% through 2026 (vs. quarterly cuts from July previously) given the growing energy shock.

Datapoints/trends we're focused on

- ECB policy; we continue to expect the ECB to remain on hold this year, although the likelihood of hikes has risen given the continued upward pressure on energy prices.

Emerging Markets (EM)

Latest GS proprietary datapoints/major changes in views

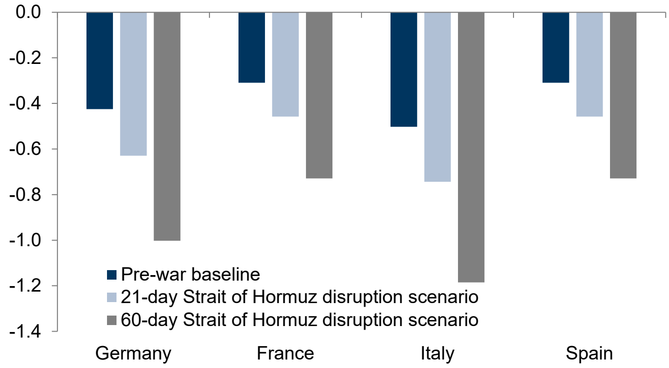
- We raised our 2026 GDP growth forecasts in Brazil and Ecuador but lowered them in Mexico, Chile, and Peru while revising inflation higher across the region amid the Iran war.
- We lowered our 2026 GDP growth forecast for China and most other Asian economies while revising inflation up across the region to reflect higher energy prices.

Datapoints/trends we're focused on

- GCC economies, some of which could see a larger decline in non-oil GDP amid the Iran war than during the pandemic.
- Hungary elections, in which Viktor Orban's Fidesz party faces its strongest challenge since returning to office.

Europe: energy price shock to growth

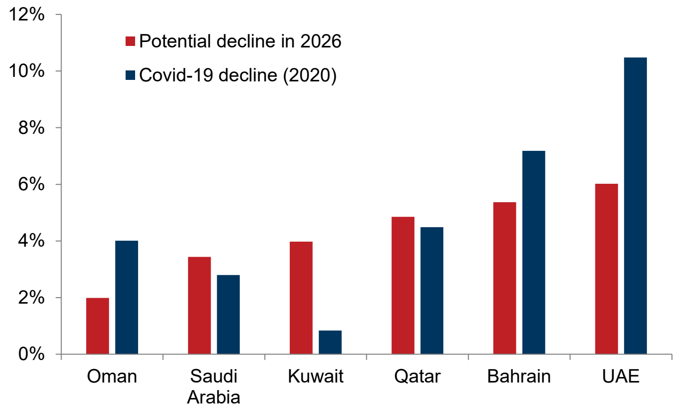
Model-implied peak effect of energy price shock on the level of real GDP vs. pre-conflict baseline, %



Source: Goldman Sachs GIR.

GCC: potentially bigger growth hit than Covid

Decline in non-oil GDP among GCC economies, %



Source: Haver Analytics, Goldman Sachs GIR.

Iran conflict: how long, and how bad?

The unprecedented US and Israeli coordinated attack on Iran has resulted in the largest energy supply disruption in history (see pg. 9), as shipping traffic in the Strait of Hormuz—through which 20% of global oil supplies typically flow—has ground to a near halt and regional energy infrastructure has become the latest target in the conflict. While the economic and market impacts have so far remained relatively muted, the longer the conflict and the resulting chokehold on the Strait of Hormuz and global energy supply last, the less likely that will remain the case. With seemingly no end to the conflict in sight as it continues to escalate, its potential duration, impact on global energy supplies, and economic and market implications are Top of Mind.

To gain insight on when and how the conflict could end, we first turn to Sanam Vakil, Director of the Middle East and North Africa Programme at Chatham House, Ambassador Dennis Ross, Fellow at The Washington Institute for Near East Policy, and Alec Phillips, GS Chief US Political Economist.

Vakil, a longtime Iran watcher, argues that Iran views the conflict as nothing less than an "existential struggle" and is prepared to use its ample asymmetric drone capabilities for as long as is required to guarantee the Islamic Republic's long-term survival, which likely entails sanctions relief. "Until Iran sees a credible path to those guarantees," she says, "it has little incentive to end this war."

Ambassador Ross, who served as a Middle East advisor in five US administrations, contends that President Trump likely would have already declared victory and ended the war were it not for Iran controlling transit through the Strait of Hormuz, which has the potential to inflict substantial pain on the global economy. Until that chokehold is broken, he says, the US cannot unilaterally end the war. Ross sees mediation—potentially facilitated by Russian President Putin—as the quickest way to end the war if the US can't wrest control of the Strait, though he doesn't see the conditions to enter a mediation in place at this point and says it isn't clear who would even mediate on the Iranian side now that Former Speaker of the Parliament Ali Larijani has been killed. So, Ross seems to agree that an end to the war doesn't look imminent.

While Phillips takes no view on when the war will end, he argues that US political considerations don't seem to be much of a constraint on the war's duration for now, but that could change if gasoline prices continue to rise. While he explains that the Trump Administration has some tools in its toolbox to lower energy prices—several of which it has already deployed—these may not be enough to completely offset the price shock.

Given the critical role the Strait of Hormuz is playing in the conflict and its potential outcome, we then speak with Vice Admiral Kevin Donegan, USN, Former Commander of the US Navy's Fifth Fleet, to better understand the potential capability and capacity to restore flows through the Strait. He's confident US and allied military convoys would be capable of ensuring safe transit through the Strait, explaining that the US and many other nations have long rehearsed how to keep vital waterways in the region open. But he clarifies that convoys would never be able to restore oil flows to normal levels, noting that potentially only 20% of flows could be restored. Returning oil flows to pre-conflict levels, he says, would ultimately require Iran to stop harassing ships, which is a matter of incentives and leverage on all sides. But, even then, he argues that restoring flows would not be a matter of simply flipping a switch, because recovering the trust that transit would indeed be safe would take time.

With a quick end to the conflict seeming increasingly elusive and global energy prices already high, GS Commodities Strategists Samantha Dart, Daan Struyven, and Alexandra Paulus then lay out potential energy supply disruption scenarios and the implications for prices, with a gradual recovery in oil supplies from here allowing Brent prices to decline close to \$70/bbl in 4Q26 and a ramp-up in Qatari LNG production starting in early April allowing TTF prices to gradually decline to €30-40/MWh in 2H26.

But, they estimate that longer disruptions could present substantial upside risk to prices, with Brent likely to exceed its 2008 all-time high if depressed flows keep the market focused on the risk of lengthier disruptions. And even if LNG supplies begin to recover in the short term, they warn that the damage to Qatari LNG infrastructure is likely to delay the significant pressure on global LNG prices that they've expected from "the largest ever LNG supply growth wave" from 2027.

So, how hard might the global economy be hit? Joseph Briggs, GS Senior Global Economist, provides some useful rules of thumb: each 10% increase in oil prices lowers global GDP by a bit more than 0.1%, raises global headline inflation by 0.2pp—with larger impacts in Asia ex-China and Europe—and leads core inflation 0.03-0.06% higher, depending on the country. For now, these rules imply moderate hits to growth and boosts to inflation, but those impacts could quickly grow should energy flows remain disrupted for a prolonged period. (Farouk Soussa, GS Senior MENA Economist, then zeros in on the hit to MENA economies in particular, finding a significant near-term hit and possible long-term scars).

And what about markets? Kamakshya Trivedi, GS Chief FX and EM Strategist, explains that markets so far have generally priced the conflict and resulting spike in energy prices as an inflationary shock, suggesting that growth downside could be the next shoe to drop if disruptions continue. If it is, he sees equities and relatively resilient cyclical assets facing increased pressure and the recent hawkish repricing in front-end yields reversing, creating an environment for a key safe haven to shine: the Japanese Yen.

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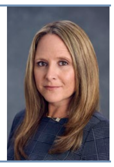


Interview with Sanam Vakil

Sanam Vakil is Director of the Middle East and North Africa Programme at Chatham House. Below, she argues that Iran views the current conflict as an existential struggle and will not seek an end until it receives guarantees that ensure the Islamic Republic's long-term survival.

The views stated herein are those of the interviewee and do not necessarily reflect those of Goldman Sachs.

Ashley Rhodes: Fundamentally, what is this conflict about for Iran?

Sanam Vakil: Tehran sees the US and Israel as waging a regime change war. So, what remains of the Iranian regime is fighting for its very survival. Its strategy is to spread the costs of the war as widely as possible, so that when off-ramps emerge, the war will have become so costly for everyone involved that Iran can obtain guarantees that it will not be repeated again whenever external powers see fit. This strategy reflects lessons from the Twelve-Day War in June 2025, which Iran felt it conceded too early on, signaling weakness and inviting future challenges. So, Iran isn't seeking a quick de-escalation but rather responding to what it perceives as an existential struggle.

Ashley Rhodes: But Iran is also bearing a tremendous cost in this conflict. How long can it continue to do so?

Sanam Vakil: In conventional terms, the US and Israel are the dominant military players and will likely succeed in destroying/degrading Iran's ballistic missile capabilities, its defense industrial complex, and the remnants of its nuclear program. But Iran isn't fighting this war on conventional terms. Instead, it's playing for time—dragging the conflict out, spreading out the costs, and forcing the US, Israel, and regional countries to drain their resources, thereby putting them on the defensive. Iran's strength lies in asymmetric capabilities like low-cost drones that can continue to inflict damage even as the US and Israel make conventional military gains. That Iran is still able to do so three weeks into the conflict underscores that it can continue to endure significant costs.

Ashley Rhodes: How has Mojtaba Khamenei's selection as the new Supreme Leader influenced Iran's strategy?

Sanam Vakil: Mojtaba Khamenei's appointment as Supreme Leader in the wake of his father's death has had little to no impact. His elevation is largely symbolic and likely reflects the regime's desire to preserve the status quo. The Islamic Republic is often presented as a personalistic theocracy. In reality, it's a bureaucratic, institutional system in which power is distributed among various factions. The Islamic Revolutionary Guard Corps (IRGC) is particularly important given its responsibility for the war effort and internal security, but the importance of Speaker of the Parliament Ghalibaf and President Pezeshkian shouldn't be discounted.

The system is prepared for leadership losses and continues to operate even with Mojtaba reportedly incapacitated. The IRGC is managing the day-to-day defensive response through a decentralized, mosaic command structure, while the technocratic civilian government runs the country. And should Mojtaba be removed, another leader would emerge, underscoring the regime's resilience and cadre of loyalists. The Islamic Republic has endured for 47 years because its leadership unites in times of crisis. This time is no different.

Ashley Rhodes: How and who could end this war?

Sanam Vakil: This war won't end simply because the US or Israel decide it should. The lesson Iran learned in the Twelve-Day War is that this conflict must end in a conclusive way that guarantees the Islamic Republic's longer-term survival, which entails securing an agreement to that effect. Iran is currently making extreme zero-sum demands, including that US military bases not be used to strike it. But Iran ultimately needs sanctions relief, which will require negotiations in which both it and the US make concessions on longstanding issues—including Iran's nuclear program and broader regional security. Until Iran sees a credible path to those guarantees, it has little incentive to end this war.

Ashley Rhodes: How likely is a domestic uprising that brings an end to the conflict?

Sanam Vakil: Historically, sustained protests or mobilizations are rare during wartime, and this conflict has allowed the regime to intensify internal security, restrict movement, and suppress dissent under the banner of national defense. In that sense, the war actually works *against* the emergence of an uprising, which no doubt factors into the regime's motivations for continuing the war, particularly with the brutal way the massive January protests have been handled on people's minds.

While most Iranians remain deeply dissatisfied with the system and fearful that the war will further entrench the Islamic Republic, they lack the organizational and military capacity to mobilize. Revolutions require unity and planning—qualities absent in the fragmented opposition. And even if the regime emerges from the war militarily weakened, it will likely remain capable of suppressing internal dissent. So, I don't see a scenario where domestic unrest hastens the end of the war.

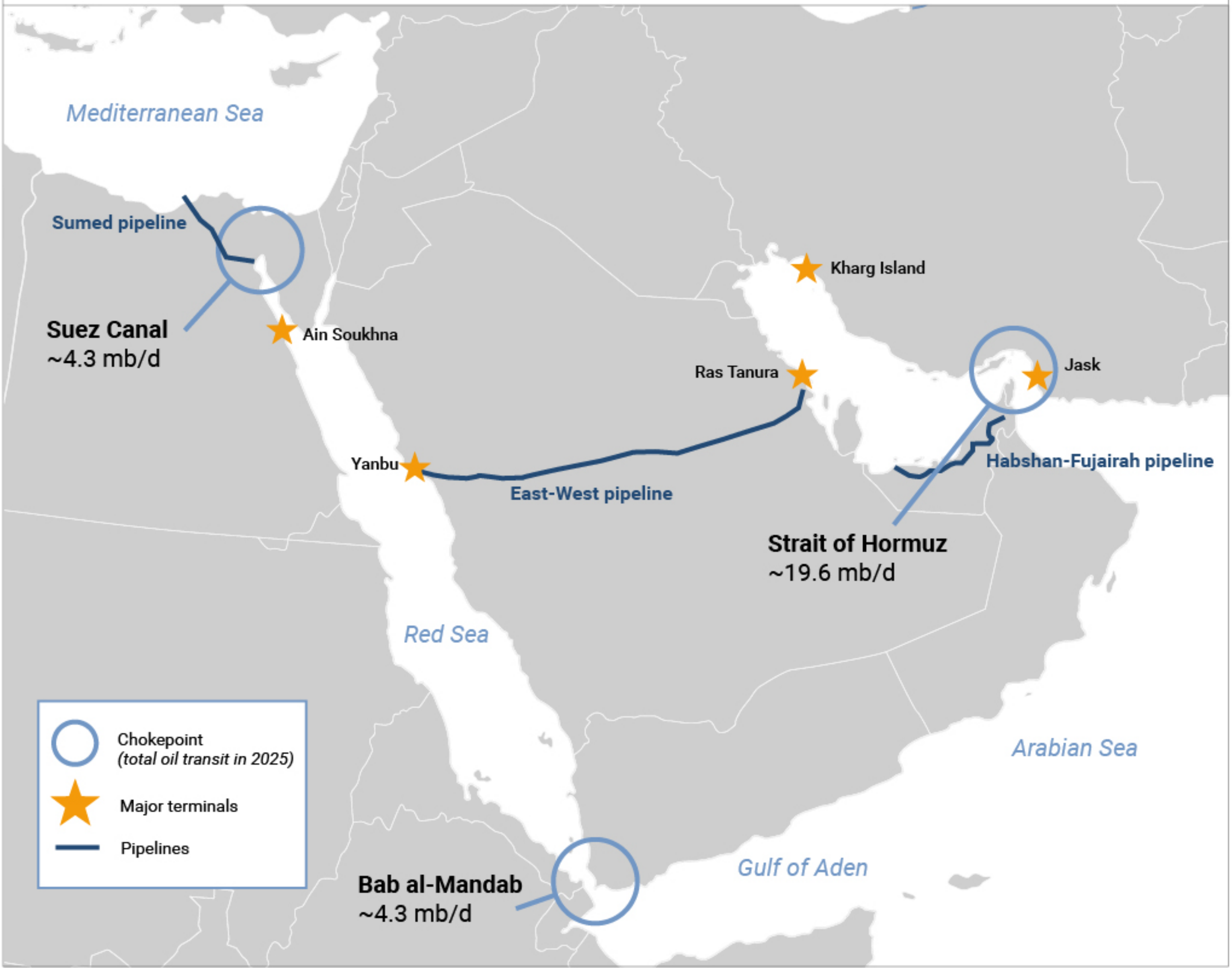
Ashley Rhodes: What are you watching most closely to determine when and how this conflict could end?

Sanam Vakil: First and foremost, I'm watching Iran's appetite for further escalation. Although Iran initially focused its efforts on disrupting Gulf economic activity and effectively closing the Strait of Hormuz, it recently broadened its approach by launching direct attacks on regional energy infrastructure.

I'm also watching for any Iranian statements that indicate interest in an off-ramp and, importantly, the specific terms it asks for. A softening of Iran's current out-of-left-field demands may indicate a shift in Tehran's cost-benefit calculus and reveal that Iran is ready to enter serious negotiations. But for now, all evidence suggests that the conflict will persist—Iran isn't yet signaling that it wants out, and the US has yet to achieve an outcome it can plausibly claim as a victory. Until one or both of those dynamics meaningfully changes, we should expect this war to continue.

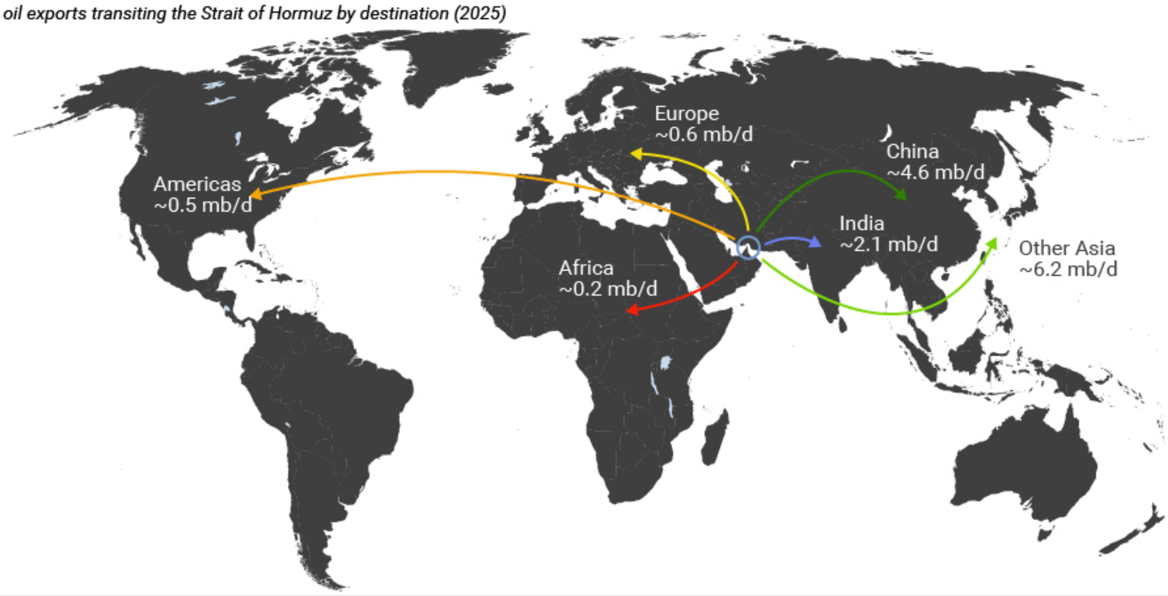
Middle East oil flows, mapped out

Although the Saudi East-West Pipeline and the UAE Habshan-Fujairah route can redirect some oil flows amid disruptions to the Strait of Hormuz, their estimated spare capacity remains low...



...and is likely insufficient to compensate for the large volume of oil normally transported through the Strait to the rest of the world

Crude oil exports transiting the Strait of Hormuz by destination (2025)



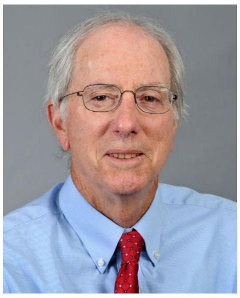
Note: Top map does not show all energy infrastructure in the region.
Source: Bloomberg, IEA, Kpler, Library of Congress, Goldman Sachs GIR.

Special thanks to GS Commodities Research analyst Alexandra Paulus for data.

Interview with Dennis Ross

Ambassador Dennis Ross is Fellow at The Washington Institute for Near East Policy. Previously, he served as Director of Policy Planning under President H.W. Bush, Special Middle East Coordinator under President Clinton, and National Security Council Senior Director for the Central Region under President Obama. Below, he argues that control of the Strait of Hormuz will remain key in determining how and when the conflict ends.

The views stated herein are those of the interviewee and do not necessarily reflect those of Goldman Sachs.



Allison Nathan: There are conflicting narratives about how the US and Israel came to go to war together for the first time. What's your view?

Ambassador Dennis Ross: On both the far-right and the far-left of the US political spectrum there is a narrative that Israel dragged the US into this war. Destroying Iranian capabilities that threaten Israel and the broader region has been a strategic priority for Israeli Prime Minister Netanyahu for 30 years. I refer to it as his "Winston Churchill issue." And he has undoubtedly made the case for this to President Trump. So, it's possible that Netanyahu influenced Trump's thinking. During Trump's first term, Israel, in a covert operation in Tehran, seized 55,000 documents—the "nuclear archive"—that detailed Iran's nuclear program. This archive demonstrated that, contrary to its denials, Iran had been developing nuclear weapons—with plans for at least five such weapons. So, perhaps Netanyahu was able to influence Trump's thinking with the intelligence that Israel provided in the past and more recently.

But Trump made the decision to go to war. It was Trump who built up a huge armada in the region and said in response to the Iranian protests in January that the US was "locked and loaded" and "help was on the way." In fact, the day after US Secretary of State Rubio explained the imminent need for the US to strike Iran because Israel was planning to do so and the US would be a target for Iranian retaliation, so better for the US to strike first, Trump disagreed, saying instead that he probably forced Israel's hand. I've never seen anybody who determines what President Trump does other than himself.

If anything, it's Netanyahu who cannot say no to Trump. Trump has all the leverage in the world on Netanyahu. He has put Trump on a pedestal for all the help and support he has provided to Israel over the years, including a readiness to attack Iran last June, which no other US president had been willing to do. So, when Trump decides, Netanyahu responds. For example, when Iran fired missiles at Israel just after a ceasefire in the Twelve-Day War was supposed to go into effect, and Israel launched 50 planes in response, Trump told Netanyahu to turn those planes around, and he did.

Allison Nathan: Iran poses an existential threat to Israel, but not to the US. So, why now for the US to strike Iran?

Ambassador Dennis Ross: I don't speak for the Administration and am humble about predicting Trump. But my interpretation is that the Administration saw this as a moment of opportunity. Although Trump said the US had obliterated Iran's nuclear capability last June, intelligence suggested that the Iranians were beginning to clean out some of their nuclear weapon sites, probably with an eye to reconstituting them. That created a perceived need to do something. But it was the absence of any significant air defense in the wake of the Twelve-Day War, which left Iran militarily vulnerable, that created a sense of opportunity. There was also the sense that the Iranian regime might be teetering given the massive January protests and the brutality with which the regime responded, which was seen as a sign of weakness. So, the Administration seemed to assume—wrongly—that taking out the Supreme Leader would lead to regime collapse and seized the opportunity to strike.

Allison Nathan: Some divergence seems to exist between the US and Israel's war strategy/goals. Is that true?

Ambassador Dennis Ross: I see more convergence than divergence between their war aims right now. Israel absolutely aims to end the current Iranian regime. That might be Trump's preference too, but this war will probably end without the regime collapsing, even if it has been terribly weakened. So, US and Israeli goals likely diverge somewhat in that sense.

But their goals converge in the shared conviction that Iran's military capabilities must be damaged sufficiently to avoid the need for another war anytime soon. As I mentioned, Iran was already beginning to return to its nuclear sites and had also recovered most of its ballistic missile production capability since the Twelve-Day War. Money and weapons were flowing back to Iran's proxy network, even as the proxies have proven largely ineffective yet expensive to support. So, the Iranians were acting as if the dramatic weakening of the proxies and the Twelve-Day war had never happened; whatever the cost, they had learned no lessons and were keeping the same strategy. Both Trump and Netanyahu wanted the Iranians to understand that strategy would end one way or the other. As such, the US and Israel's key shared objective in this conflict is to weaken Iran to the point that even if it wanted to reconstitute its proxies, nuclear program, and ballistic missile and drone capability, the cost would be simply too great. Bottom line: the goal is to ensure Iran cannot practically pose a threat to Israel and others in the region for at least five years, if not longer.

Allison Nathan: Iran seems to be pursuing a strategy of inflicting as much pain as possible in the hopes of securing a better outcome for itself. Is there a point at which the costs of the war will be too great for the US/Israel to bear?

Ambassador Dennis Ross: This is an area where I see potential for a real divergence between the two countries. The Israelis can continue this conflict for as long as it takes. While Netanyahu doesn't have strong support in the country writ large, he has almost 100% support for dealing with the Iran threat. Remember, Ayatollah Ali Khamenei had a clock ticking down to the end of the Israeli state in downtown Tehran. The Israeli public has been sleeping in shelters every night for fear of missile strikes from Iran or its proxies. The attitude is do enough that this won't be necessary again for a long time.

That's not the case in the US. This conflict marks a rare instance of the US going to war without the support of the majority of the public. So, Trump isn't benefiting from a "rally around the flag" effect, which may owe partly to the lack of any serious effort on the Administration's part to explain what it's doing and why. That said, the President likely feels he has some license to continue for now. I'm not sure that those around him who are worried about affordability given surging energy and fertilizer prices, which will lead to higher gasoline, airline, food prices, and beyond, agree. But my sense is that the President will care more about the inflationary implications of the conflict as the midterm elections approach. So, he has some license but probably also some endpoint in mind.

Allison Nathan: Can the US/Israel end the war unilaterally?

Ambassador Dennis Ross: The key to the US being able to walk away is the Strait of Hormuz. On his own, Trump probably would have already declared victory, pointing to the amount of damage the US and Israel have done to Iran's key targets, naval, missile, and drone capabilities, and nuclear weapons program. He likely can rightly claim today that Iran won't be able to pose a conventional threat to its neighbors for at least five years. But he can't say he's won and stop as long as Iran controls who can and can't export oil and sail through the Strait.

At this point, the Iranian threat has been severely diminished; it's firing a relatively small number of ballistic missiles and even drones every day. Some of this is Iran holding back to preserve an ongoing capability to fire missiles and drones to try to outlast the US. But Iran also doesn't have the capability to fire much more in no small part because the US can act preemptively once launchers emerge. Given the amount of firepower the Iranians still retain, they are likely capable of continuing to fire at their current rate for at least several more weeks, if not longer. So, Iran's capacity to threaten hasn't disappeared, but has been severely reduced. If this, combined with a convoy system that helps protect ships, enables the US to wrest control of the Strait away from Iran, Trump would be able to declare an end to the war. And even if the Iranians don't agree, there wouldn't be much they could do about it. Of course, many questions remain about whether the US can achieve such control. It may not be possible to stop Iran from threatening ships without seizing territory along its coast. A US Marine Expeditionary Unit is currently en route to the Middle East, which may open the door for island seizures, and the US and some allies have seemingly stepped up their efforts to reopen the Strait.

Allison Nathan: If the US can't gain control of the Strait, would a quick end to the war still be possible?

Ambassador Dennis Ross: Barring wresting control of the Strait, the only way the conflict resolves soon is through mediation. Russian President Putin seems well-placed to serve as the mediator since Iran doesn't really have any other friends, so it has incentive to respond to him, and Putin also has a direct line into Trump, who also seems to be somewhat responsive to Putin. Both sides have something to offer in a negotiation. Iran could cede control of the Strait and pledge to not close it again and hand over its 440 kilograms of highly enriched uranium—or more likely allow the Russians to excavate it from its current position, buried deep under rubble, making it a massive project for anyone—which would allow Trump to say he's put an end to the Iranian nuclear program for a very long time. In return, Iranians—as well as the Russians, if they do help—would seek guarantees of no new attacks on Iran as long as they continue to comply with the agreement as well as some sanctions relief.

But the conditions to enter a mediation don't seem to be in place at this point as both sides further harden their positions. My sense is Trump would still need to show a changed reality in the Strait before any mediation could occur to avoid looking like the one who needs a way out. He is sending even more forces to the region in the hopes of gaining the whip hand to that end. But Iran also wants to show it's not the one who needs to end the war, even as the costs continue to grow. So, mediation seems more likely than a unilateral end as long as the Strait is in play, but it doesn't seem imminent.

Allison Nathan: Who would Putin mediate with in Iran?

Ambassador Dennis Ross: If one person embodied the leadership in the wake of Ali Khamenei's death, it wasn't Mojtaba Khamenei but Former Speaker of the Parliament Ali Larijani, who was recently also killed. Larijani had relationships with every regime faction, including the Islamic Revolutionary Guard Corps (IRGC). If anyone could orchestrate a move to end the war through mediation, it would have been him.

Although Mojtaba Khamenei has received much attention, he seems unlikely to be able to speak for the regime. His physical state remains unclear. Even if we assume he is physically ok, he lacks authority. He was his father's son but had no public persona or religious credentials that would have made him an ayatollah. He ran the Supreme Leader's office, which had significant power and tentacles in every institution and facet of Iranian society, but he was just an implementer. President Pezeshkian also doesn't have the same access as Larijani did to the different factions nor his authority or credibility. Perhaps Mohammad Bagher Ghalibaf, the current Speaker of the Parliament, could provide some weight, but that's unclear. So, a hole currently exists in the Iranian leadership lineup, which reduces the prospect of a mediated outcome in the near term.

Allison Nathan: So, what are you watching from here?

Ambassador Dennis Ross: I am watching the Russians and the Chinese. As we discussed, the Russians can play a major role as mediator. The Chinese, though, can pressure the Iranians. The Iranian position has always been that if they can't export, nobody can. But right now, the Iranians are saying that they can export and will determine who else can. The minute the US takes the stand that Iran won't be allowed to export if it continues to prevent others from doing so, China would likely strongly object given its dependency on Middle Eastern energy supplies and would pressure the Iranians to change course. Again, at the end of the day, this all comes down to the Strait of Hormuz. Trump cannot end the war as long as Iran controls it. So, either the US does something dramatic to deny Iran that leverage or waits until conditions shift enough to allow for a mediated outcome. That's what we should be watching for.

Iran war: US political pressures

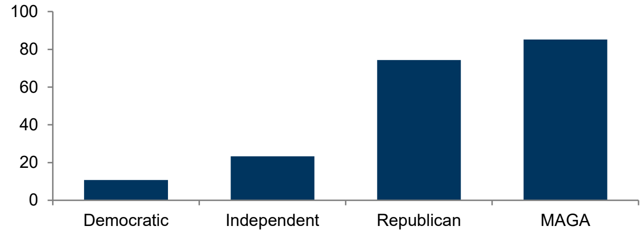
Alec Phillips explores US political pressures amid the Iran war and the Administration's response to the energy price shock

President Trump and his cabinet have consistently indicated the Iran war will likely last 4-6 weeks. On March 15, Energy Secretary Wright said it would end "in the next few weeks" and on March 17, Trump described the timetable as "a couple more weeks, not much longer." While US military objectives—elimination of Iran's nuclear program, missile capabilities, and naval assets—will likely drive US decisions, the economic and political costs of the war could become constraints.

So far, the lack of broad public backing for the war does not appear to be weighing on support for the Administration or congressional Republicans. Polls conducted before and after the first US strikes show Trump's approval rating holding steady at 42.6%, while support for Republicans in this year's midterm election is also roughly unchanged (Democrats still lead by 4pp). Importantly, President Trump's base appears to broadly support the war, with self-reported MAGA supporters viewing the war even more favorably than Republicans overall.

President Trump's base appears to broadly support the war

Support for the war/strikes on Iran, %



Source: YouGov, NBC, CNN, Angus-Reid, Goldman Sachs GIR.

But public opinion may become a constraint if retail gasoline prices, currently hovering around $4/gallon (national average), rise further. Financial and commodity markets could also play a role. On March 17, President Trump indicated that markets influenced his thinking on striking Iran, citing record equity market prices as providing an opening to enter the conflict.

A soft check on the war's duration might come from the War Powers Resolution of 1973, which requires the president to obtain congressional authorization for the use of military force within 60 days. While unlikely to force a change in military strategy, nearing the 60-day mark could raise public pressure to clarify objectives and endpoints.

Trump also signaled that his planned early-April meeting with Chinese President Xi would be postponed by around six weeks. With this meeting seemingly unlikely to occur while the conflict remains ongoing, the timing might also provide some insight into how Trump is thinking about the duration of the conflict.

In the meantime, the White House has taken a few steps to lower energy prices:

- **Release from the Strategic Petroleum Reserve (SPR):** In coordination with IEA members, the US announced a draw of 172mb over 120 days, or an average of 1.4mb/d. In theory, the total amount or daily rate could increase, as the SPR holds 415mb with a theoretical maximum draw rate of around 4mb/d for 30 days and around 2mb/d after 60 days. On March 19, Treasury Secretary Bessent suggested US SPR releases could exceed the recently-announced amounts. However, we expect the Administration to be cautious for now, as US SPR stocks are already below 60% of capacity following the 2022-2023 drawdowns and would drop to 33% by mid-year under the recently-announced plan.
- **Sanctions relief:** Treasury waived sanctions on purchases of Russian crude and refined products on the water as of March 12 (estimated at 130mb). On March 19, Bessent suggested the US might similarly waive sanctions on Iranian oil on the water (estimated at 105mb). Treasury also provided a general license to purchase Venezuelan oil that, unlike the Russia-related waiver, is not explicitly temporary and applies to new production as well as to oil on the water.
- **Jones Act waiver:** The 1920 Jones Act prohibits shipping between US ports by foreign-built/flagged/owned vessels. The Administration waived this rule for 60 days, which could ease oil and refined products transport from the Gulf Coast (including the SPR) to other US regions (particularly the East Coast). This might incrementally reduce fuel prices and also signals the Administration's willingness to temporarily break with policies it supports to address higher energy costs.

Further steps are possible. The Administration seems likely to waive limits on gasoline's ethanol content in summer, allowing for a greater share of lower-cost ethanol in gasoline blends.

The President also likely has the authority under IEEPA to ban the export of crude and refined products and might become more inclined to use it if disagreements with allies reemerge over military assistance in and around the Persian Gulf. That said, Wright said on March 12 that an export ban was not under consideration, and media reports indicate White House officials reiterated this to oil executives on March 19. It is also far from clear that a ban would benefit US consumers, and it could discourage domestic shale oil production. So, while very plausible, we don't view such a ban as the base case.

Other policy changes would require congressional support. The most obvious would be a suspension of the $0.18/gallon federal gas tax, which raises around $30bn/yr. We think the odds lean against it, as a good chance exists that prices would have started to fall by the time Congress could agree. While we expect a few states to consider suspending their gas taxes, which collectively raise $42bn/yr, as some did in 2022, a broad suspension seems unlikely as states no longer have the fiscal cushion they did following Covid-related federal fiscal relief.

Beyond fuel taxes, the recent developments look likely to restart talk of a second party-line fiscal package passed through the reconciliation process, but we remain skeptical that major legislation along these lines will pass this year. Trump looks likely to soon request $100-200bn in supplemental defense funding, but Democrats will likely object and thin Republican majorities make passage of a party-line reconciliation bill difficult. However, if such a spending package gains political momentum, it could become a vehicle for other items like a gas tax holiday or affordability-focused measures. For now though, we do not expect such a package to pass this year.

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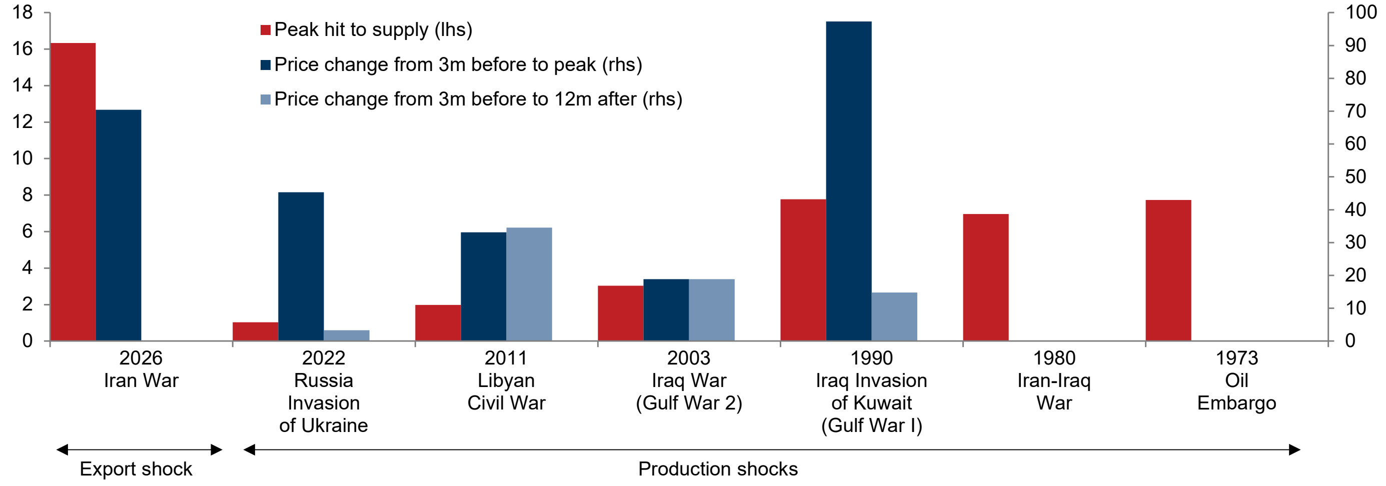
The oil supply disruption, visualized

The size of the current hit to oil flows from the Persian Gulf marks the largest oil supply shock in history…

Historical oil supply hit events (% of global supply, lhs) vs. corresponding price change (%, rhs)

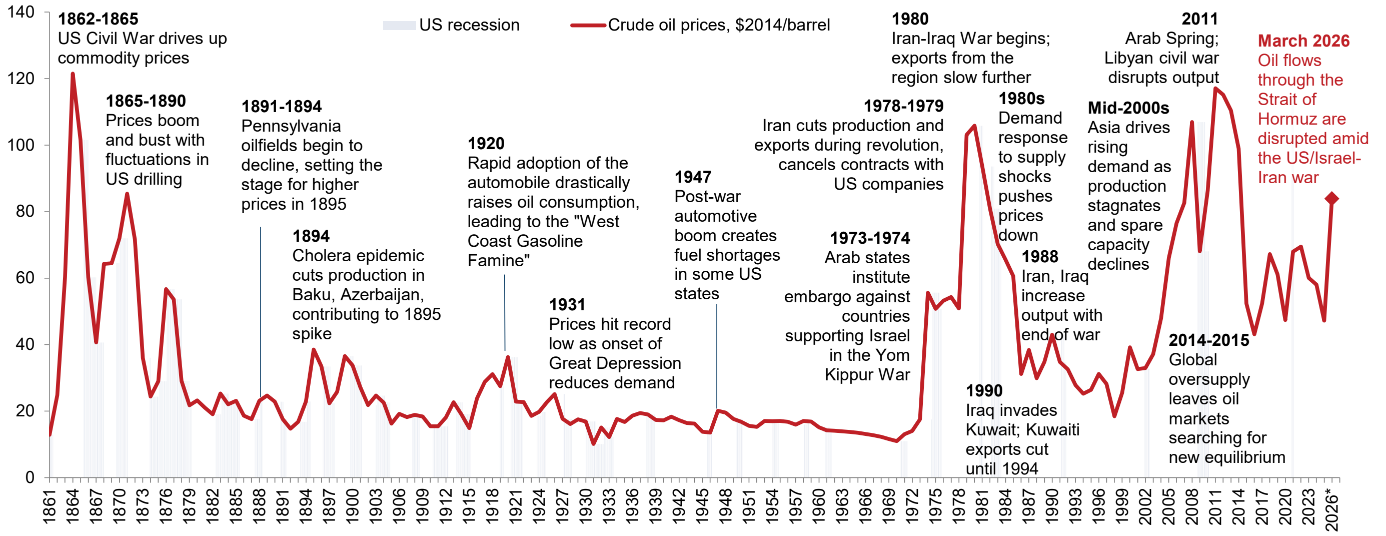


Note: Peak price is the highest daily close price. For the 2026 Iran War, peak hit to supply is defined as the estimated latest hit to Persian Gulf oil exports based on reported vessel count; peak price is taken to be the highest realized daily close price since February 28.

Source: IEA, ICE, Goldman Sachs GIR.

…which has led oil prices sharply higher, although the oil price response is so far smaller than during the 2003 Iraq War and 1980 Iran-Iraq war…

Crude oil prices, $2014/bbl

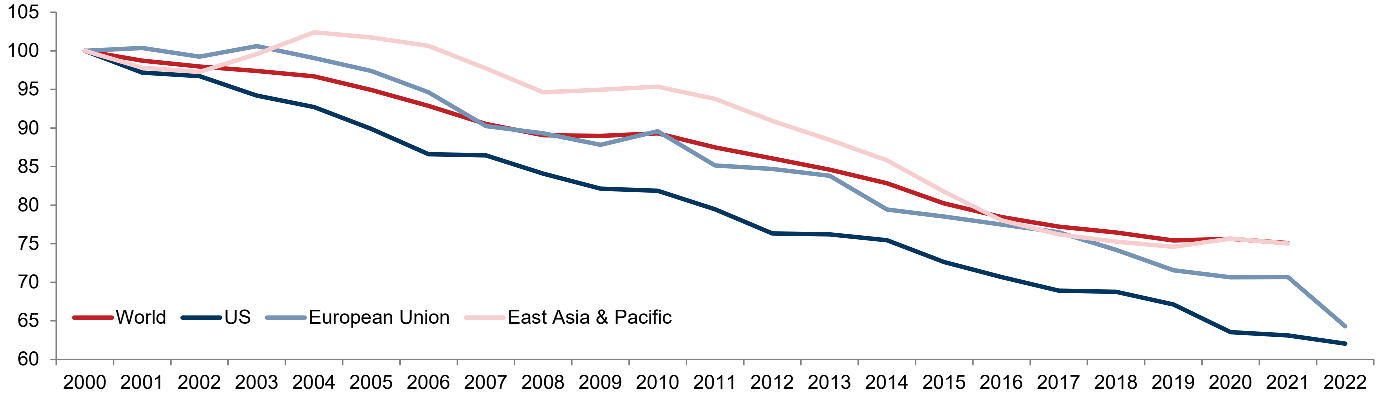


Note: 2026 is year-to-date data.

Source: Bloomberg, BP, Federal Reserve, Haver Analytics, various news sources, Goldman Sachs GIR.

…potentially as the world is much less vulnerable to oil price shocks now than it was in prior decades amid declining energy intensity

Energy intensity (megajoules per $2021 PPP GDP), indexed, 2000=100



**Data as of 2022 for EU and US; data as of 2021 for World and East Asia & Pacific.*

Source: World Bank, Goldman Sachs GIR.

Special thanks to GS Commodities Research analyst Alexandra Paulus for data.

Interview with Kevin Donegan

Vice Admiral Kevin Donegan, USN (ret) served as Commander of the US Navy's Fifth Fleet, Commander of the 32 Nation Combined Maritime Forces in the Middle East, and Director of the Navy Staff and Deputy Chief of Naval Operations for Operations, Strategy and Planning at the Pentagon. Below, he argues that the US has made significant progress in achieving its military objectives in the Iran conflict, but restoring Strait of Hormuz oil flows to normal levels won't be achieved through military convoys alone.

The views stated herein are those of the interviewee and do not necessarily reflect those of Goldman Sachs.

Jenny Grimberg: What are the US military objectives in the Iran conflict?

Vice Admiral Kevin Donegan: Despite all the political back and forth, the US military campaign's objectives have always been clear and stable: destroy the Iranian national security apparatus' ability to project power outside their borders by degrading Iran's extensive ballistic missile and attack drone capabilities and the naval capabilities used to hold the economic chokepoint of the Strait of Hormuz hostage. The US has unsurprisingly done a good job in defanging Iran and degrading those capabilities. The US' targeting of Iran's mobile launchers and command-and-control nodes and use of Low-cost Uncrewed Combat Attack Systems (LUCAS) has significantly reduced Iran's ability to launch ballistic missiles and drones—Iran went from launching 1,000 missiles on day one of the conflict to a few hundred each day after during the first week to now only around several dozen missiles a day. The US Navy has also sunk or damaged almost all Iranian Navy vessels.

Iran has retaliated in several ways, most of which were expected—the US always knew Iran would hold the Strait of Hormuz hostage and attack regional US military bases and Gulf allies' territory. What has been unexpected is Iran's willingness to attack Gulf nations' regional civilian infrastructure—hotels, airports, etc.—in an effort to drive the Gulf states to pressure the US to end the conflict. But the Gulf states haven't done so, either publicly or behind the scenes, and are instead more worried about the prospect of the US ending its military campaign too early. The last thing they want is for the conflict to end with a half-defanged Iran.

> "Despite all the political back and forth, the US military campaign's objectives have always been clear and stable: destroy the Iranian national security apparatus' ability to project power outside their borders."

Jenny Grimberg: Can the US ultimately "finish the job" without putting boots on the ground?

Vice Admiral Kevin Donegan: Yes. The US undoubtedly has enough firepower to accomplish its military objectives. The US initially used advanced at-range weapons like Tomahawk Land Attack Missiles (TLAMs) and extended-range missiles to gain air superiority over Iran. Now that it has air superiority and can conduct land, sea, and air operations without Iranian interference, it has shifted to using its ample arsenal of lower-tech and lower-cost weapons. The US' ability to achieve its objectives using its military hardware has never been in doubt.

What's likely impossible to achieve without boots on the ground is regime change. The first wave of US/Israeli strikes changed the regime's leadership, not the regime itself, which remains even after successive waves of strikes. The Islamic Revolutionary Guard Corps (IRGC), the core pillar of the regime, is deeply woven into Iranian society, from internal security to banking institutions to construction companies. It also has an internal security apparatus that oppresses the population and its own intelligence arm. And it controls Iran's missiles, drones, and sea mining capabilities. So, the IRGC isn't a traditional military, and the US would have to dismantle most of the Iranian government system to remove the IRGC, which would require ground forces. But again, regime change has never been a US military objective. President Trump and the Israelis have mentioned it, but the Commander of US Central Command (CENTCOM) Admiral Brad Cooper and Chairman of the Joint Chiefs of Staff General Dan Caine were never given this as an objective.

> "The US has unsurprisingly done a good job in defanging Iran and degrading [Iran's] capabilities."

Jenny Grimberg: Even if the US has gone a long way toward achieving its military objectives, won't Iran's asymmetric capabilities—its low-cost drones and sea mines—present ongoing challenges?

Vice Admiral Kevin Donegan: While the threat from ballistic missiles will likely be constrained as Iran remains leery of firing them for fear of exposing launch and storage sites, prolific weapons will remain a threat. Iran manufactures attack drones by the tens of thousands. This drone threat will be severely degraded but will never be zero. As a result, we can expect this threat, which is most effective against soft targets, to linger even after the US military operation ends.

Iran also has plenty of sea mines, ranging from the simple to the sophisticated, that it can threaten to lay in vital waterways. But the US and Israeli military campaign has destroyed many Iranian minelaying vessels and storage bunkers. So, Iran can no longer deploy sophisticated, anchored mines in specific locations, instead having to resort to deploying drifting mines from non-military vessels like fishing boats. These mines are unpredictable because they are a slave to the strong currents in the Strait of Hormuz and may not stay in the areas that are problematic for ships.

Laying sea mines in the Strait of Hormuz is also ultimately counterproductive to Iran's needs because the vast majority of Iran's oil exports flow through it and up to 60% of Iran's government budget comes from oil exports, mostly to China. For that reason, the rhetoric that has come out of Iran is not that it has closed the Strait, but that it *controls* the Strait.

> "While the threat from ballistic missiles will likely be constrained as Iran remains leery of firing them for fear of exposing launch and storage sites, prolific weapons will remain a threat."

Jenny Grimberg: If Iran wants to continue causing trouble in the Strait of Hormuz, is it practical to think that convoys could successfully transit through it?

Vice Admiral Kevin Donegan: I'm confident the US and its allies have the capability to escort commercial ships through the Strait. The US has long rehearsed how to keep the Strait of Hormuz and the Bab al-Mandab Strait open, and under much more challenging conditions than exist today in terms of Iranian capabilities. The most recent such exercises took place in early 2025 with 30 nations participating.

The US can obviously deploy meaningful capabilities, but so can many other countries, whether it's ships, staff, or experience. The US 5th Fleet Commander knows exactly what each nation can contribute because this exercise has been rehearsed annually for the past 15 years. France and Australia have very capable ships, and France has experience owing to its fight against the Houthis in the Red Sea. Greece has also received substantial training because of the Houthis in particular. The UK, Germany, Japan, South Korea, Turkey, Italy, Spain, and Egypt are also equipped to participate to varying degrees.

The basic idea would be to create a mobile sanctuary of security as the ships transit the Strait, which would include air cover, ships, drones, satellite assets, and intelligence. Securing a convoy is actually more achievable today than what's been planned for because of the US' air superiority over the Iranian coast and the severe degradation of Iranian military capability. And while Iran, even with no radar capabilities, will be able to locate the convoys, which would certainly be targets, the US will likely have little tolerance for hostile boats in the water near the convoys and is well-equipped to deal with the residual Iranian capabilities launched from shore to attack the convoys.

Whether the US and other countries have the political will and desire to participate in a convoy is a different question, and one we're currently seeing play out. The Trump Administration said that restoring the free flow of commerce through the Strait of Hormuz should be everybody's job. While many countries initially refused or were noncommittal, the leaders of the UK, France, Germany, Italy, the Netherlands, and Japan have since issued a joint statement expressing their "readiness to contribute to appropriate efforts to ensure safe passage through the Strait." Caine has also said that some allies are using Apache attack helicopters to "handle one-way attack drones" as part of a stepped-up effort to reopen the Strait. How this issue evolves bears close watch.

Jenny Grimberg: Even if convoys could allow some ship passage through the Strait, could they do so at the scale required to restore the flows the global economy needs?

Vice Admiral Kevin Donegan: No. It is important to distinguish between capability and capacity. Convoys would never be able to restore oil flows through the Strait to normal levels. They don't scale well because ships must be escorted out as well as back in.

So, convoys may only be able to restore potentially 20% of normal oil flows, with overland pipelines adding another 15-20%, though this will depend significantly on how the conflict ends and how Iran reacts to the first few escorts. The bottom line is that no on-off switch exists here—the conflict won't stop one day and oil flow normally the next. A negotiated ceasefire would be the fastest path to restoring normal flow. And even then, shipping, insurance, tanker, and LNG companies would need time to gauge Iran's behavior and gain trust before resuming transit.

> "I'm confident the US and its allies have the capability to escort commercial ships through the Strait... [but] it is important to distinguish between capability and capacity. Convoys would never be able to restore oil flows through the Strait to normal levels."

Ultimately, restoring oil flows to pre-conflict levels will require Iran to stop harassing ships in the Strait. It may not be incentivized to do so now, especially if the IRGC is currently making decisions as opposed to more seasoned political figures like the late Ali Larijani that tend to think more about the longer-term picture. But my sense is that eventually the Iranian regime will, because it needs cash and sanctions relief to appease its population and wants to ensure its survival, which is its singular most important goal.

The US also has cards to play. It has the capability to hold crucial Iranian flows from Kharg Island hostage. The US could also stop the "shadow fleet" carriers on either side of the Gulf that are moving Iranian oil. And regional Gulf nations like Oman and Qatar can also apply pressure. So, restoration of flows is not so much a question of convoys but of incentives and leverage on all sides, and no switch exists to just turn on flows when the conflict ends.

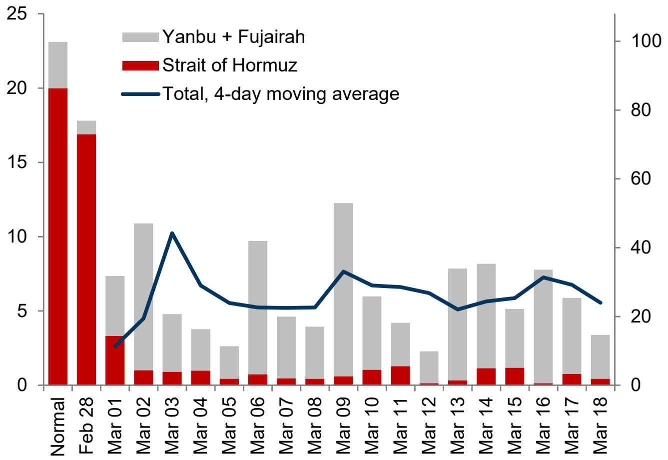
Sizing the energy shock

The GS commodity research team explores energy disruptions amid the Iran war

As the world continues to digest the largest shock to energy supplies in history, we assess the impact on prices, finding that price risks are skewed to the upside both in the short term and over the longer term given the potential for lengthier disruptions and persistent supply losses.

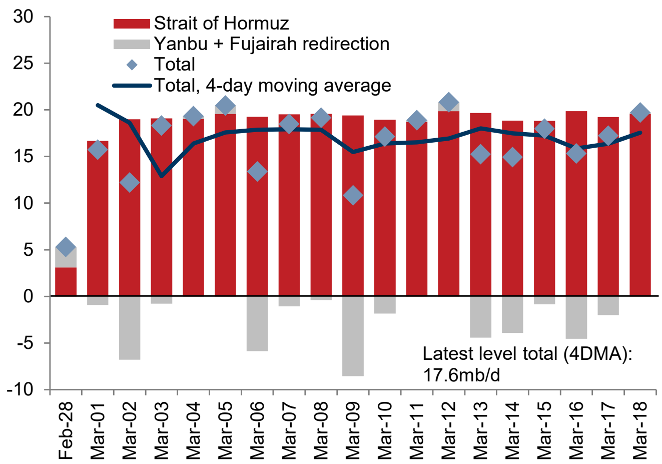
The estimated hit to oil flows from the Persian Gulf is 17.6mb/d, making today's shock the largest in history

Est. oil flows from Persian Gulf based on reported vessel count, mb/d (lhs), % of normal (rhs)



Note: Normal flows are assumed to be 20mb/d for Strait of Hormuz and 2025 average for Yanbu (1.4mb/d) + Fujairah (1.7mb/d).
Source: S&P Global Commodities at Sea, Kpler, Goldman Sachs GIR.

Est. hit to Persian Gulf oil exports based on reported vessel count, mb/d



Note: Hit is calculated relative to normal flows.
Source: S&P Global Commodities at Sea, Kpler, Goldman Sachs GIR.

Current state of the oil shock: the largest in history

We estimate that the hit to total oil flows from the Persian Gulf stands at 17.6mb/d (17% of global supply), making today's shock the largest in history and 18 times larger than the peak April 2022 hit to Russian oil production. Oil flows through the Strait of Hormuz are down 97% from their normal levels (0.6mb/d versus 20mb/d normal) as the physical risks of crossing the Strait remain very high, with the latest confirmed attack on an oil tanker occurring three days ago. Some flows are being redirected to pipeline-adjacent ports outside of the Persian Gulf, most notably Yanbu (Red Sea, Saudi Arabia) via the Saudi East-West pipeline and Fujairah (Gulf of Oman, UAE) via the Habshan-Fujairah pipeline, but the extent of redirection is capped by the combined spare capacity of the two pipelines—our latest estimate of net redirections stands at just 1.8mb/d.

Near-term oil prices: high risk premium required

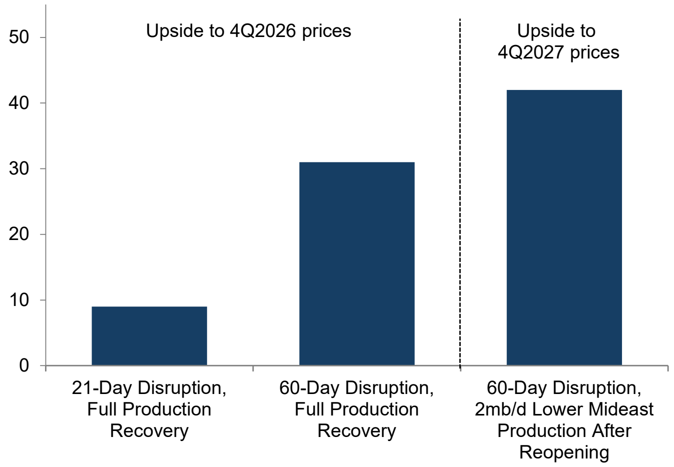
We expect oil prices to continue trending higher in the short term, until the market gains more certainty about how long the disruption will last, especially in light of increasing strikes on energy infrastructure. This risk premium is required to generate enough precautionary demand destruction to hedge against the risk that commercial oil inventories might fall to critically low levels. Brent is likely to exceed its 2008 all-time high if depressed flows keep the market focused on the risk of lengthier disruptions. Any rise in market-perceived risks of US export restrictions would likely further widen the Brent-WTI price gap.

Medium-term oil prices: the longer, the higher

Given the fluidity of the situation, we consider three scenarios to assess the potential direction of oil prices over the medium term.

Given uncertainty around the longevity of the conflict and associated oil disruptions, we consider three potential scenarios

GS estimated impact on 4Q2026/4Q2027 Brent oil prices under different Strait of Hormuz disruption and recovery scenarios, $/bbl



Source: Goldman Sachs GIR.

For the first two scenarios, we estimate 2026Q4 prices, relative to a no-Iran shock counterfactual of $62/bbl, as a function of 1) cumulative hits to production and commercial oil inventories, 2) global policy responses, and 3) a boost from positioning.

In a (most optimistic) first scenario where flows through the Strait of Hormuz gradually recover from here and return to pre-war levels within a month, we would expect prices to average $71/bbl in 4Q2026. This supply shock would translate into a 6% (617mb) hit to global commercial oil inventories. The release of strategic petroleum reserves (SPR) by IEA member states and China, as well as the uptake of Russian oil on water, would likely offset 50% of the hit, translating into $6/bbl upside to fair value prices. We assume positioning would provide an additional $3/bbl boost, bringing our estimate of 4Q2026 price upside to $9/bbl.

In a second scenario where flows remain disrupted for 60 days (through April 28th), we think 4Q2026 prices would average $93/bbl, as the supply shock would lead to a significantly larger nearly 20% (1816mb) hit to global commercial oil inventories. Even with increased releases of global SPR (although constrained by logistical speed limits) and Russian oil on water, policy responses would likely only offset about 30% of the hit, translating into upside of $26/bbl. Combined with an additional $5/bbl boost from positioning, we estimate 4Q2026 price upside at $31/bbl.

Finally, in a third, more extreme scenario where Hormuz flows remain very low for 60 days and significant scarring to long-run supply occurs (specifically, that Middle Eastern production remains 2mb/d lower after the Strait opens), we estimate that prices would reach about $110/bbl in 4Q2027, consisting of over $40/bbl of upside relative to a no-Iran shock counterfactual of roughly $70/bbl.

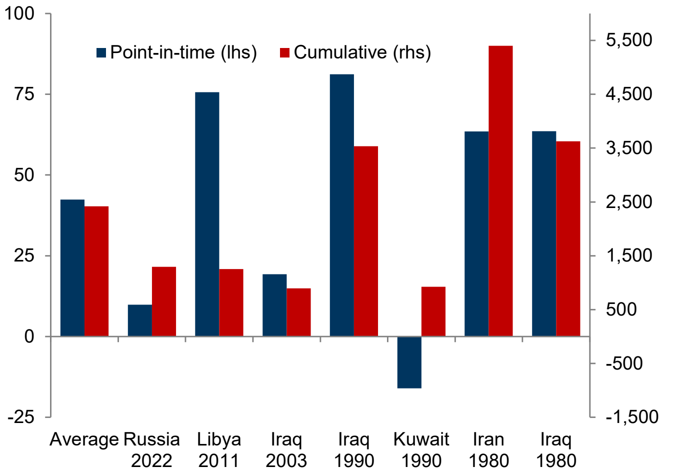
Risks to long-term oil prices

We also consider four risks to long-term oil prices—two on the supply side and two on the demand side.

On the supply side, the primary long-term risk to prices is the potential for oil supply to remain low for longer if production potential is damaged, which would pose upside price risk. The five prior largest supply shocks (Iran-Iraq War, Gulf War I, Gulf War II, Libyan Civil War, and Russian invasion of Ukraine) suggest that production four years after the shock remains over 40% below normal levels, largely driven by damage to oil infrastructure and low investment. This risk could be particularly significant to today's shock given that Iran and the seven other Persian Gulf countries collectively produced 30% of the world's crude oil in 2025. Additionally, roughly 25% of the region's crude oil is produced offshore, where engineering and geological complexity requires significantly longer ramp-up times than for onshore production.

The five prior largest supply shocks suggest that production four years after the shock remains over 40% below normal levels

Hit to oil production in affected country 4yrs after shock, % (lhs), mb (rhs)



Note: We estimate point-in-time production hit as percent decrease in a country's production four years after the shock relative to the pre-shock oil production. We estimate cumulative production hit as a total decrease in a country's production over the four years of the shock relative to the pre-shock oil production.
Source: IEA, EIA, Goldman Sachs GIR.

But oil supply could be higher than expected if OPEC deploys spare capacity, i.e., production that can be brought online quickly and sustained for a few months, to help stabilize the market, which would limit the upside to prices.

On the demand side, faster strategic stock building from 2027 on low end-2026 reserves and higher SPR targets in response to lingering uncertainty would pose an upside risk to prices. But slowing demand should high prices accelerate fuel efficiency gains and shifts to other fuels, or the energy shock weigh meaningfully on GDP growth, would pressure prices in the other direction.

On net, we think risks to oil prices are skewed to the upside in the short term and over the longer term. The persistence of several prior large supply shocks underscores the risk that oil prices may stay above $100/bbl for longer in risk scenarios with lengthier disruptions and large persistent supply losses.

European natural gas: risk of a tighter longer-term outlook

Similar to oil, on the natural gas side the recent escalation of events and the associated increased risk that supply disruptions linger have increased the risk premium embedded in European natural gas prices, which are now more than 90% above pre-war levels, at €61/MWh ($21/mmBtu). These price levels incentivize max gas-to-coal switching in power generation, but are not high enough to incentivize switching from gas into oil products.

Assuming Qatari LNG production begins to ramp up from early April, near-term prices would likely rise modestly, with the Apr/May TTF contract rising to €75/70MWh to reflect a higher risk premium and higher front-loaded demand destruction (gas-to-oil switching), until the market sees an actual increase in supply. As this early demand destruction creates a greater supply cushion, natural gas prices should be able to price gradually lower vs coal, with 2H26 TTF in the €30-40/MWh range.

However, the longer it takes for the ramp-up in Qatari LNG production to start, the longer European gas prices would likely stay in the oil switching range, above €70/MWh, to generate incremental demand destruction. A particularly longer disruption, with Qatari LNG production at zero for two months or longer, could lift prompt TTF contracts further still, likely close to €100/MWh ($35/mmBtu), to broaden industrial demand destruction across Europe and Asia in order to offset these supply losses ahead of the winter. And the long-term (3-5 years) damage to Qatari production capacity inflicted by recent attacks, worth 3% of global LNG supply, along with announced delays to the construction of Qatar's new LNG trains, would likely slow the largest ever LNG supply growth wave from 2027 we've expected, and delay the resulting significant downside pressure on European gas prices.

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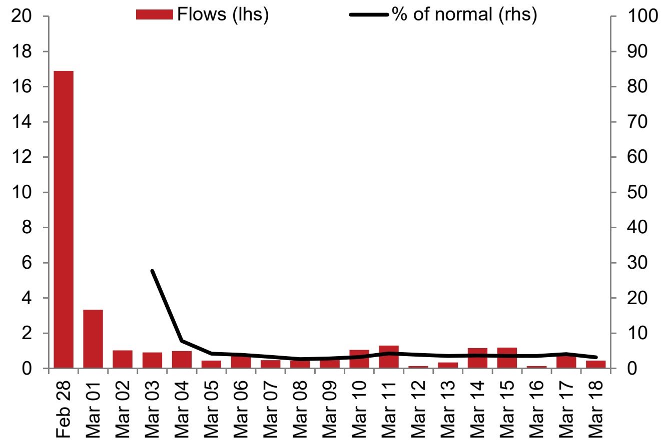
Disruptions caused by the Iran conflict...

Estimated daily oil flows through the critical Strait of Hormuz are down 90%+ from normal levels...

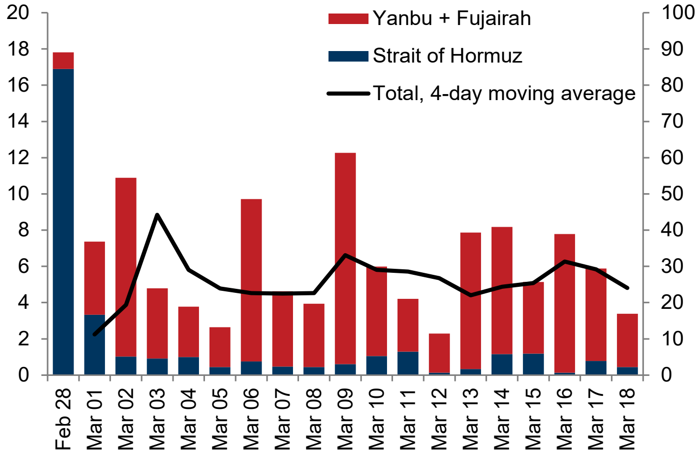
Estimated oil exports through the Strait of Hormuz based on reported vessel count, mb/d (lhs), % of normal (rhs)



Note: Vessel count as a % of normal is based on an average of S&P and Kpler data on the daily # of oil tankers crossing the Strait of Hormuz relative to the Jan 1-Feb 14 average for S&P and Feb 28 for Kpler. Flows are estimated by applying this proportion to assumed normal flows of 20mb/d (average over last ~8 years).
Source: S&P Global Commodities at Sea, Kpler, Goldman Sachs GIR.

...leading to increased redirection via the ports in Yanbu (Red Sea, Saudi Arabia) and Fujairah (Gulf of Oman, UAE)

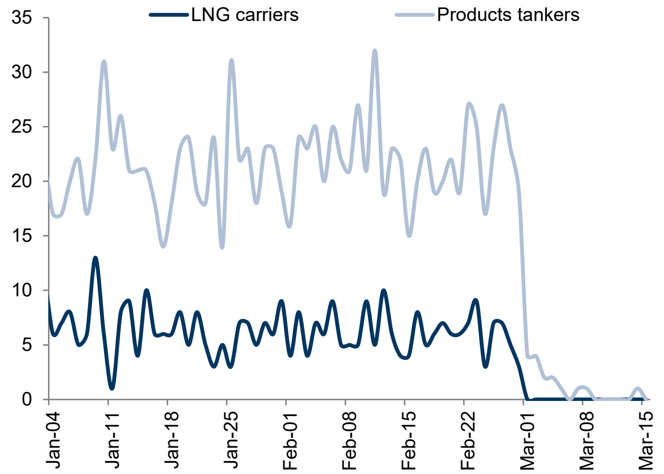
Estimated oil flows from Persian Gulf based on reported vessel count, mb/d (lhs), % of normal (rhs)



Note: Normal flows are assumed to be 20mb/d for the Strait of Hormuz and 2025 average for Yanbu (1.4mb/d)+Fujairah (1.7mb/d).
Source: S&P Global Commodities at Sea, Kpler, Goldman Sachs GIR.

Gas and refined product tankers have also largely stopped transiting the Strait of Hormuz amid the conflict...

Number transiting the Strait of Hormuz



Source: Clarksons Research, Goldman Sachs GIR.

...and shipments of chemical products from the Middle East, where sizable amounts of such products are produced, have been disrupted according to major trade publications

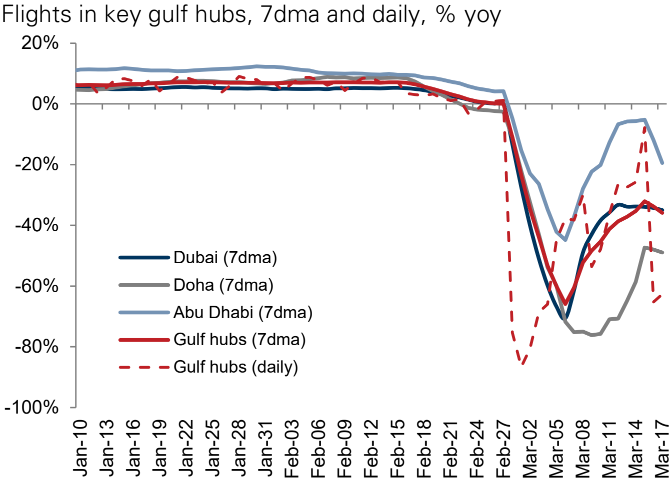
Regional exports by chemical chain as a share of global exports

Chemical product	Iran	UAE/Kuwait/Qatar	Eastern Saudi Arabia
Ammonia	~5% of global exports	~4% of global exports	>10% of global exports
Urea	>5% of global exports	>10% of global exports	>5% of global exports
Phosphate	N/A	N/A	>10% of global exports
Sulfur	<5% of global exports	>20% of global exports	>10% of global exports
Polyethylene	>5% of global exports	>5% of global exports	>5% of global exports
MDI	Small	N/A	>10% of global exports
Caustic	Small	<5% of global exports	Small
PVC	N/A	N/A	N/A
MTBE	<5% of global exports	<5% of global exports	>15% of global exports
TiO2	N/A	N/A	N/A
Methanol	>15% of global exports	Small	>15% of global exports

Source: ICIS, Argus, IFA, COMTrade, CMA, S&P Global, Goldman Sachs GIR.

The conflict initially resulted in a significant reduction in flights to and from the Gulf, though some of that has now recovered

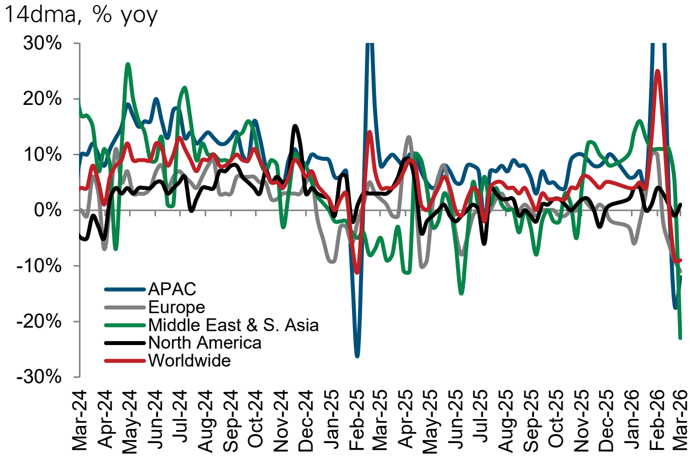
Flights in key gulf hubs, 7dma and daily, % yoy



Source: OAG, Flight Aware, Goldman Sachs GIR.

Global air cargo volumes, which Middle East airlines account for 13% of, have also fallen amid the conflict

Global top-20 forwarders' chargeable air cargo weight by origin, 14dma, % yoy



Source: World ACD, Goldman Sachs GIR.

...have led to increased prices

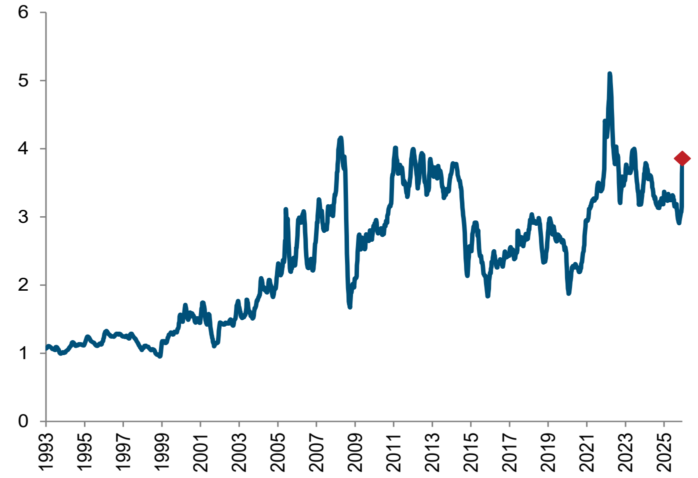
Oil prices have significantly increased, now exceeding $100/bbl, though they remain below the highs of the energy crisis

Brent crude oil prices, $/bbl

Source: Bloomberg, Goldman Sachs GIR.

US gasoline prices have also risen, from around $3/gallon before the conflict began to nearly $4/gallon currently

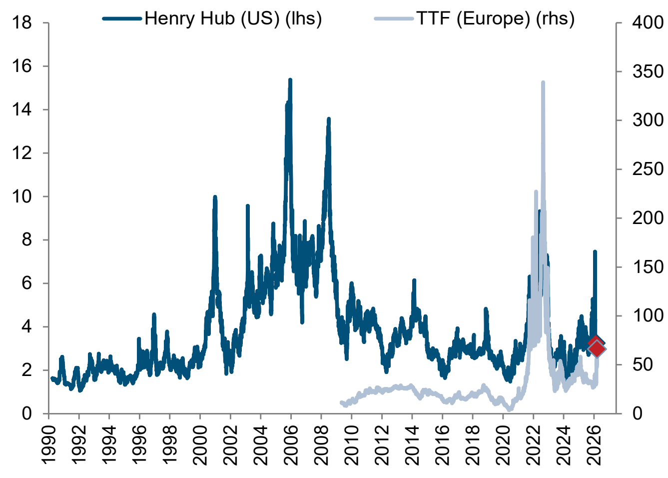
US all grades all formulations retail gasoline prices, $/gallon



Source: EIA, Haver Analytics, Goldman Sachs GIR.

European natural gas prices have nearly doubled since the conflict began, while US prices have risen only ~11%

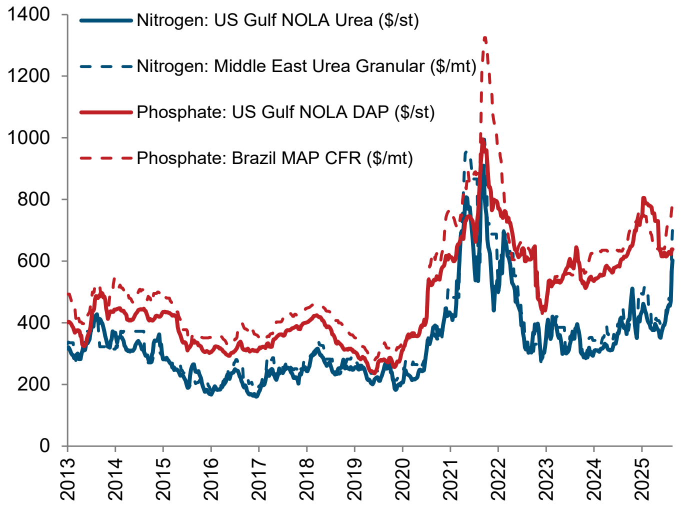
Henry Hub prices, $/MMBtu (lhs), TTF prices, €/MWh (rhs)



Source: Bloomberg, Goldman Sachs GIR.

Fertilizer product prices have increased

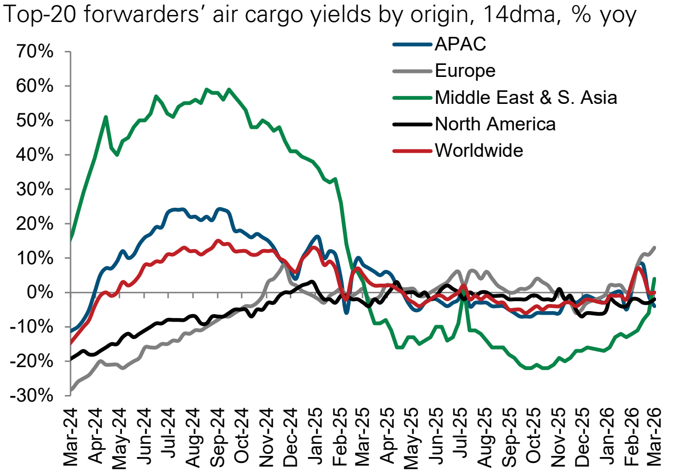
Spot prices of fertilizer products (key benchmarks for each)



Source: Bloomberg, Goldman Sachs GIR.

Air cargo rates have risen for cargo from Asia and Europe

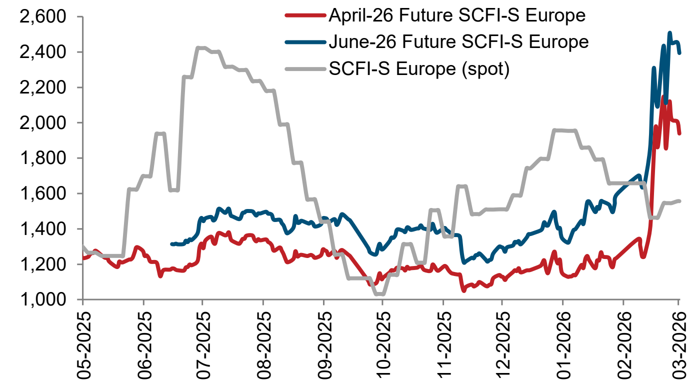
Top-20 forwarders' air cargo yields by origin, 14dma, % yoy



Source: World ACD, Goldman Sachs GIR.

In container shipping, future freight rates have increased as carriers have implemented fuel and emergency surcharges

Future SCFI-S (Shanghai Containerized Freight Index based on Settled Rates) Europe (futures contracts) (index, base=1,000 points as of 6/1/2020)



Source: Clarksons Research, Bloomberg, Goldman Sachs GIR.

Special thanks to GS Commodities Research team (Alexandra Paulus), US Chemical Equity Research team (Duffy Fischer and Jordan Lee), and European Transportation Equity Research team (Patrick Creuset and Nathan Arnaud) for data.

Iran conflict: growing economic impacts

Joseph Briggs assesses the global economic impacts from the ongoing Iran conflict

As the Iran conflict ends its third week, we continue to think that the biggest economic impacts will come through higher energy prices, which will slow growth and boost inflation in most countries, with the risks skewed toward more growth downside and inflation upside. But we see limited risk that the large second-round inflation effects that were a prominent feature of the 2021-2022 pandemic shock will re-emerge, as—unlike in 2021-2022—the major risk to global supply and inflation today is mostly confined to energy.

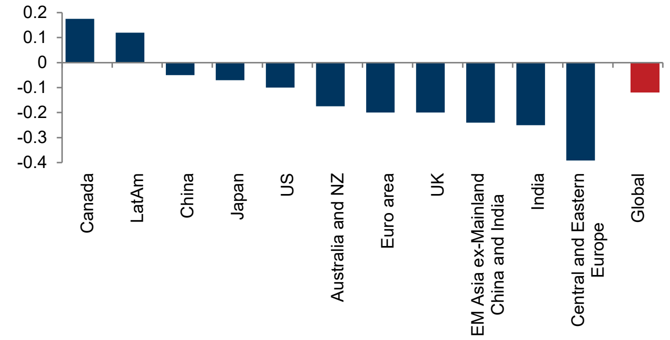
Economic impacts: rules of thumb

To gauge the economic impacts of the conflict, we rely on rules-of-thumb estimates of how higher oil prices affect GDP and consumer prices. On the growth front, our rules-of-thumb imply that each 10% increase in oil prices lowers global GDP by a bit more than 0.1%, reflecting real income and spending drags that are only partially offset by boosts to income and spending in oil exporters such as Canada and some LatAm economies.

On the inflation front, our rules-of-thumb imply that each 10% rise in oil prices raises global headline inflation by 0.2pp, with larger impacts in Asia ex-China and Europe, owing to increases in energy costs and indirect effects on downstream production. The core inflation impacts are smaller, ranging from 3-6bp for each 10% increase in oil prices depending on the country.

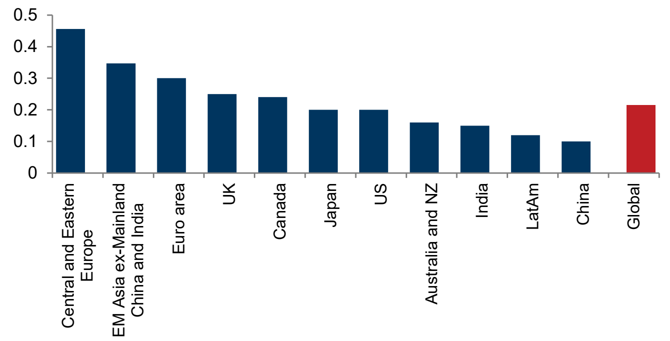
Each 10% increase in oil prices lowers global GDP by 0.1%...

GS rules of thumb effect of a sustained 10% rise in oil prices on real GDP, %



...and boosts global headline CPI by 0.2%

GS rules of thumb effect of a sustained 10% rise in oil prices on headline CPI, %



Source: Goldman Sachs GIR.

As such, a sustained rise in oil prices will likely have a sizable impact on the global economy. To level set, prior to the conflict, we expected global GDP growth of 2.9% this year on an annual average basis and headline inflation of 2.3% on a Q4/Q4 basis.

Our rules of thumb imply that the current three-week disruption will result in a 0.3% drag to global GDP and 0.5-0.6% boost to global prices. In a 30-day disruption scenario, we estimate a 0.5% headwind to global GDP and 0.9% boost to global prices. And in a 60-day disruption scenario, we estimate a 0.9% headwind to global GDP and 1.7% boost to global prices. And the longer the Strait of Hormuz disruptions extend, the greater the hit to growth and boost to inflation.

Even on top of these direct macro implications of higher oil prices, we see macro risks further skewed toward growth downside and inflation upside for two reasons. First, the disruption of Qatari gas supplies, which account for around 20% of global LNG supply, poses additional challenges for Asia and Europe, especially after the recent attacks on energy infrastructure. While the US remains relatively insulated owing to its status as the world's largest natural gas producer, Asia and Europe's heavy reliance on LNG leaves prices and industrial production more exposed. Second, the recent tightening in financial conditions—our global FCI has tightened by 51bp since the conflict began—could slow growth further if sustained.

Not your brother's supply disruption

The recent selloff in front-end rates signals an expectation that global central banks will be especially sensitive to higher inflation given the memory of post-pandemic supply disruptions and associated inflation surge. While the size of this energy shock will likely prompt some caution among policymakers on rate cuts, a key difference from 2021-2022 is that today's supply shock appears concentrated in the energy sector, whereas the energy price increases of 2022 were only one aspect of a much broader global supply chain crisis and inflation surge. We see three reasons why the impacts on non-energy supply will likely be less severe this time around.

First, non-energy trade with Gulf economies accounts for only 1% of global trade, with major DMs and EMs even less exposed. By contrast, the 2021-2022 China and East Asia trade disruptions affected 20% of global trade.

Second, while Gulf exporters account for a large share of global production for certain chemical and metal products—including helium, sulfur, methanol, polyethylene, aluminum, and fertilizers—Gulf production of these products accounts for less than 0.02% of global GDP. By contrast, disruptions to semiconductor production in 2022 affected trade worth ~1% of global GDP. Even more importantly, these product disruptions are unlikely to create shortages in critical, non-substitutable inputs that would lead to broader production shutdowns.

Third, the problematic shipping disruptions of 2021-2022 appear less likely to repeat. Gulf economies are not major re-exporters of most products (aside from boats), non-tanker ocean shipping costs have ticked down since the conflict began, and the recent rise in airfreight prices, while notable, implies only a modest uplift to global inflation (<5bp). Taken together, this limits the risk that the large second-round inflation effects observed in 2021-2022 will re-emerge.

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Markets: growth, the next shoe to drop?

Kamakshya Trivedi argues that growth downside could be the next shoe to drop in markets if the war extends and escalates

The Iran war and resulting spike in energy prices have reverberated sharply across global markets. Two particular features of the price action stand out. First, apart from positioning unwinds, global markets have priced this as an inflationary shock across DM and EM economies, with hawkish policy implications that have pushed rates higher and to a more limited extent, equity prices lower. Second, energy price shocks produce winners and losers, and the relative performance of assets, including FX and regional equities, have reflected those terms-of-trade (ToT) differences. Markets have been slower to focus on the growth implications of the energy shock, suggesting that growth downside could be the next shoe to drop.

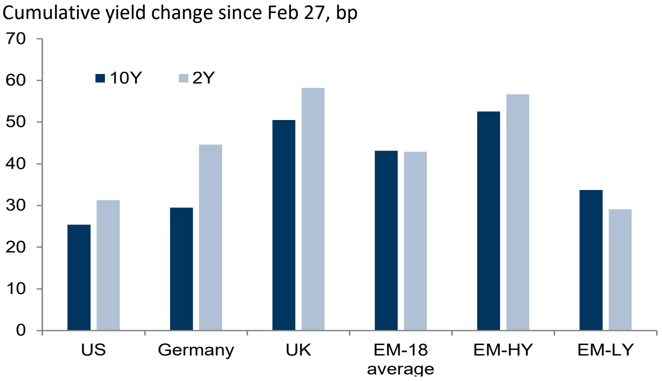
Interest rates: inflation fears drive a hawkish repricing

The ~50% increase in oil prices as a result of the effective shutdown of energy transit through the Strait of Hormuz was quickly reflected in inflation fears and interest rate pricing. That shift is understandable, and generally in line with the direction of revisions in our economic forecasts toward higher headline inflation, smaller increases in core inflation, and later interest rate cuts across the US, UK, and some EMs.

That said, the initial reactions across interest rates markets were even more acute, probably reflecting a couple of factors. First, coming into this shock, we and most investors expected a relatively benign inflation outlook for this year (vs. 2025) given soft wage inflation and structural considerations such as China's export disinflation and potential AI impacts. So, market participants were likely wrong-footed going into this shock. Indeed, the largest reactions occurred in markets that were priced for the most rate cuts, including UK front-ends in G10 and Hungary in EM. Second, the trauma of the 2022 inflation shock, during which interest rates rose much more than initially anticipated, is still fresh in market participants' minds. While it's certainly possible that the current energy shock can worsen, the sharp post-Covid economic uncoiling amid large fiscal support as well as the supply chain bottlenecks that led to higher and more sustained inflation in 2022 are absent today (see pg. 16).

Front-end yields have risen sharply as inflation fears erode rate cut pricing across G10 and EM

Cumulative yield change since Feb 27, bp



Source: Bloomberg, Goldman Sachs GIR.

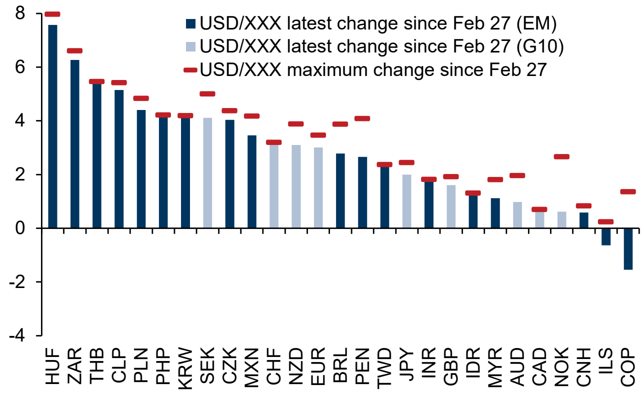
FX and equities: an energy ToT shock

In contrast to the broadly hawkish rates response, the FX response has been more differentiated. The Dollar has appreciated broadly, as expected during a global risk-off event that also fuels a sharp rise in energy prices, as the Dollar sits on the right side of the ToT shock. By contrast, most European and Asian currencies are on the wrong side of that shock, and as the war has progressed, global FX pairs have increasingly reflected those differences. Within G10, the currencies of energy exporters like Norway and Canada have outperformed. Likewise, within EM, the currencies of energy producers Malaysia and Brazil have outperformed, and the Colombian Peso has appreciated in nominal terms versus the Dollar. We expect the ToT shock to remain the dominant driver of FX markets for now and, as such, see these trends continuing over the near term.

In equities, positioning unwinds dominated the initial selloffs, with European and Asian markets that had seen the strongest performance in the past year underperforming the most. But the price action has since become more fundamentally driven, with markets such as India and CEEMEA, where earnings are more fundamentally vulnerable, experiencing only limited rebounds while North Asia and China markets, where earnings are more resilient to the shock, have modestly recovered. Looking forward, we expect this type of differentiation to extend as long as energy prices remain elevated.

Relative FX weakness versus the Dollar lines up along the ToT axis

USD/XXX changes since Feb 27, 2026, %



Source: Bloomberg, Goldman Sachs GIR.

Growth downside: the next shoe to drop?

While the market has been quick to reflect the inflation and ToT implications of the energy shock, it has been slower to appreciate the growth downside that could follow. That reflects a belief—also evidenced in the downward-sloping oil and gas futures curves—that the war will be relatively short-lived. If that confidence is misplaced and the energy price increases prove more durable, markets will need to price in a more significant hit to global growth and earnings. In that scenario, DM and EM equities as well as other cyclical assets such as copper and the Australian Dollar that have been quite resilient so far would face increased pressure, with some of the hawkish pressure on front-end yields also reversing. Such simultaneous pressure on equities and yields should also lead the Japanese Yen to rival the Dollar as the safe-haven currency of choice.

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MENA economies: big hit, possible scars

Farouk Soussa lays out the near-term hit, and possible long-term scars, to MENA economies

The Iran war is unambiguously bad for business in the Gulf, resulting in sharp declines in oil and gas production and revenues as well as broader economic activity. Beyond the near-term hit, the war could also leave lasting scars on regional economies, though much will depend on the regional energy landscape and geopolitical order that emerges once the dust settles, which remains far from clear.

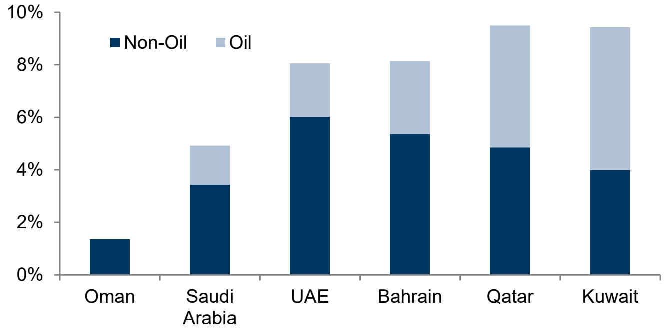
Hits to the oil and non-oil economies

The hits to oil and gas production are significant but uneven across the GCC. In a scenario in which the conflict depresses energy production at current levels until the end of April before gradually recovering to pre-war levels over a two-month period, we find that Kuwait and Bahrain would lose over a fifth of oil production this year. In Qatar, missile damage to the 77mtpa Ras Laffan LNG plant will shut in 17% of the country's LNG production for the next 2-3 years, according to Qatar Energy CEO Saad Al-Kaabi. We estimate that total production there could be down as much as one-third this year. Saudi Arabia and the UAE, which both have capacity to divert flows away from the Strait of Hormuz, would lose around 10% and 15% of production, respectively. Oman, which sits outside the Strait, is largely unaffected from an oil production perspective.

Beyond oil and gas, the war is significantly negatively impacting near-term economic activity, including tourism, aviation, logistics, hospitality, domestic trade, manufacturing, and financial services. While the final impact remains uncertain, we estimate that the economic hit could be as large as, if not larger, than that experienced during the Covid pandemic. The combined effect of oil and gas production losses and a broader decline in economic activity could see GCC economies contract this year by more than any other time in the past 30 years.

GCC economies will see large hits to oil and non-oil output

Total potential decline in 2026 real GDP, %



Source: Haver Analytics, Goldman Sachs GIR.

Serious financial implications

The financial implications of the war are also serious. We estimate that the GCC collectively is losing roughly $700m in oil revenues alone each day as the conflict endures, bringing total losses so far to over $15bn. Two months of disruption at current levels and losses would approach $80bn. And losses in natural gas and other energy supplies will only add to these figures. While such a loss would be manageable for GCC economies given their wealth, it would necessitate even more borrowing than we had anticipated prior to the war, which could have implications for credit spreads in the medium term. Much, though, will depend on how oil and gas prices evolve after the end of hostilities. Should prices adjust slowly to their pre-war path, then higher earnings may compensate Gulf producers for the losses endured during the war. If, on the other hand, prices correct sharply downward, these losses will crystallize over the course of the year.

More questions than answers over the longer term

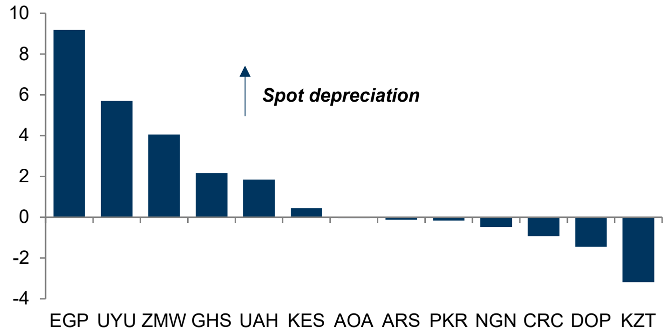
Beyond the immediate hit to energy production, revenues, and growth, markets are also mulling the longer-term implications of the war for GCC economic prospects. In particular, questions have arisen about potential scarring to consumer and investor confidence in the region; whether populations, the majority of which are expat workers, would recover to pre-war levels and over what time period; and whether plans to grow and diversify economies through increased investment in non-oil sectors will be affected, especially around AI/data centers which were one of the early casualties of Iran's retaliatory attacks against its GCC neighbors. The answers to these questions will shape the longer-term implications of the war for GCC economies, though much will depend on the regional geopolitical order that emerges once the dust settles, which remains highly uncertain.

Spillovers beyond the GCC

The impact of the conflict extends beyond the GCC, affecting other countries across the MENA region. The kinetic action is directly impacting Iraq, Jordan, and Syria, with each experiencing varying levels of economic disturbance. Ongoing conflict between Israel and Hezbollah in Lebanon as well as fears of further escalation casts a dark shadow over the country's economic future and the recovery prospects for Lebanon's defaulted Eurobonds. Egypt, a key focus for international investors given the popularity of the local market carry trade over the past two years, is experiencing considerable financial contagion. A combination of heavy pre-war positioning, generalized risk-off behavior, and overblown fears about the impact on the Egyptian economy have led to a significant portfolio retrenchment among investors, leaving the Egyptian Pound (EGP) as the worst performer among frontier FX since the start of the war. However, despite the short-term volatility, we remain constructive on EGP over the medium term as we believe that a combination of robust FX buffers and a (so far) refreshingly orthodox policy response will likely underpin investor and local confidence.

EGP has been the worst performer among frontier currencies

USD/XXX change since the start of the war, %



Source: Bloomberg, Goldman Sachs GIR.

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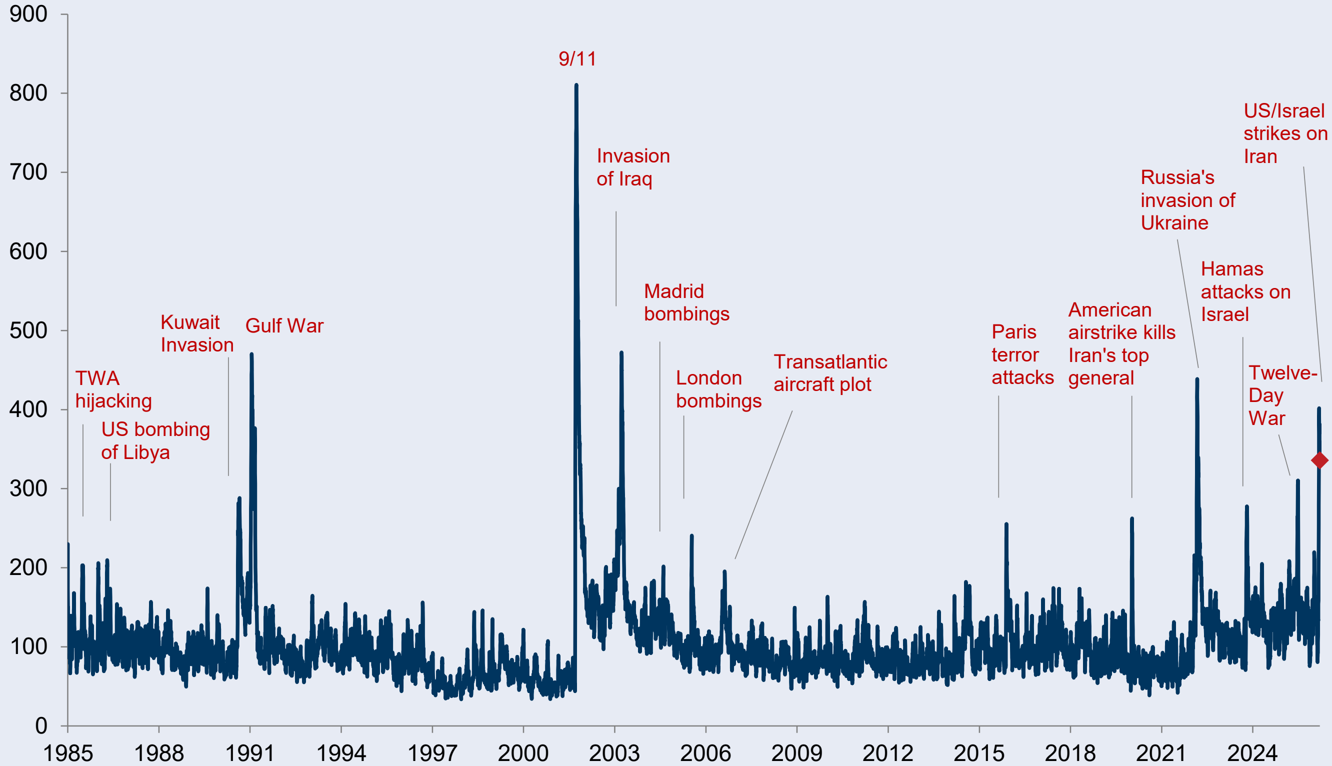
Geopolitical risk through the years

Geopolitical tensions, which take many different forms, are difficult to measure. One proxy for assessing the geopolitical environment is the news-based Geopolitical Risk Index developed by economists from the Federal Reserve Board.

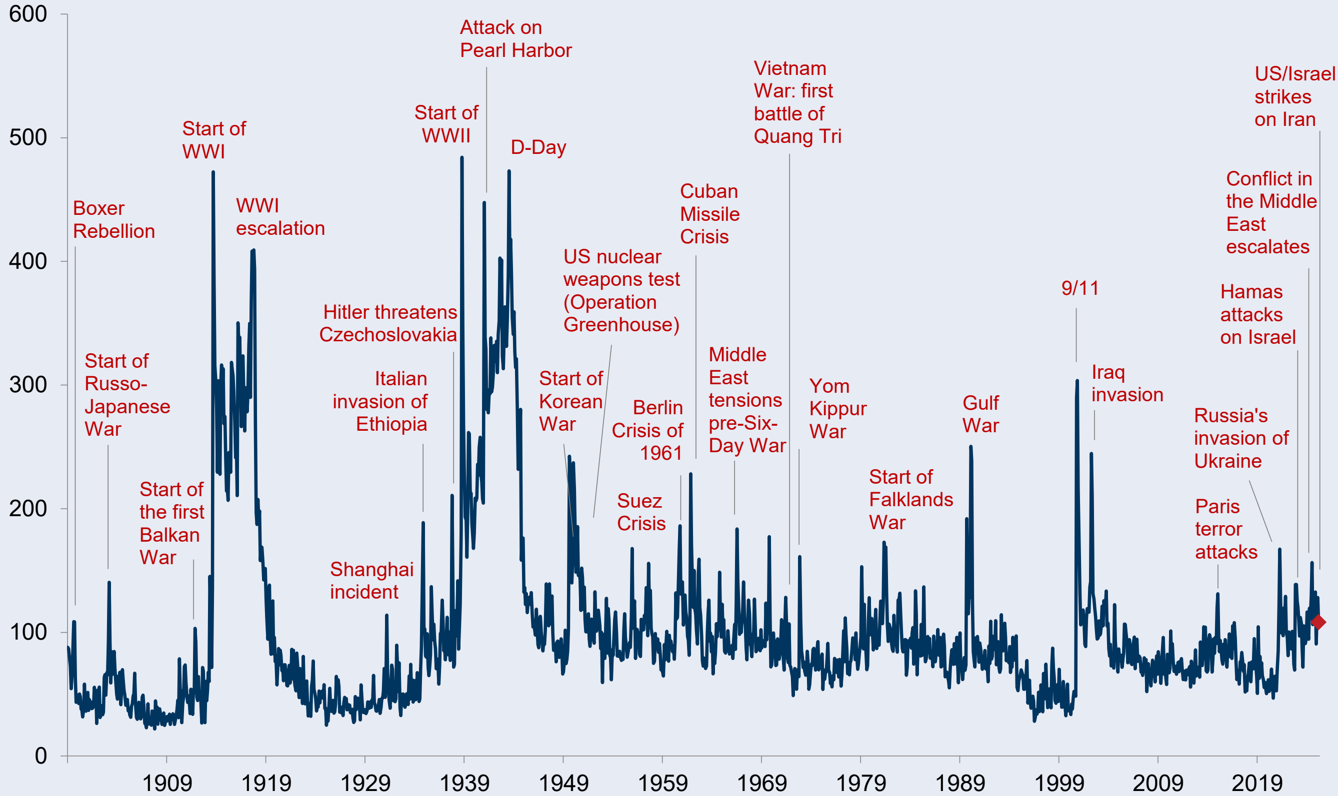
1985-Present

Daily Geopolitical Risk Index, 1985-2019=100



1900-Present

Monthly Geopolitical Risk Index, 1900-2019=100



Note: Daily index as of March 18 and monthly index as of March 1. The index from 1985 on counts the number of articles in 11 US, UK, and Canadian newspapers mentioning phrases related to geopolitical tensions. The index from 1900 on performs the same analysis using the archives of three newspapers: the New York Times, the Wall Street Journal, and the Financial Times. The choice of newspapers for both indices implies a measure of geopolitical risk as covered by the Anglo-Saxon press. See here for more information.

Source: Dario Caldara and Matteo Iacoviello, Federal Reserve Board, Goldman Sachs GIR.

Summary of our key forecasts

GS GIR: Macro at a glance

Watching

- **WATCH THE IRAN WAR.** The severity and longevity of the Iran conflict remain highly uncertain and further escalation that results in more prolonged energy disruptions could have significant impacts on economies and markets.
- **Globally,** we expect real GDP growth of 2.5% yoy in 2026 amid rising real income growth but growing headwinds from higher energy prices owing to the Iran war. We expect global core inflation to decline to 2.4% by the end of 2026, reflecting a fading tariff boost and further normalization in shelter and wage inflation but a boost from higher energy prices.
- **In the US,** we expect real GDP growth of 2.2% on a Q4/Q4 basis in 2026, reflecting a boost from tax cuts but a drag from higher oil prices, tighter financial conditions, and elevated geopolitical risk. We expect core PCE inflation to fall to 2.4% yoy by the end of 2026, as the tariff boost fades and wage and shelter inflation continue to cool, though higher oil prices provide some upward pressure. We expect the unemployment rate to end 2026 at 4.6%.
- **We expect the Fed** to deliver two 25bp cuts this year in September and December for a terminal rate range of 3-3.25%.
- **In the Euro area,** we expect real GDP growth of 1.0% on a Q4/Q4 basis in 2026, reflecting real income gains and expansionary German fiscal policy but a drag from higher energy prices and more intense export competition with China. We expect core inflation to fall to 2.2% by end-2026 as energy price pressure is partly offset by cooling wage inflation, a stronger Euro, and the potential for a negative inflation effect from Chinese trade rerouting.
- **We expect the ECB** to remain on hold this year, although the likelihood of hikes has risen given the continued upward pressure on energy prices.
- **In China,** we expect real GDP growth of 4.7% yoy in 2026 as resilient export growth, continued government policy easing, and a decreasing drag from the ongoing property market downturn offset headwinds from higher energy prices and sluggish domestic demand. We expect CPI/PPI inflation to rise to 0.9%/0.8% yoy this year owing to government policy efforts, a low base, and rising global commodity prices.

Goldman Sachs Global Investment Research.

Growth

Source: Haver Analytics, Goldman Sachs Global Investment Research.
Note: GS CAI is a measure of current growth. For more information on the methodology of the CAI please see "Technical Updates to Our Global CAIs," Global Economics Comment, Sep. 01, 2025.

Forecasts

Economics

GDP growth (%)	2026 GS (Q4/Q4)	2026 Cons. (Q4/Q4)	2026 GS (CY)	2026 Cons. (CY)	2027 GS (CY)	2027 Cons. (CY)
Global	2.4	--	2.5	2.7	2.7	2.6
US	2.2	2.1	2.4	2.2	2.0	2.5
China	4.6	4.7	4.7	5.0	4.7	4.6
Euro area	1.0	1.3	1.0	1.4	1.2	1.2

Policy rates (%)	2026 GS	2026 Mkt.	2027 GS	2027 Mkt.
US	3.13	3.47	3.13	3.28
Euro area	2.00	2.45	2.00	2.47
China	1.30	1.40	1.20	--
Japan	1.00	1.16	1.50	1.59

Markets

Interest rates 10Yr (%)	Last	E2026	E2027
US	4.25	4.10	4.15
Germany	2.96	3.00	3.00
Japan	2.26	2.00	2.00
UK	4.73	4.25	4.00

FX	Last	3m	12m
EUR/$	1.15	1.14	1.20
GBP/$	1.34	1.33	1.33
$/JPY	158	160	155
$/CNY	6.87	6.85	6.70

Commodities	Last	3m	12m
Crude Oil, Brent ($/bbl)	109	75	71
Nat Gas, NYMEX ($/mmBtu)	3.17	3.75	4.00
Nat Gas, TTF (EUR/MWh)	61.98	43	29
Copper ($/mt)	12,046	13,000	12,400
Gold ($/troy oz)	4,600	4,805	5,445

Credit (bp)		Last	2Q26	4Q26
USD	IG	86	85	90
	HY	294	295	315
EUR	IG	102	95	99
	HY	310	298	315

Equities

S&P 500	E2026 GS	E2026 Cons.	E2027 GS	E2027 Cons.
Price	7,600			
EPS	$309	$319	$342	$372
Growth	12%	16%	10%	16%

Returns (%)	12m	YTD	E2026 P/E
S&P 500	15.0	-3.5	21.4x
MXAPJ	18.1	5.6	13.8x
Topix	19.1	5.9	17.9x
STOXX 600	6.8	-1.2	14.7x

Consumer	2026 CPI (%, yoy)	2026 Unemp. Rate	2027 CPI (%, yoy)	2027 Unemp. Rate	Wage Tracker 2026 (%) Q1	Q2	Q3	Q4
US	2.7	4.6	1.8	4.5	3.5	–	–	--
Euro area	2.5	6.3	1.6	6.2	–	–	–	--
China	0.9	--	0.9	--	–	–	–	--

Source: Bloomberg, Goldman Sachs Global Investment Research. For important disclosures, see the Disclosure Appendix or go to www.gs.com/research/hedge.html.

Market pricing as of March 19, 2026

Glossary of GS proprietary indices

Current Activity Indicator (CAI)

GS CAIs measure the growth signal in a broad range of weekly and monthly indicators, offering an alternative to Gross Domestic Product (GDP). GDP is an imperfect guide to current activity: In most countries, it is only available quarterly and is released with a substantial delay, and its initial estimates are often heavily revised. GDP also ignores important measures of real activity, such as employment and the purchasing managers' indexes (PMIs). All of these problems reduce the effectiveness of GDP for investment and policy decisions. Our CAIs aim to address GDP's shortcomings and provide a timelier read on the pace of growth.

For more, see our CAI page and Global Economics Comment: Technical Updates to Our Global CAIs.

Dynamic Equilibrium Exchange Rates (DEER)

The GSDEER framework establishes an equilibrium (or "fair") value of the real exchange rate based on relative productivity and terms-of-trade differentials.

For more, see our GSDEER page, Global Economics Paper No. 227: Finding Fair Value in EM FX, 26 January 2016, and Global Markets Analyst: A Look at Valuation Across G10 FX, 29 June 2017.

Financial Conditions Index (FCI)

GS FCIs gauge the "looseness" or "tightness" of financial conditions across the world's major economies, incorporating variables that directly affect spending on domestically produced goods and services. FCIs can provide valuable information about the economic growth outlook and the direct and indirect effects of monetary policy on real economic activity.

FCIs for the G10 economies are calculated as a weighted average of a policy rate, a long-term risk-free bond yield, a corporate credit spread, an equity price variable, and a trade-weighted exchange rate; the Euro area FCI also includes a sovereign credit spread. The weights mirror the effects of the financial variables on real GDP growth in our models over a one-year horizon. FCIs for emerging markets are calculated as a weighted average of a short-term interest rate, a long-term swap rate, a CDS spread, an equity price variable, a trade-weighted exchange rate, and—in economies with large foreign-currency-denominated debt stocks—a debt-weighted exchange rate index.

For more, see our FCI page, Global Economics Analyst: Our New G10 Financial Conditions Indices, 20 April 2017, and Global Economics Analyst: Tracking EM Financial Conditions – Our New FCIs, 6 October 2017.

Goldman Sachs Analyst Index (GSAI)

The US GSAI is based on a monthly survey of GS equity analysts to obtain their assessments of business conditions in the industries they follow. The results provide timely "bottom-up" information about US economic activity to supplement and cross-check our analysis of "top-down" data. Based on analysts' responses, we create a diffusion index for economic activity comparable to the ISM's indexes for activity in the manufacturing and nonmanufacturing sectors.

Macro-Data Assessment Platform (MAP)

GS MAP scores facilitate rapid interpretation of new data releases for economic indicators worldwide. MAP summarizes the importance of a specific data release (i.e., its historical correlation with GDP) and the degree of surprise relative to the consensus forecast. The sign on the degree of surprise characterizes underperformance with a negative number and outperformance with a positive number. Each of these two components is ranked on a scale from 0 to 5, with the MAP score being the product of the two, i.e., from -25 to +25. For example, a MAP score of +20 (5;+4) would indicate that the data has a very high correlation to GDP (5) and that it came out well above consensus expectations (+4), for a total MAP value of +20.

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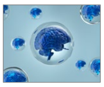
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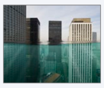
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Reg AC

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TOP of MIND

WILL AI EAT SOFTWARE?

For over a decade, software has come to "eat" the world. But the emergence of new agentic AI tools for software development in recent months has fueled concerns that AI could in turn "eat" software, triggering a sharp re-rating of the software sector as well as other sectors vulnerable to AI disruption. So are these fears about the future of software (and other industries) warranted, or overdone? GS' Gabriela Borges, Sherlund Partners' Rick Sherlund, and Cohesity's Sanjay Poonen generally agree that AI won't eat software but diverge on how radical a transformation legacy software firms must undergo to remain competitive and how durable incumbent moats are. With all this uncertainty, we then explore what it will take to stabilize share prices (earnings stability), whether stress in software-exposed credit can catalyze a turn in the credit default cycle (unlikely), and how investors should be positioned, with Sherlund and GS analysts and strategists agreeing: investors should be selective, in software and beyond.

"Legacy software companies aren't standing still. They are innovating as fast followers… That, combined with their moats, could ultimately leave incumbents in a better place from a soup-to-nuts platform perspective.

\- Gabriela Borges

Moats buy incumbents time to adapt, as seen in prior software disruption cycles… But these moats aren't insurmountable—AI will very likely erode them over time.

\- Rick Sherlund

Agentic AI has already proven to be an incredibly powerful tool with the potential to be enormously disruptive... However, just like any technology wave, you must surf this tsunami, or it will demolish you.

\- Sanjay Poonen"

WHAT'S INSIDE

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Macro news and views

We provide a brief snapshot on the most important economies for the global markets

US

Latest GS proprietary datapoints/major changes in views

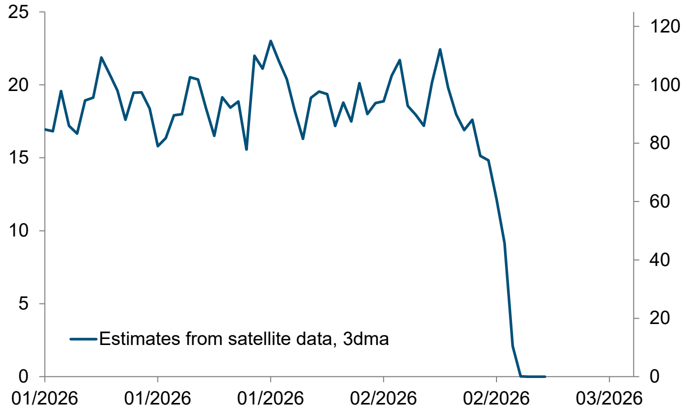
- No major changes in views.

Datapoints/trends we're focused on

- Oil disruptions related to the US-Iran conflict, which pose downside risks to growth and upside risks to inflation, depending on the longevity and severity of the conflict.
- Tariffs; we expect the global tariff that has replaced the IEEPA tariffs to lower the increase in the effective tariff rate since the start of 2025 to 9pp from 10pp.
- Labor market; the Feb jobs report suggests a higher risk that labor demand remains too weak to support the labor market stabilization we had seen in previous months.
- Concerns about nonbank lending, which have risen.

Strait of Hormuz oil transport grinds to a near halt

Oil exports via the Strait of Hormuz, mb/d (lhs), % of normal (rhs)



Source: Kpler, Goldman Sachs GIR.

Japan

Latest GS proprietary datapoints/major changes in views

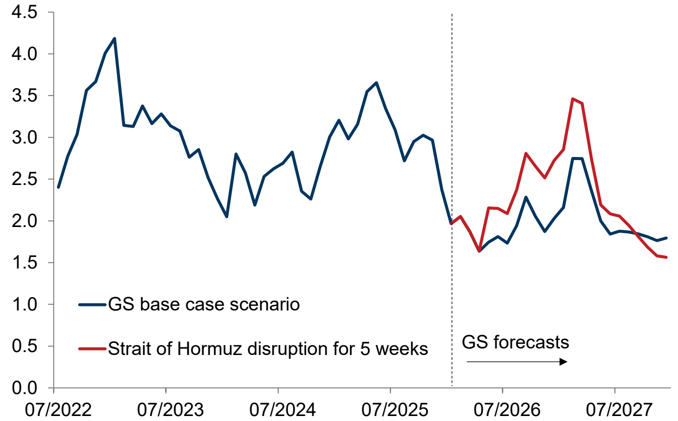
- We recently raised our FY26 average core CPI inflation forecast to 2.1% (from 1.6%) to reflect our higher energy price and USD/JPY forecasts.

Datapoints/trends we're focused on

- LDP's landslide victory in Japan's Lower House elections, which we think will make it easier for the Takaichi Administration to implement its proactive fiscal agenda, posing increased risk to Japan's fiscal position.
- BoJ policy; we continue to expect the next BoJ rate hike in July, and think two recent board member nominations have lowered the probability of an earlier rate hike.
- *Shunto* negotiations; we expect a base pay rise of 3.2-3.5%.

Rising Japanese inflation

Core CPI inflation rate by scenario, % chg., yoy



Source: Goldman Sachs GIR.

Europe

Latest GS proprietary datapoints/major changes in views

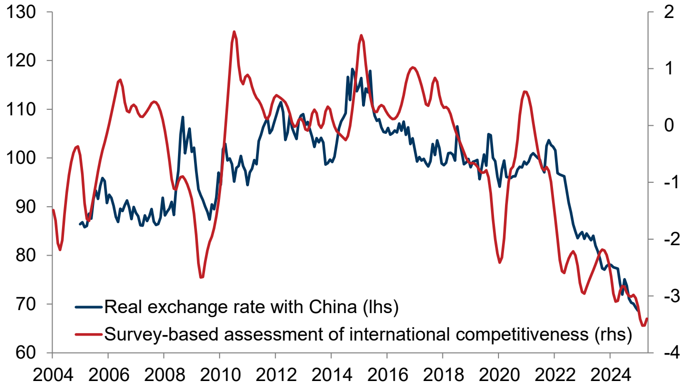
- We lowered our 2026 EA growth forecast and raised our 2026 EA headline inflation forecast to reflect higher energy prices amid the US-Iran conflict, with growth risks skewed further to the downside and inflation risks to the upside.
- We lowered our 2026 UK growth forecast to 1.0% yoy (from 1.1%) and raised our headline inflation forecast to 2.4% yoy (from 2.3%) to reflect higher energy prices; as such, we now expect the next BoE cut in April (vs. March before).

Datapoints/trends we're focused on

- European structural reforms; recent political momentum behind further reform steps to counter Europe's waning international competitiveness is promising, in our view.

Europe: still losing competitiveness

Real exchange rate with China (lhs, index, 2019=100) vs. survey-based assessment of int'l competitiveness (rhs, z-score)



Source: Haver Analytics, Goldman Sachs GIR.

Emerging Markets (EM)

Latest GS proprietary datapoints/major changes in views

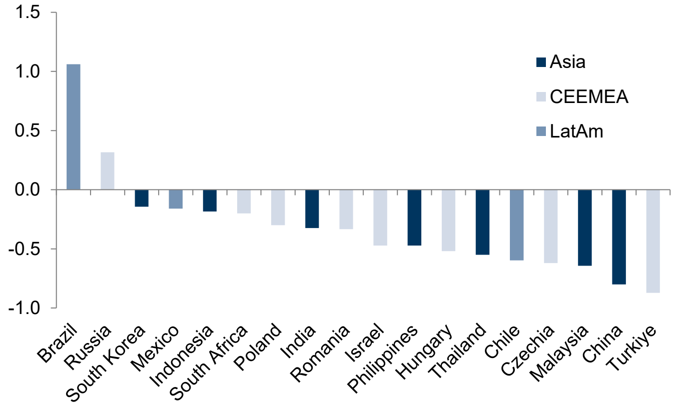
- We lowered our 2026 GDP growth forecasts for Saudi Arabia, Oman, Qatar, Bahrain, Kuwait, and UAE to reflect Strait of Hormuz disruptions and a fall in non-oil economic activity amid the US-Iran conflict, with risks skewed to the downside.

Datapoints/trends we're focused on

- US-Iran conflict impacts beyond the ME; we find that a 10% rise in oil prices would be especially negative for Türkiye and oil importers across Asia, while oil exporters Brazil and Russia are likely to be the biggest beneficiaries.
- China growth; we expect real GDP growth to fall to 4.8% yoy this year (from 5% in 2025), in line with the recently-announced official growth target of 4.5-5% this year.

Higher oil prices: negative for most EMs

Real GDP sensitivity to a 10% rise in oil prices, pp



Source: Goldman Sachs GIR.

Will AI eat software?

For over a decade, software has come to "eat" the world, dominating an ever-wider range of industries. But the emergence of new agentic AI tools for software development in recent months (see pg. 11) has fueled concerns that AI could in turn "eat" software, triggering a sharp re-rating of the software sector (see pg. 9) as well as other sectors vulnerable to disruption from these increasingly powerful and accessible AI tools. So are these fears about the future of software (and other industries) warranted, or overdone?

We ask three long-time software industry watchers: Gabriela Borges, GS US Software analyst, Rick Sherlund, Founder and CEO of Sherlund Partners, and Sanjay Poonen, CEO and President of data security and management firm Cohesity.

Borges explains that "AI *is* software" because it is code designed to perform tasks. So, rather than eat software, she expects AI to *expand* the software market. But she also expects competition to intensify as AI lowers the cost of code. The main risk for legacy software firms, she says, is that AI-native entrants capture the new AI opportunity while the incumbents are left with the old opportunity—namely, being the "system of record"—that shrinks as a share of the software stack. But Borges notes that legacy firms aren't standing still but rather "innovating as fast followers." This, together with moats such as domain experience that new entrants can't quickly replicate, could ultimately lead incumbents to deliver better outcomes for customers, in her view. But she stresses that they must prove they can leverage their moats to do so, which hasn't happened yet.

Sherlund shares Borges' conviction that AI won't eat software, viewing the current moment akin to past transformational technology shifts that were highly disruptive but ultimately propelled the software industry onto a new growth curve. In his view, software "is being reborn around AI", necessitating a fundamental re-architecture of legacy systems around large language models (LLMs) and AI agents. While Sherlund acknowledges that incumbent moats can slow disruption, he is skeptical of their long-term durability, arguing that they will buy legacy software firms time rather than guarantee their survival.

And Poonen, for his part, agrees that new AI tools for software development mark a transformative moment for the industry, warning that "just like any technology wave, you must surf this tsunami, or it will demolish you." In his view, how legacy software firms fare in the AI era will depend crucially on whether they can sustain their current price per user. Firms unable to defend pricing in their core offerings will need to expand and improve their products. But, like Sherlund, Poonen is explicit that just bolting AI agents onto legacy systems may not be sufficient—incumbents must ultimately be willing to rebuild, in some cases from scratch, to remain competitive.

So, with this uncertainty, what will it take to stabilize the sector and the firms most exposed to it? Ben Snider, GS Chief US Equity Strategist, argues that stabilization is unlikely to occur anytime soon, with history suggesting that share prices in industries facing structural disruption (e.g. newspapers) and an associated deep rethink of firms' terminal value tend to stabilize only when earnings stabilize, underscoring Borges' point that "the numbers need to contradict the narrative."

Shamshad Ali, GS senior credit strategist, also expects pressure to persist in the broadly syndicated loan (BSL) market—which software comprises a meaningful share of—as impairment in software-exposed loans will likely be difficult to avoid. But he doesn't see this as sufficient to catalyze a turn in the credit default cycle, assuming the benign macro backdrop and lower funding cost environment continue.

GS US Financials analysts Alexander Blostein and Michael Vinci are somewhat more positive on alternative asset managers, who have been hit hard given the meaningful software exposure in their private equity and direct lending portfolios. Blostein and Vinci argue that despite this exposure, firmwide risk remains manageable as software lending represents a relatively small share of management fees, software loans are largely senior secured and short duration, and managers have been highly selective in their software investments. But they caution that sentiment and headline risk will likely persist, with slower retail fundraising and accelerating redemption requests potentially weighing on growth despite the durability of underlying earnings streams.

So, how should investors be positioned? Sherlund and Borges agree: this is not an environment for binary bets on software's survival or collapse, but one that demands selectivity. Sherlund argues that "investors must carefully assess which companies are most exposed to disruption and which possess complex workflows that are more difficult to encroach upon" and views the most promising companies as "those doing the heavy lifting to re-architect to put LLMs at their core and build robust agentic frameworks." Borges, for her part, is focused on fast followers and firms with defensible moats, seeing select opportunities in applications, infrastructure, and cybersecurity. And GS US Software analysts Adam Hotchkiss and Matthew Martino see value in Vertical Software, Data Infrastructure, and Physical-to-Digital firms.

GS Head of Asset Allocation Research Christian Mueller-Glissmann then digs into how to manage AI risk in portfolios more broadly given the outsized role the technology sector plays in driving US equity valuations and returns. To achieve more balance between innovation and inflation risks, he believes investors may need to continue rotating from capital-light sectors, like technology, to capital-heavy sectors that AI is less likely to disrupt (think telecoms, industrials, and utilities), though he favors being more selective given the aggressive rotations that have already occurred.

Finally, with the "AI scare trade" having moved beyond just software to many other sectors—from consumer internet and information services to insurance, CRE services, financials, and transportation—GS equity analysts assess how new AI tools could impact these industries (see pgs. 21-23). Their key message: while AI could disrupt some firms, many may also benefit, reinforcing the idea that investors should be selective as AI technology continues to evolve.

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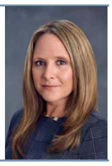


Interview with Gabriela Borges

Gabriela Borges is Senior Equity Research Analyst at Goldman Sachs covering US Software. Below, she argues that AI won't eat software and some of the recent price action has gone too far, but legacy software firms will need to prove themselves through better fundamentals.

Jenny Grimberg: How does the current software correction compare to past corrections?

Gabriela Borges: The YTD correction is squarely anchored to terminal value debates: instead of analyzing a change in the near-term demand trendline or the mathematical impact of higher interest rates—as was the case in the 2022 correction—the market is questioning software moats and business models. While this debate is centered primarily on the application software firms, investors are also questioning whether the infrastructure /security stack could be impacted, and the ROI tied to hyperscaler capex. This is sparking a return to first principles questions about the value the industry provides and the nature of competition. And this is all happening against a backdrop of elevated uncertainty given the rapidly changing technology. So, the end state is difficult, if not impossible, to predict.

Jenny Grimberg: Generative AI technology has been around for several years. So, what's changed recently?

Gabriela Borges: The technology is now beginning to impact enterprises. Most AI chatbot applications that have emerged since ChatGPT's debut have primarily impacted the consumer-facing world. Concerns about how data is maintained, cleaned, and validated have been a huge limiting factor for enterprise AI adoption, and regulatory, compliance, and security frameworks are in the process of adapting to make enterprises comfortable adopting these mission-critical yet invasive technologies. At the same time that progress has been made on these fronts, tools more relevant for enterprise use cases have emerged.

For example, Anthropic's recently-released Claude Cowork product makes its Claude Code product accessible to all knowledge workers, not just coders. And its new large language model (LLM) Claude Opus 4.6 has several domain-specific applications, including in cybersecurity, legal, and financial services. OpenAI debuted Frontier, an enterprise platform for building, deploying, and managing AI agents and is strategically leaning into enterprise monetization. And personal AI assistant OpenClaw launched with capabilities much more powerful than a typical AI chatbot's; usage today is primarily limited to technically-savvy consumers, but this may expand as security and usability challenges are solved. So, AI use cases and the power to execute them has progressed, leading the technology to intersect more with the traditional software world and creating a competitive bake-off dynamic between new entrants and today's software leaders at enterprises now better equipped to utilize the technology.

Jenny Grimberg: So, what's your response to those who argue that AI will "eat" software?

Gabriela Borges: The key question is, what do you have to believe to be bullish on software? You have to believe that the total addressable market (TAM) will grow more than enough to offset the rise in competition. We feel comfortable that the TAM will grow because, despite the current nomenclature, AI *is* software—it is code designed to perform a set of tasks, which is in many ways simply an extension of the software category. But the competition part of the question is currently the bigger unknown as the rapid pace of technological change is attracting new entrants and a wave of VC money. At the same time, the cost of code is falling, which also catalyzes more competition. In our view, the biggest risk for today's software leaders is that the new opportunity AI creates accrues to these new competitors, and the incumbents are left with outsized exposure to the old opportunity that compresses as a percentage of the total software stack.

One way to illustrate this is with the concept of a "system of record." Many of today's application software leaders are systems of record. These systems, such as Enterprise Resource Planning, Customer Relationship Management, and Human Resources, are the digital backbone of the business, storing the foundational data upon which it operates. AI may find it hard to disrupt these for two reasons. First, the data and historical context sitting in the system of record cannot be extracted holistically and in a way that captures ongoing data updates owing to nuances with meta context and data labeling as well as with stream processing that are proprietary to the system of record and break when extracted, even when using sophisticated APIs. Second, using generative AI technology to replace deterministic logic is expensive and ineffective. Put simply, it makes more strategic and economic sense for new AI entrants to leverage existing systems of record for high quality data and context than to replace them. The question then becomes whether the incumbents can also capture the new opportunity because of this domain experience and context.

Jenny Grimberg: So, is it possible to be bullish on agentic AI and *not* bearish on legacy software?

Gabriela Borges: Yes, because some legacy software firms will likely capture the upside opportunity of AI. We expect some incumbents will be able to prove to customers that their domain experience drives better AI outcomes. It's worth highlighting that incumbents are not static—many have world-class engineering teams that benefit from the same AI productivity gains as start-ups, founder involvement, and strong cash balances to respond to new competitive threats.

But the playbook to capture AI upside is three-fold, and companies must succeed on all three fronts, which won't be easy but is possible. First, companies must minimize any technical debt in their tech stack. That means ensuring all acquisitions integrate nicely into the company's core technology backbone and data flows across different modules and is centralized and uniformly cleaned. Second, companies must understand the holes in their roadmaps. One misconception is that a software leader must be a leader in technology. It can instead be a fast follower, letting the VC ecosystem fund next-gen technologies, seeing which have durability and the best product market fit, and then acquiring them or using internal R&D to capture that product market fit, which is arguably a better use of capital than spending heavily to develop every new technology in-house.

Third, companies must figure out how to monetize AI, which may require a change in pricing model. For example, instead of charging per customer service representative, a software company could instead charge per successfully completed customer service call. Another type of pricing model is enterprise license agreements (ELAs), which allow customers to access many different products for one price. ELAs have existed in software for some time. Salesforce recently shifted to agentic ELAs (AELAs), giving customers access to their AI and other products. This is important because it shifts the software pricing model more holistically to include agentic value and allows customers to have fluidity between seats and outcomes. Importantly, we don't think customers can reduce seat count and maintain/enhance productivity unless software tools are picking up the slack and delivering more value.

Jenny Grimberg: What will allow the sector to stabilize?

Gabriela Borges: The numbers need to contradict the narrative. The software sector has been in a four-year-long depression; for 14 consecutive quarters, revenue growth has decelerated and the cost of acquiring new customers has risen. So, the numbers need to stabilize. It would be hard to argue that software companies are being disintermediated if they put up better fundamentals, because the only way they can do so in the AI era is by having exposure to the new opportunities and budgets that AI creates. Evidence of this would be clean beat-and-raised quarters and improved unit economics, either on an industry or, more likely, company-specific basis.

Jenny Grimberg: But couldn't that just push out the bear case for software disruption rather than prove it wrong?

Gabriela Borges: This is where moats become important. A company must prove its fundamentals are holding up because it has a defensible moat. One example is data architecture and collection moats, which are a feature of some security software companies and are outside the realm of AI's current and likely future capabilities. Cloudflare, for example, has a physical edge compute network co-located in telecom data centers, meaning that it owns physical computing infrastructure deployed close to end-users. Palo Alto Networks and CrowdStrike directly collect data, either through physical hardware deployed at the network edge or, in CrowdStrike's case, through an agent installed on an end-user device. These are classic examples of network effects that have compounded over 10+ years.

Another defensible moat is domain experience. Software firms with domain experience understand how knowledge workers like salespeople, accountants, and financial analysts work and use their tools owing to years of being embedded in an enterprise. That usage also allows these firms to collect proprietary data, some of which is very nuanced and difficult to abstract. Beneath the surface of Microsoft Copilot, for example, is a highly nuanced graph showing how knowledge workers in a specific enterprise collaborate, which Microsoft has the intelligence to create because it has observed how these workers interact with its software for many years. That is a powerful type of domain experience. Microsoft is also vertically integrated, operating across all three layers of software—the platform, infrastructure, and application layers—which allows it to observe and optimize how enterprise work gets done.

Jenny Grimberg: But couldn't AI native companies replicate those moats over time?

Gabriela Borges: Yes, but time matters, as does whether it makes strategic sense to do so. If everything were frozen in place right now and AI-native firms were given two years to figure out how to replicate domain experience, for example, they probably would. But, again, legacy software companies aren't standing still. They are innovating as fast followers. The incumbents have access to the same AI technology as the AI firms themselves—the LLMs, the coding productivity tools, etc.—and so are benefitting from the same productivity improvements. That, combined with their moats, could ultimately leave incumbents in a better place from a soup-to-nuts platform perspective. They already have the system of record, domain experience, and user interface. Layering AI on top of that gives incumbents everything they need to build a successful agentic solution more powerful than any one new competitor can offer. But ultimately, incumbents will have to demonstrate they can leverage their moats to deliver such an outcome, and so far, that hasn't happened in a broad enough way. So, the pressure is on for these companies.

Jenny Grimberg: Given everything we've discussed, has the recent sharp selloff in software gone too far?

Gabriela Borges: The pace of competition and the level of uncertainty is undoubtedly high. However, two areas of discovery value allow us to be bullish on many software assets. First is the dynamic we've discussed on incumbents being able to leverage domain experience to drive better AI outcomes. Second is the potential for more companies to manage to GAAP earnings, which provide a level of valuation support for the stocks. The level of competitive noise is likely near a local maximum, and the perception vs. reality gap should narrow ahead for the majority of software firms. Looking at it from that lens, some of the recent price action has likely gone too far.

Interview with Rick Sherlund

Rick Sherlund is Founder and CEO of Sherlund Partners LLC, an investment advisory firm focused on the rapidly evolving AI software sector and traditional enterprise SaaS space and is a Senior Advisor at Wedbush Securities. Below, he argues that the software industry is being reborn around AI, necessitating a complete re-architecture of existing software systems.

The views stated herein are those of the interviewee and do not necessarily reflect those of Goldman Sachs.

Allison Nathan: Does the emergence of new agentic AI tools for software development mark a transformative moment for the traditional software industry?

Rick Sherlund: Very much so. The software industry undergoes a major platform shift roughly every 10-20 years as new technologies emerge. While these shifts can be highly disruptive, they also serve as powerful catalysts for growth, propelling the entire software industry onto new growth curves. We are at such a juncture today, with over 10,000 venture-backed AI-native companies poised to encroach on and disrupt the ~$400bn global Software as a Service (SaaS) market.

Allison Nathan: How does this compare to past periods of transformation?

Rick Sherlund: Some technology shifts, such as the move to mobile, allowed companies to adapt their existing architecture without fundamentally altering business processes. However, other shifts created true discontinuities that forced companies to completely rebuild their systems from the ground up. The move to relational databases in the 1980s/1990s is a prime example. Companies could not simply tweak their existing frameworks. Instead, they had to reconstruct their architectures entirely—a task akin to demolishing and rebuilding a skyscraper from its foundation. Once-dominant players that were unable to reinvent their platforms to compete with Oracle, which was a leader in promoting relational database architecture, were ultimately left behind. Similar upheavals occurred during the transitions to client-server models and later to cloud computing.

The rise of AI marks a similarly transformative shift, enabling processes that legacy system architectures cannot readily accommodate. Faced with falling valuations, incumbent SaaS providers must overhaul their systems. Remaining competitive in the AI era requires software companies to do much more than add incremental AI capabilities to automate narrow legacy human tasks. While such bolt-on functions may be quicker and look good in the marketing literature, they are superficial and risk missing the more critical and transformative opportunity to rearchitect systems around large language model (LLM) intelligence with reasoning at the core to orchestrate evolving autonomous agents, which can power a new generation of business processes. This shift is not only disruptive for the software industry, but for businesses and sectors globally, representing a fundamental change in technology platforms.

Allison Nathan: What new processes will such AI-enabled platforms be able to achieve?

Rick Sherlund: Traditional SaaS platforms rely on hard-coded workflows and human-centric, menu-driven interfaces. By contrast, AI-based systems feature more conversational interfaces, embedding LLM intelligence at their core to orchestrate agents capable of reasoning, decision-making, and executing tasks previously handled by humans. This enables the integration of entirely new processes. For example, legacy customer support consists of menu-based interactive voice responses (IVR) and irritating cascading menus. Users may frequently scream "representative" in frustration to exit the menu system to connect with a human to address their issue, but even that representative may not be able to do so. And customer support remains costly.

So, companies are turning to AI-powered virtual agents that are available 24/7, multilingual, and trained on comprehensive support materials. These agents run on hyperscaler systems that avoid problems of availability and scalability in periods of peak demand. Agents may also identify that a customer has, say, an older version of a product that may no longer be supported. The agent can then pivot from a cost center support function to a revenue function by recommending a new product. So, AI agents can seamlessly shift from support to sales and billing, orchestrating complex, multistep workflows across disparate systems with minimal human intervention.

As infrastructure improves to provide governance and guardrails, AI agents will increasingly handle autonomous, agent-to-agent processes, with humans overseeing compliance and exceptions. This shift will unlock even more processes and expand automation as systems become more intelligent and capable of advanced reasoning.

Allison Nathan: So, are concerns that AI could "eat" software, which have sparked a major selloff in legacy software companies, warranted, or overblown?

Rick Sherlund: At the end of the day, AI is expressed in software, so declaring software "dead" is misguided. Software will endure, even if the industry looks different than it does today. Certain segments of the software market—particularly applications—will experience considerable disruption, but the broader industry will be reincarnated, adapting to new processes and reaping substantial benefits from the widespread integration of AI capabilities across all business functions. This will lead to a step function increase in productivity that was not possible with traditional SaaS architectures.

Allison Nathan: Won't the rise of "vibe coding" eventually make software replaceable, as some argue?

Rick Sherlund: Concerns that "vibe coding" will make it so simple to write software that software itself becomes trivial are largely unfounded. Coding accounts for only around 15–20% of writing enterprise systems. So, while automating code generation is a major efficiency gain, it is just one aspect of a far more complex process. Designing effective workflows, defining data structures, and establishing secure access controls are all critical and complex tasks. Developers must decide how databases are organized, what data is stored, and how sensitive information, like payroll data, is protected. These challenges extend well beyond basic coding. So, easier coding won't sound the industry's death knell. On the contrary, the industry will likely become more vibrant than ever as firms leverage the new technology to remain competitive. So, the software market is not dead; it is being reborn around AI.

Allison Nathan: How defensible are legacy software moats amid the AI transition?

Rick Sherlund: Complex workflows are a powerful moat, demanding deep domain expertise to design effective interactions with vendors and suppliers in the supply chain. This expertise extends not only to vertical markets but also across horizontal complexities. Data access is another powerful moat, as software companies fiercely protect their customers' data and resist third-party access, recognizing its tremendous value. Scale and distribution moats are powerful as well—building and maintaining an enterprise sales force is extremely costly, so established distribution networks further protect incumbents.

Together, these moats buy incumbents time to adapt, as seen in prior software disruption cycles. For example, larger companies with complex workflows like SAP and Oracle were slower to adopt cloud architecture but eventually did so, leveraging their moats to buy time. So, firms with these protective moats, like the large SaaS incumbents, have more time to adjust amid the AI transition. But these moats aren't insurmountable—AI will very likely erode them over time.

Allison Nathan: How are incumbent software companies responding?

Rick Sherlund: Incumbent software companies are working hard to fundamentally re-architect their systems. Large firms like Salesforce, Workday, and ServiceNow—who were disruptors in the shift from client-server to the cloud—are now integrating LLMs into the core of their systems and developing agentic frameworks to support intelligent, autonomous workflows. Having been disruptors themselves, these vendors recognize that major technological shifts require comprehensive transformation, not just superficial bolt-ons.

These companies are also making significant efforts to address broader issues such as governance, security, and the need for advanced data structures like knowledge and context graphs. Such measures are vital for understanding data and workflow intent, ensuring that systems can operate autonomously while maintaining compliance and security. While these efforts do not guarantee that the incumbent SaaS companies will "win" against emerging AI-native competitors, they represent significant steps toward adapting and maintaining relevance in an evolving market. Whether they can move fast enough and reposition to leverage their market presence, distribution, and scale advantages to be net beneficiaries of the AI era remains to be seen.

Allison Nathan: By contrast, how can AI-native firms compete and win in the space?

Rick Sherlund: Disruptors have historically succeeded by entering at the lower end of the market, where there is less invested in incumbent systems and companies are more open to adopting innovative solutions. This was the case with Salesforce and Workday, who began by serving small- and medium-sized businesses rather than directly competing with SAP.

Over time, as new entrants enhance their offerings and scale their distribution, they can move gradually upmarket, competing with larger incumbents—adopting a "land and expand" strategy. Salesforce, for example, began with sales automation before expanding into marketing and customer service, while Workday started in Human Resources before adding financial capabilities. Ultimately, the path to market disruption starts at the bottom where the protective moats are fewer and fresh ideas can gain traction, and scales upward.

Allison Nathan: Given all that, how do you foresee the competitive landscape for software evolving?

Rick Sherlund: As more AI-native companies enter the lower end of the market, the software industry will likely become increasingly fragmented. We saw this with the move from client-server systems to the cloud/SaaS model, where specialized "best-of-breed" software solutions were embraced over traditional monolithic suites from providers like SAP and Oracle. Now, with the rise of agentic AI systems, we will likely see a further disaggregation as agents enable a fragmentation of functions and an unbundling of existing SaaS functions. This is already evident in the customer support segment, with AI agents transforming Contact Center as a Service (CCaaS) and IVR systems, as well as in the emergence of dedicated sales and marketing agents.

Allison Nathan: How can investors distinguish between software companies that will likely remain resilient in the face of AI-driven disruption, and those that may be at risk of becoming obsolete?

Rick Sherlund: Investing in software has now become riskier, making it important to distinguish between layers of the software stack—applications are probably more at risk than infrastructure software, while areas such as governance, security, and data management stand to benefit most from AI advances. Investors must carefully assess which companies are most exposed to disruption and which possess complex workflows that are more difficult to encroach upon, even if these firms will eventually need to undergo significant transformation to remain competitive.

Ultimately, the most promising companies are those doing the heavy lifting to re-architect to put LLMs at their core and build robust agentic frameworks. At the end of the day, only those with the right architecture will be able to adapt and capitalize as AI capabilities rapidly advance.

The software ecosystem, at a glance

The software industry can be thought of in three layers: Applications, Platforms, and Infrastructure. Companies operating within each layer are considered software companies because their core offering is the creation, deployment, or sale of software. A company can appear in multiple layers of the software stack depending on its product offerings, in a strategy known as "vertical integration".

Applications

The application layer represents the software products that end-users directly interact with to perform specific tasks. Applications include an array of software such as enterprise resource planning (ERP) systems, customer relationship management (CRM) tools, productivity suites, and industry-specific applications.

Adobe, (365 applications and Copilot), SAP, Intuit, Workday, Salesforce, ServiceNow (SaaS products)

Platforms

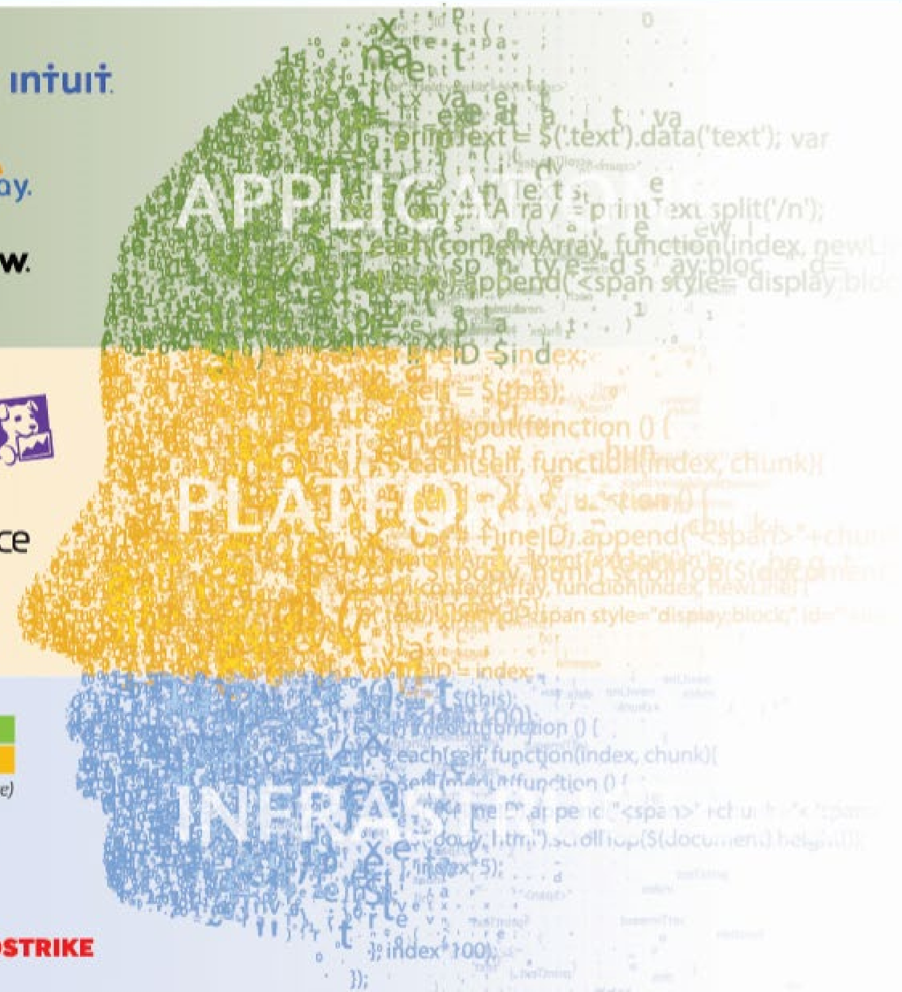
The platform layer provides the software environment, tools, and services that enable developers to build, deploy, and manage applications. Platforms include operating systems, databases, development frameworks, cloud Platform-as-a-Service (PaaS) offerings, and API platforms.

(Foundry platform), Snowflake, ServiceNow (the Now Platform), Dynatrace, Rubrik

Infrastructure

The infrastructure layer comprises the core computing resources, networking capabilities, and security and storage solutions that underpin all software. Infrastructure includes physical and virtual servers, data centers, cloud Infrastructure-as-a-Service (IaaS) offerings, databases, and cybersecurity essentials.

Oracle, Palo Alto Networks, Google (Cloud), (Azure), Amazon (AWS), Nebius, CrowdStrike



Note: List of companies in each layer is not exhaustive.
Source: Goldman Sachs GIR.

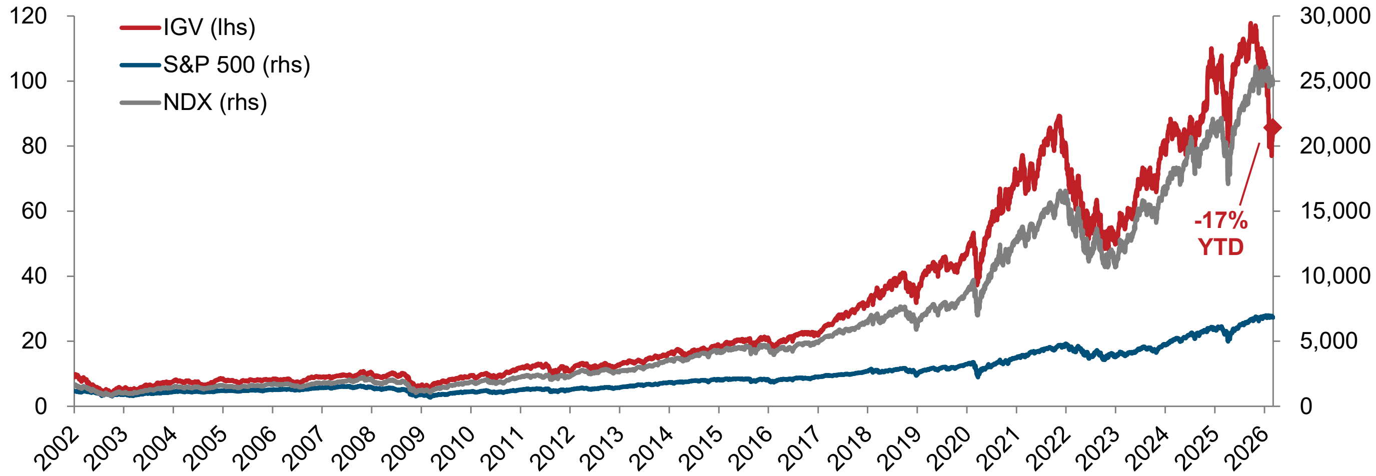
Special thanks to GS US Software Analyst Matthew Martino for data and guidance.

Illustration created using random AI generated code.

The software selloff, visualized

The iShares Expanded Tech-Software Sector ETF (IGV), which tracks the performance of around 120 North American stocks in the software industry, has declined by around 17% since the beginning of the year and 26% from its fall 2025 highs

Price

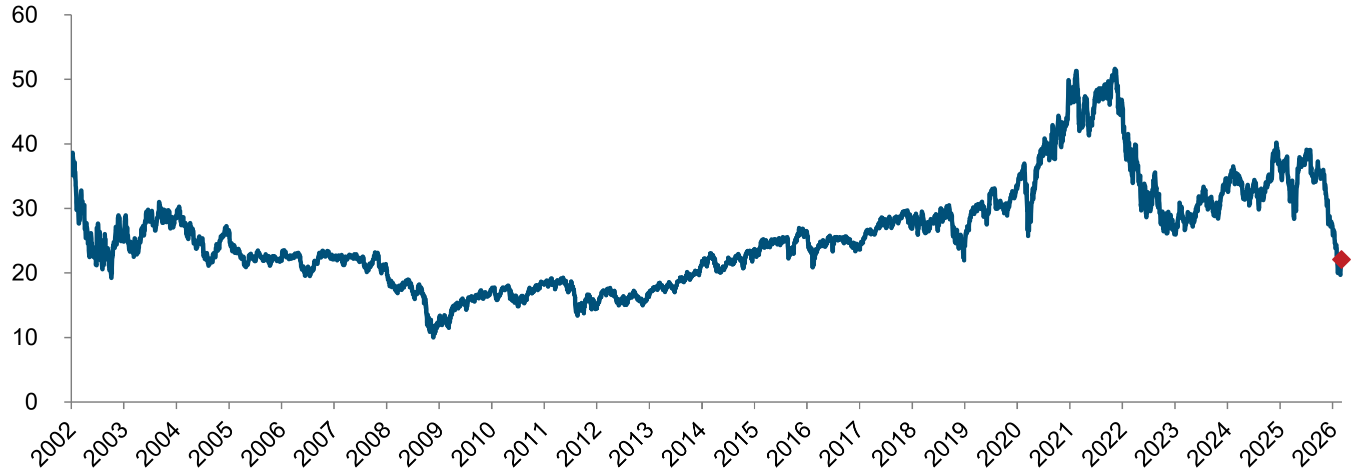


See here for more details about the IGV ETF.

Source: FactSet, Goldman Sachs GIR.

Valuations have also fallen sharply, with the companies in the ETF trading at roughly 22x, well below their Covid-era peak of 51x

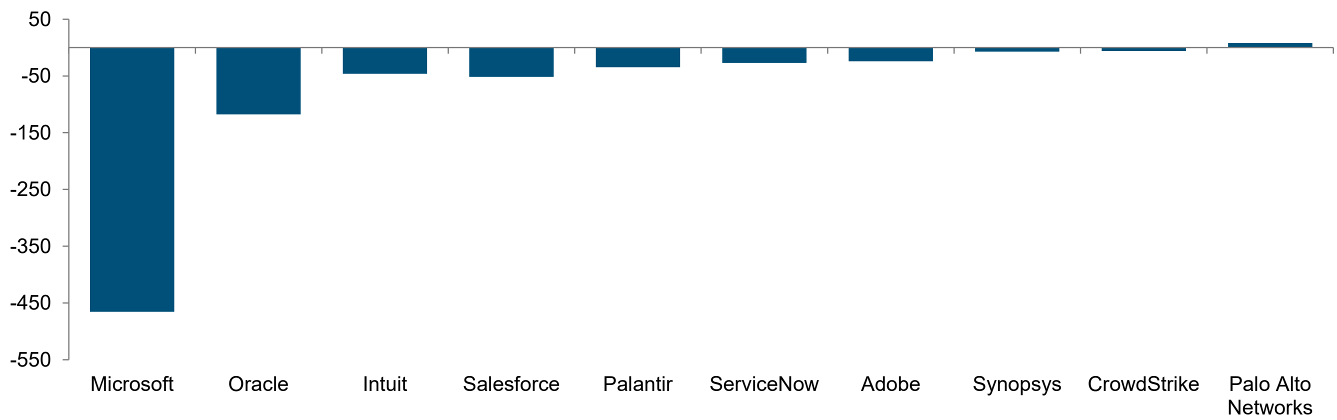
IGV NTM P/E



Source: FactSet, Goldman Sachs GIR.

The 10 largest stocks in the IGV ETF have in total lost nearly $800 billion in market capitalization since the beginning of the year

Market cap change for each of the 10 largest stocks in IGV since January 2, 2026, $bn



Source: Bloomberg, Goldman Sachs GIR.

Special thanks to GS US Portfolio Strategist Christophe Sung for data.

AI disruption: software subsector impacts

Q: Vertical Software, Data Infrastructure, and Physical-to-Digital are key subsectors of the software space. To what extent are legacy firms in these subsectors vulnerable to AI disruption?

Vertical Software

Adam Hotchkiss,
US Emerging Software Analyst

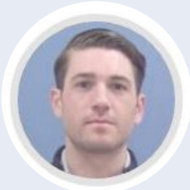
A: Vertical software refers to products built for specific industries rather than general-purpose systems like email or accounting tools.These systems store critical data, reflect industry-specific workflows, and are deeply embedded in companies' daily operations. Vertical vendors include firms like Guidewire Software (insurance), Tyler Technologies (public sector), and Veeva Systems (life sciences). As AI technology continues to rapidly evolve, the key question is whether new agentic technologies will make vertical systems less valuable or even replace them. Our view is nuanced but ultimately constructive. First, we believe vertical software systems will remain the core "systems of record" (see pg. 4-5) for their industries. The data and context embedded in these platforms are both unique to complex, industry-specific networks and essential for AI to generate meaningful, reliable outputs. Second, AI gives vertical software vendors an opportunity to enhance connectivity between complicated industry workflows and get customers more value out of their data. Over time, this will likely drive synergies between products and result in higher revenue, even if competition grows.

The important nuance here is execution. Management teams that are fast followers—those that actively integrate new AI capabilities and maximize the value of available LLM technology—are better positioned to limit value abstraction. Conversely, less innovative vendors, rather than be valued as an immersive software ecosystem, could see their value erode to that of a database, which would likely weigh on pricing power and growth.

Competition in this layer will likely intensify. We are already seeing early signs of this with businesses that deploy engineers directly within a customer's environment to implement AI (often called forward deployed engineering), as well as early-stage tools launched by LLM companies themselves (i.e., Claude plugins). That said, incumbents that already own the underlying data and industry context layer start from a position of strength. Customers are also likely to be relatively patient with these vendors given the strength of their relationships and deep embeddedness of the software.

These dynamics could ultimately drive more M&A. Incumbents seeking to remain competitive may increasingly turn to M&A to keep pace with AI-native innovation, while private equity firms may find vertical businesses with clear data and context moats increasingly attractive given the compression of public market valuations.

Data Infrastructure and Physical-to-Digital



Matthew Martino,
US Software Analyst

A: Data infrastructure refers to platforms that store, govern, and secure data so it can be accessed and used across an organization, such as Snowflake, MongoDB, and Rubrik. We view AI as a force multiplier in this space. As organizations deploy AI copilots and agents, they will create and query more data, demand faster response times, and tolerate less downtime. This raises the value of this foundational layer, because it determines whether AI can run reliably and securely in production.

The primary risk this subsector faces is that, over time, workloads could shift to newer or cheaper platforms—a familiar pattern in technology transitions—and more integrated AI stacks could reshape enterprises' spending priorities. However, we see these as competitive dynamics to monitor rather than immediate existential threats.

The disruption debate is more pointed for physical-to-digital companies. These are software firms that sell digital products to physical asset-intensive industries like construction and transportation. On the surface, these products can look like everyday applications. In reality, they sit inside mission-critical workflows with governance and compliance baked in, which raises the bar for AI-native challengers.

The long-term risk is not that AI replaces these platforms outright, but that a general-purpose agent layer could sit above them and reduce their standalone importance. But our view is that AI adoption will gravitate toward systems that control proprietary data and link directly to real-world outcomes. For such systems, AI will likely become a source of reinforcement rather than disruption.

AI tools for enterprise

Tool	Summary
	Core enterprise AI platforms
Anthropic Claude Cowork	An AI productivity tool that operates as an AI agent that can perform tasks directly on a user's computer, including automating repetitive tasks such as organizing files, extracting data from PDFs into spreadsheets, and generating presentations.
OpenAI Frontier	An enterprise-grade platform designed to build, deploy, and manage autonomous AI agents that act as "digital co-workers." AI agents can perform complex, multi-step tasks across business systems like CRM and data warehouses.
Microsoft Copilot	An AI digital assistant integrated across Microsoft 365 apps (Word, Excel, Teams), Windows, and web browsers, designed to boost productivity and streamline workflows.
Google Gemini for Workspace	An AI digital assistant integrated directly into Gmail, Docs, Drive, Slides, Sheets, and Meet that helps users boost productivity through drafting, organizing, summarizing, and brainstorming.
OpenClaw	An open-source autonomous AI agent that acts as a personal digital assistant capable of managing files, running terminal commands, and automating tasks across platforms like Discord, Slack, and WhatsApp.
Adept AI	Uses autonomous AI agents that can execute manual, repetitive, end-to-end workflow tasks across existing software applications.
	AI for customer service and sales
Sierra AI	An AI-powered customer service agent that can handle complex conversations across digital channels including around tracking orders or resolving issues.
Ada CX	An AI customer service platform designed for enterprises to automate support across chat, email, and voice channels.
Gong AI	A revenue intelligence platform using AI to analyze customer interactions to improve sales performance and identify risks.
Clari AI	Provides AI-driven revenue forecasting and deal intelligence.
	AI for HR
HR Plugin for Claude Cowork	Allows HR professionals to deploy specialized agents that operate across local files and enterprise apps. It can draft offer letters, build onboarding plans, and run compensation analyses by pulling data from multiple spreadsheets.
Paradox	A tool for high-volume recruitment that handles candidate screening, answers FAQs, and automatically schedules interviews based on recruiter availability.
	AI for financial services
AlphaSense	An AI-powered market intelligence platform that aggregates millions of data points from equity research, filings, and news to provide real-time insights for analysts.
Alteryx	Automates repetitive data workflows and analytical modeling, allowing finance teams to run various risk scenarios and consolidate disparate data sources without manual effort.
Prophix One	A finance-native platform embedding AI agents directly into budgeting, forecasting, and "continuous close" workflows to reduce human error in reporting.
Trullion AI	An AI accounting platform that automates lease accounting and revenue recognition.
Anaplan (PlanIQ)	An AI platform that detects anomalies and generates predictive forecasts across finance, sales, and supply chain operations.
	AI for manufacturing and supply chains
Siemens & NVIDIA (Industrial AI OS)	A joint operating system that transforms "passive" digital twins into active intelligence, allowing real-time simulation and optimization of factory floors.
Blue Yonder	An AI-driven supply chain platform that provides cognitive analytics to predict disruptions and automatically adjust production and logistics schedules accordingly.
Cognex Vision AI	Leverages deep learning for high-speed machine vision quality control, detecting subtle defects that traditional rule-based systems miss.
Interos	An AI risk management platform that monitors global supply chains for geopolitical, financial, and ESG-related disruptions.
Phaidra	Provides autonomous AI control systems for industrial manufacturing and mission-critical facilities, such as data centers.
	AI for legal and professional services
Harvey AI	A domain-specific AI platform used by major law firms for regulatory research, document analysis, and drafting documents.
Luminance	An AI platform specialized in M&A due diligence and autonomous NDA negotiation, capable of identifying anomalies across large document sets.
Ironclad	A contract lifecycle management tool that uses AI to automate the drafting, negotiation, and tracking of legal obligations at scale.
Casetext (CoCounsel)	An AI legal assistant that performs case law research, summarizes depositions, and reviews documents for relevance using advanced reasoning.
	AI for retail and e-commerce
Salesforce Einstein	An integrated AI layer for CRM that generates personalized product descriptions, predicts customer churn, and provides merchandising insights.
Shopify Magic	Suite of AI tools that automates inventory management, generates marketing copy, and personalizes product recommendations.
ConversionBox	An AI shopping assistant that uses conversational AI to guide customers through product discovery, acting as a digital salesperson on e-commerce sites.

Note: Table does not constitute an exhaustive list of all AI tools for enterprise.
Source: Company data, various news sources, compiled by Goldman Sachs GIR.

Interview with Sanjay Poonen

Sanjay Poonen is CEO and President of Cohesity, an AI-powered data security and management firm, and serves on the boards of Philips and Snyk. Previously, Poonen was COO at VMware and President of SAP. Below, he argues that agentic AI has the potential to be enormously disruptive for the software industry, and legacy firms must adapt by leveraging the technology to improve their product offerings or risk being demolished by it.

The views stated herein are those of the interviewee and do not necessarily reflect those of Goldman Sachs.

Allison Nathan: Based on your vast experience in the software industry, does the emergence of new agentic AI tools for software development mark a transformative moment for the industry?

Sanjay Poonen: Agentic AI has already proven to be an incredibly powerful tool with the potential to be enormously disruptive. As an engineer who began my career writing code, this is a truly exciting and transformative moment—back in the 1990s, I often stayed up late trying to debug code, hoping to find another engineer also working late whom I could consult. Today, that "person" is an AI assistant.

However, just like any technology wave, you must surf this tsunami, or it will demolish you. If software engineers aren't leveraging AI technology to write code and improve productivity today, it's a crime. The best software engineers I know, both at Cohesity and beyond, say that the technology is revolutionizing the way they code. So, we've thoroughly embraced it at Cohesity. But every business should be thinking about how these tools can support and improve not only their software engineers, sales reps, and human resources personnel, but also their products and services. I see limitless potential to leverage generative AI in almost every industry, from medtech to aeronautics and even space.

> "Agentic AI has already proven to be an incredibly powerful tool with the potential to be enormously disruptive... However, just like any technology wave, you must surf this tsunami, or it will demolish you."

Allison Nathan: You paint a very positive picture of the benefits of AI for software and productivity. But the whole software sector has re-rated substantially lower because of concerns that legacy software companies won't be able to compete even if they are embracing AI technology themselves. So, are these concerns warranted?

Sanjay Poonen: With such transformative technology, every software company must assess whether they can sustain their current price per user or will instead be threatened by AI-native or other competitors leveraging the technology. At the core of every Software as a Service (SaaS) application is a CRUD—Create, Read, Update, Delete—database that maintains data, application logic, and workflows, which AI tools have the potential to radically disrupt. If a company can sustain its price per user amid this disruption, it will likely be fine. But some companies may need to lower their price per user to remain competitive. That doesn't necessarily mean the company will go bankrupt or disappear, but it does have important implications for revenue growth and terminal value. So, every CEO, analyst, and investor is rightly assessing this risk.

> "With such transformative technology, every software company must assess whether they can sustain their current price per user or will instead be threatened by AI native or other competitors leveraging the technology."

Allison Nathan: Is it realistic to think that these companies will be able to effectively respond to this pricing pressure and stay competitive?

Sanjay Poonen: Yes, but it requires innovation. Firms whose core products are under pressure may need to add a second or third module with enhanced capabilities that garners a higher price per user to compensate for the lost revenue. For example, a company could offer a small, medium, and large packaging of its product, with the price per user of the small, base product coming down, but the price per user of the larger product offerings higher.

Alternatively, firms can expand into new areas to grow their total addressable market (TAM) and sustain revenue growth even as price per user declines. When I was at SAP, we recognized that a price per user ceiling existed for our ERP software and responded by adding a system of engagement on top of the system of record by acquiring analytics companies and building new business objects. This allowed us to raise the effective price per user for the combined product offering. Every company facing disruption of their core product will need to make these types of adjustments.

SaaS companies can also attempt to shift from a price per user to a price per outcome model, where fees are based on the measurable value they provide to the customer. That pricing model is harder to sustain because tracking outcomes can be challenging and an IT buyer typically has a fixed budget and is often unable or unwilling to enter into a vendor contract with fees that depend on the broader company's performance. Because of that, most companies price per user. The only company I am aware of that uses a price per outcome model is Palantir, which often sells its product at the CEO level, where the power to make these decisions resides.

Allison Nathan: So, is the indiscriminate selloff across the software space overdone?

Sanjay Poonen: The answer to that will ultimately be determined by the revenue growth of these companies. As both a student of software and the CEO of a software company, I will be closely watching whether growth is holding up or slowing down. That said, in the security world Cohesity lives in, the large market reaction to Anthropic's announcement that it had built a code scanning tool for security was akin to Anthropic sneezing and everyone coming down with pneumonia. But the market seemed to quickly realize that this was an overreaction and that this tool won't enable users to build a firewall or a data or endpoint security product overnight.

Allison Nathan: Even if that's true today, are you concerned that it won't be in the future given the rapid rate at which generative/agentic AI technology is advancing?

Sanjay Poonen: Former Intel CEO Andy Grove said it best: "only the paranoid survive." Nobody in this space can afford to be cavalier about the potential for disruption. As a CEO, I have to assume that someone could use AI tools to build a competitor. In fact, I recently asked an AI assistant to build a Cohesity competitor to see what it would do. Unsurprisingly, its first response was that this was a "tall order." But it was able to think about the challenge in logical steps, which revealed an understanding of how to build a competitor. So, it's only safe to assume that, given time, these tools will eventually be able to do just that. For that reason, it is incumbent upon myself and every other CEO in a similar position to take advantage of these same tools to expand and improve our own product offerings. Again, we must ride the tsunami to our benefit rather than let it demolish us.

> "Former Intel CEO Andy Grove said it best: "only the paranoid survive." Nobody in this space can afford to be cavalier about the potential for disruption."

Allison Nathan: But is just bolting AI agents onto legacy layers ever going to be as successful and powerful as native AI product offerings, or will legacy companies eventually have to start from scratch?

Sanjay Poonen: Software is typically built in architectural layers, with the top layers easiest to refresh and the core, bottom layers much harder to. But even timeless software evolves every decade or two, so engineers must be comfortable with the idea of completely rewriting part of or even the whole architectural stack. The good news is that using new tools and approaches to rewrite code from scratch will probably take a small fraction of the time it took to write the original code. Even then, disrupting the whole stack is undoubtedly hard, even if necessary. In such instances, companies might decide to shelter in place and keep the business and cashflow running while simultaneously building or buying the next generation of the company. I see this as akin to living in your old house while building a new one next door and only tearing down the old house once the new one is ready.

We dealt with this at every organization I've worked at. At SAP, as new companies that were building a SaaS version of ERP in the cloud threatened to disrupt our on-prem product, we had to consider whether we could build a cloud-native product fast enough without disrupting our core platform. The best companies find a way to constantly evolve their stack to remain competitive.

Allison Nathan: While large players like a Microsoft or Salesforce may be able rebuild, is there an inherent disadvantage for the smaller companies that have less resources to do so?

Sanjay Poonen: It's wrong to assume that reinvention is easier at a Microsoft or Salesforce, and in fact the opposite may be true. It took new leadership for Microsoft to transform from a Windows company to a cloud company and now an AI company. In my experience, the bigger the company, the greater the molasses, dinosaur mindsets, and resistance to change. Smaller companies may not have as many resources, but they have the advantage of speed. It's a bit like a speedboat vs. an aircraft carrier—small firms can usually make decisions faster and are less likely to have a large legacy install base that they must worry about transforming. Microsoft pivoting away from Windows to Azure was a big and highly risky shift. A company starting completely from scratch has no legacy to deal with. I'm reminded of the joke that God created the world in seven days because he didn't have an install base. The bigger the install base, the tougher it is to pivot and change.

All that said, whether small, medium, or large, as long as a company maintains a mindset and culture that is open to innovation, none of these challenges are too daunting to achieve. Companies with such a mindset can start things from scratch and build on them without destroying the house they live in, which allows them to create S-curve upon S-curve of continued innovation.

Allison Nathan: So, what could the software industry look like in five or 10 years?

Sanjay Poonen: I expect more vertical industry software capabilities in industries like space, healthcare, material sciences, and robotics. Consider SpaceX's vision of data centers in space, where cooling isn't a problem. That endeavor will undoubtedly require software. The same could be said for biotech, medtech, and robotics, which will almost certainly increasingly attract top software talent that previously went to the Googles and Salesforces of the world, driving innovation that could help people live healthier and longer lives. So, my hope is that the software industry spreads its skillset into other vertical industries with positive implications for society and the world. And AI will undoubtedly be a part of every one of those industries.

Software stocks: earnings stability needed

Ben Snider argues that a rebound in the stocks most exposed to AI disruption risk likely won't come until earnings stabilize

The AI narrative has shifted. After three years of equity investor enthusiasm regarding the tailwinds from AI, recent equity market rotations reflect a growing focus on the risk of AI disruption. Software stocks have been at the center of this shift, falling by 18% from their YTD highs and now sitting roughly 26% below late 2025 levels. Stocks in a range of data-intensive industries beyond software, including insurance, media, and business services, have also declined sharply in recent weeks. While the moves have been swift and severe, we think a rebound in the stocks most exposed to AI disruption risk will take time, as disproving the disruption narrative won't be easy or quick.

Re-rating: a decomposition

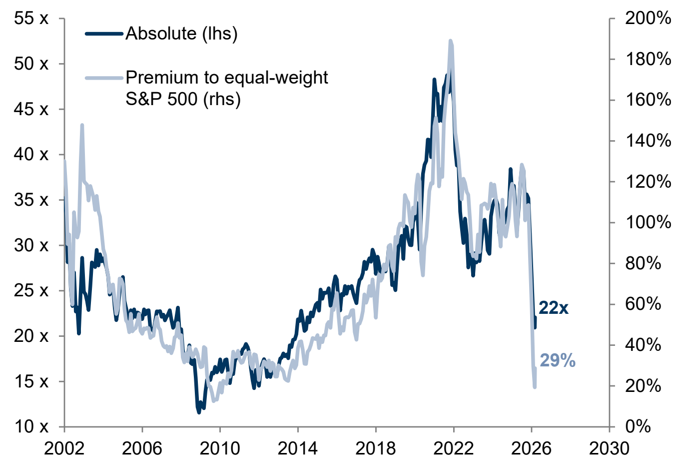
After entering 2026 at extremely elevated valuations, growth expectations, and weights in investor portfolios, the software industry has sharply re-rated to reflect declining expectations for future earnings growth as AI disruption risk grows.

The industry's forward P/E multiple has fallen from roughly 35x in late 2025—consistent with 2028 consensus revenue growth in the 15-20% range, over twice the roughly 6% growth expected for the median S&P 500 stock and the highest expected growth in at least two decades—to around 22x currently, the low end of the multiple's recent range and the lowest absolute level since 2014, consistent with 5-10% earnings growth.

Given the high starting point of valuations, software continues to trade at a valuation premium to the equal-weight S&P 500 despite the recent selloff, and its 9% S&P 500 weight is roughly double its weight a decade ago.

Software continues to trade at a valuation premium to the equal-weight S&P 500 despite the recent selloff...

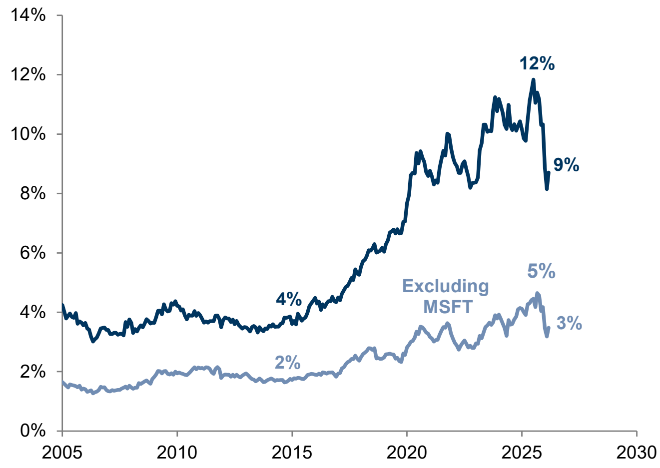
Software forward P/E, X (lhs), % (rhs)



Source: FactSet, Goldman Sachs GIR.

...with its weight in the S&P 500 roughly double that of a decade ago

Software weight in the S&P 500, %



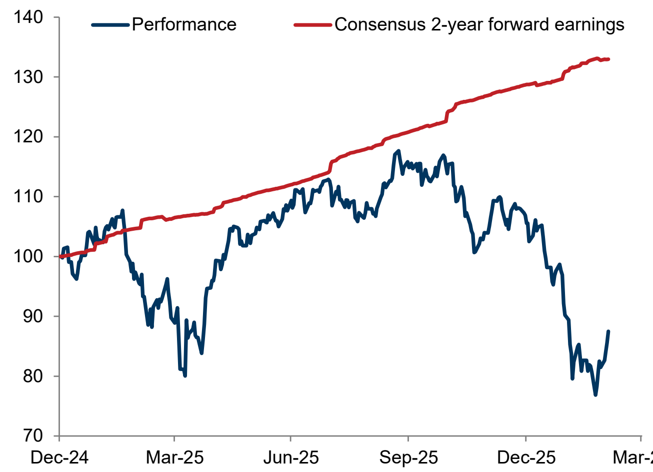
Source: FactSet, Goldman Sachs GIR.

In addition to lower growth expectations, the recent re-rating reflects rising uncertainty around the long-term trajectory of the industry and the "terminal value" of stocks previously viewed as long-duration growth assets. In other words, share prices now reflect not just slower expected growth, but also doubts about the durability of business models as AI disruption raises questions about pricing power and competitive barriers.

The impact of this long-term uncertainty can be seen in the juxtaposition of recent share price volatility and 4Q25 earnings results. In contrast with falling share prices, recent earnings results have reflected strong profitability—including double-digit earnings growth last quarter, with profit margins for the industry sitting at record highs and roughly twice those of the rest of the market—and have driven upward revisions to near-term analyst earnings estimates.

The performance and earnings of software stocks have diverged

Indexed performance and earnings of software stocks



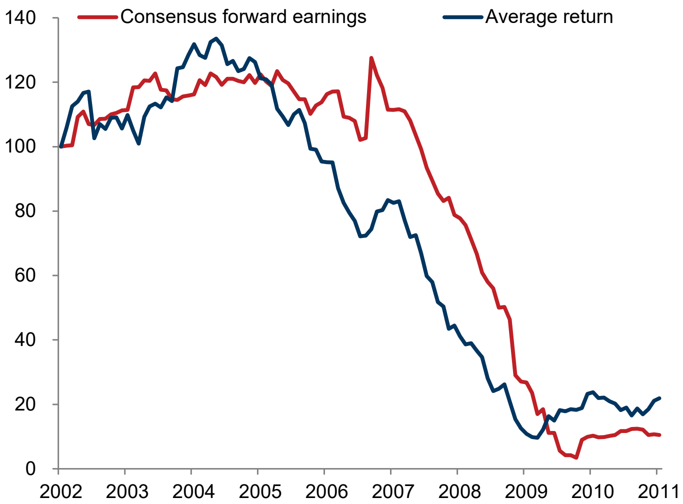
Source: FactSet, Goldman Sachs GIR.

Lessons from history: don't expect a V-shaped rebound

Historical episodes of disruption risk suggest that share prices in affected industries will likely stabilize only when earnings stabilize, and that these periods of uncertainty can last for years. In the early 2000s, US newspaper stocks fell by an average of around 95% over the course of five years, and the group bottomed out only slightly ahead of the stabilization in forward earnings estimates. Tobacco stocks in the late 1990s experienced disruption risk driven by litigation, rather than technology, but the pattern was similar. In that episode, share prices declined by over 50% in 1997-1998 and troughed only as the litigation settlement reduced uncertainty and the near-term earnings outlook stabilized. Given the rapid pace of AI advancement today, disproving the narrative of disruption risk is difficult when near-term earnings outlooks in the vulnerable industry have barely begun to weaken, much less stabilized.

US newspaper stock prices stabilized only when earnings stabilized

Indexed avg return and consensus forward earnings of US newspaper stocks



Source: FactSet, Goldman Sachs GIR.

For investors, the lesson is not to expect a V-shaped rebound in the parts of the equity market most affected by recent disruption fears. For these industries, near-term earnings can be an important signal of business resilience, but in many cases will be insufficient to disprove long-term downside risk. For the group to re-rate higher, investors will likely require more clarity on AI's eventual business impact, which will take time, or a substantially lower starting point for valuations and ownership. This may be an easier lift for the parts of the software and business services industries with particularly resilient business moats, but those moats, too, will need to be proven.

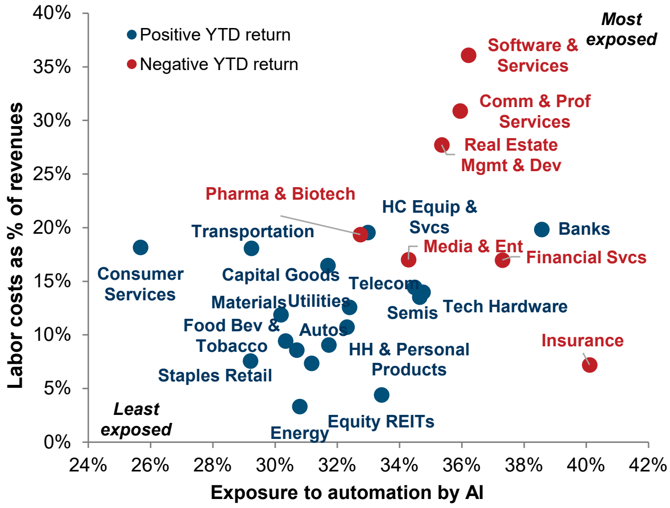
Of course, in any broad-based selloff there will liely be some proverbial "babies thrown out with the bathwater" that represent value opportunities for stock pickers. In the current episode, the key is the speed with which companies can disprove disruption risk.

Avoiding AI disruption risk: look to "real world" assets

Another consequence of the recent focus on disruption risk is a rotation of investor portfolios toward "real world" assets with perceived insulation from AI disruption risk. Recent outperformers in this vein include a handful of cyclical industries like industrials and materials that have also enjoyed tailwinds from an improving economic growth backdrop. GS forecasts for above-trend economic growth have lent fundamental support to cyclicals' recent strength, and we entered 2026 recommending investors own many of these stocks. However, the tailwind from cyclical economic acceleration also highlights the broader equity market vulnerability if the growth backdrop proves less robust than expected or industry-specific AI disruption fears spread to broader concerns about labor market disruption. This risk helps explain the recent outperformance of more defensive "AI-insulated" sectors, including consumer staples and biopharma, and we think the appropriate strategy is to balance cyclical exposures with some relatively defensive equity positions.

The recent focus on AI disruption risk has triggered a rotation toward "real world" assets

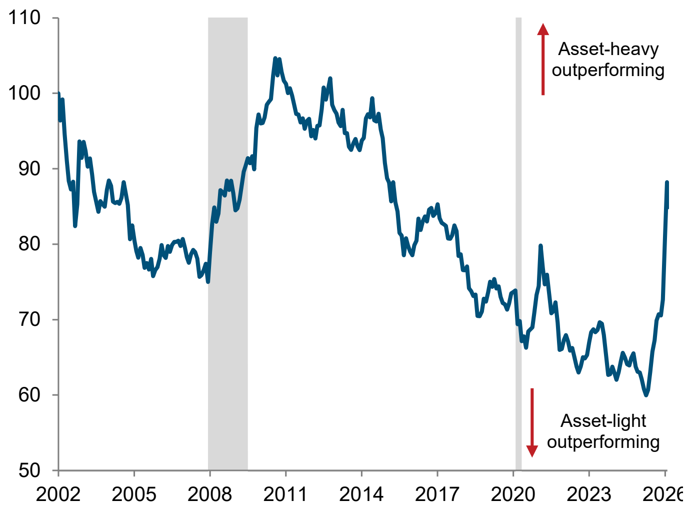
Russell 1000 industry median



Source: Revelio, company filings, Goldman Sachs GIR.

Increased investor desire for "AI insulation" has led asset-heavy stocks to sharply outperform

Asset-heavy vs. asset-light indexed performance



Note: Grey bars indicate recessions.
Source: Goldman Sachs GIR.

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Managing AI risk in portfolios

Christian Mueller-Glissmann discusses the why and how of managing AI risk in portfolios amid growing AI disruption fears

Investor concerns about AI risks for the global technology sector have grown amid ongoing worries about hyperscalers' massive AI capex spend and, most recently, rising fears that new AI tools will disrupt tech incumbents, especially legacy software firms. Such concerns matter because global portfolios are highly exposed to the technology sector, especially within US equities, where software stocks have a relatively large weight, with substantial exposures to technology and software specifically also a feature of private markets.

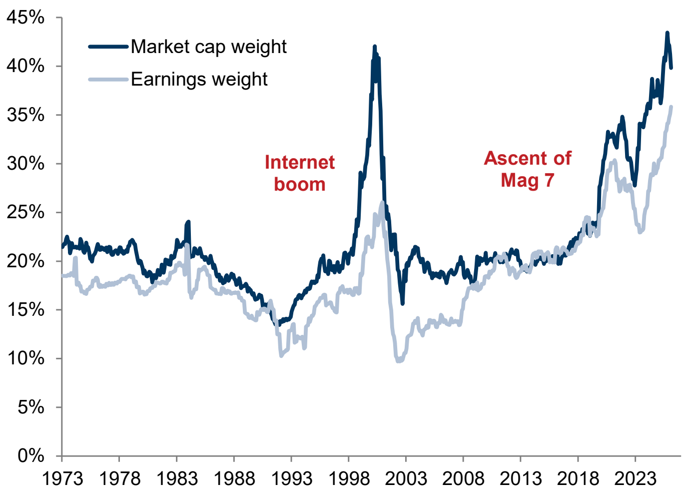
But the fate of the tech sector also matters more broadly given how closely tied structural cycles in US equity valuations and outperformance have been to the performance of tech stocks. To achieve more balance between innovation and inflation risks within portfolios, we think investors may need to continue rotating from capital-light to capital-heavy businesses that AI won't easily disrupt, though the aggressive rotations we've already seen warrant being more selective.

A tech-dependent equity market...

What happens to the tech sector has broader implications for the equity market for several reasons. The sector has historically been a key driver of structural cycles for US equities, with major waves of technological innovation often significantly impacting the broader market. During the 1960s PC boom, 1990s Internet boom, and the post-Global Financial Crisis (GFC) ascent of the Magnificent 7, tech's share of total market cap rose sharply, and significant market corrections often followed these periods. Today, the TMT sector once again commands the largest weight within US equities, comparable to Tech Bubble levels, but the sector's contribution to earnings is now substantially greater compared to then.

Tech's weight in the US equity market has risen sharply during major waves of technological innovation

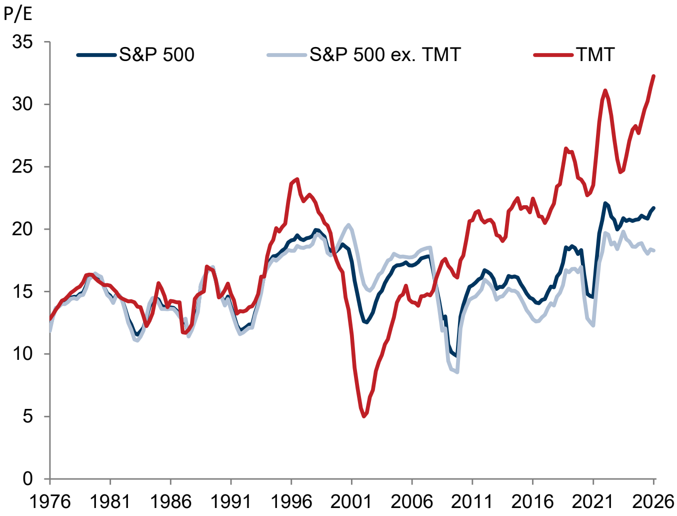
Tech market cap and earnings weight within US equities, %



Source: Datastream, Goldman Sachs GIR.

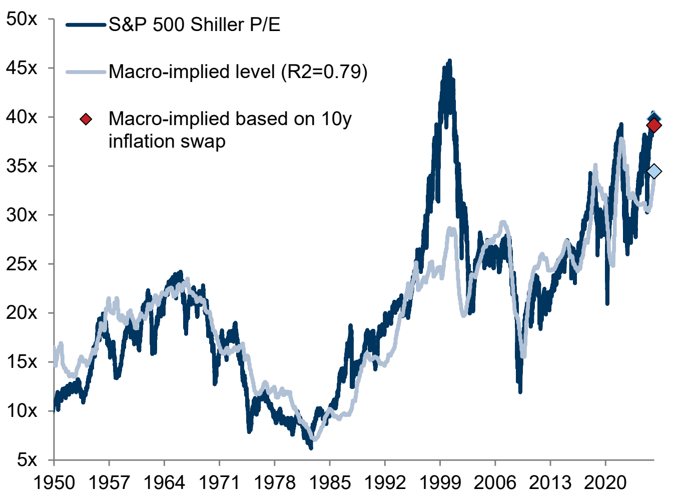
Tech has also played a key role in driving structurally higher S&P 500 profitability and valuations. Since the GFC, the return on equity (ROE) across the TMT sector has nearly doubled, significantly lifting the S&P 500's overall profitability. This trend accelerated during the pandemic as technology adoption surged. Higher profitability has, in turn, supported higher equity valuations. Indeed, the rise in the S&P 500's Shiller P/E over the past two decades reflects two interconnected elements: improved corporate profitability—driven by the growing weight of the highly profitable, capital-light US tech sector—and persistently low and stable inflation. And the strong performance of large cap tech stocks has driven a large amount of the US equity outperformance since the GFC.

The ROE for the TMT sector has nearly doubled since the GFC, significantly boosting S&P 500 profitability



Source: Compustat, Goldman Sachs GIR.

High profitability, coupled with low inflation expectations, have fueled a rise in US equity valuations



Source: Haver Analytics, Kenneth French, Robert Shiller, Goldman Sachs GIR.

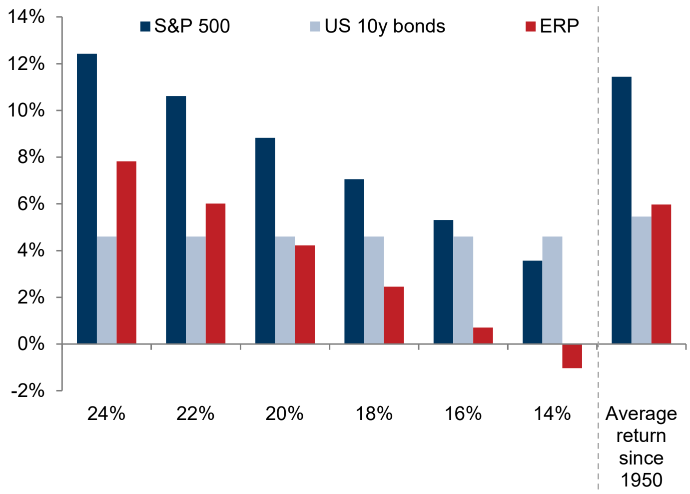
...meets AI disruption fears

Investors have extrapolated the TMT sector's high ROE, even as these firms pay out relatively little to shareholders, essentially betting that major tech conglomerates will maintain their profitability as they make substantial reinvestments. However, last year anxieties around AI capex began to impact some of the largest spenders in the sector, particularly the hyperscalers. Beyond concerns around potential future AI revenues, the big spenders are also becoming more capital-heavy, which might weigh on ROE in the medium term.

More recently, rising concerns about AI disruption have weighed on parts of the TMT sector, especially software stocks, with investors increasingly worried that the tech sector may end up disrupting itself. Advances in AI-powered task automation, for example, could undermine the pricing power of some of the most profitable large-cap software firms. While AI adoption may also boost other businesses' ROEs, a decline in the profitability of the market's largest and most lucrative segment could adversely affect the S&P 500's overall ROE and dampen long-term return expectations, which currently rely on the compounding of those ROEs. And markets have been pricing negative AI disruption faster than potential tailwinds from AI adoption.

Lower corporate profitability could portend lower long-term equity returns and less attractive equity risk premia

S&P 500 ROE in 10 years scenarios, % (x-axis) vs. annualized nominal returns for stocks and bonds and equity risk premium (ERP), % (y-axis)*



**See this note for more detail about the long-term return calculations.*
Source: Haver Analytics, Goldman Sachs GIR.

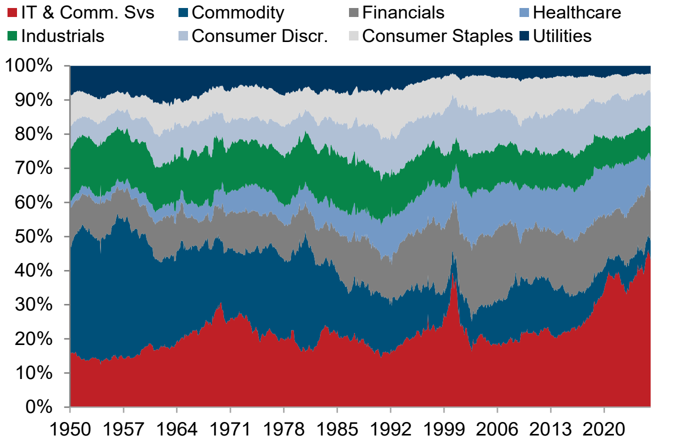
Balancing innovation and inflation risks

We expect an AI-driven rise in productivity to eventually support corporate profitability. But concerns about AI disruption may continue to be priced faster than the benefits of AI adoption. On top of that, the AI application firms driving disruption are mostly private companies and, as such, not easily investable, which makes it difficult to diversify AI disruption risk. Investors may therefore demand more risk premium for at-risk businesses, which has already driven a rotation and value convergence between capital-light growth sectors like TMT and capital-heavy value sectors like telecoms, industrials, and utilities that AI is unlikely to disrupt.

Owing to a boost from innovation, the TMT sector has become a much larger part of the S&P 500 since the GFC, while capital-heavy sectors have underperformed, partly owing to a lack of inflation. As investors continue to worry about AI disruption, this may lead them to demand a higher equity risk premium, which could result in some de-rating of equities vs. bonds, with bonds potentially also benefitting from disinflation due to AI disruption.

The TMT sector holds a dominant position within the US equity market, with capital-heavy sectors holding a relatively small one

Share of S&P 500 by sector, %



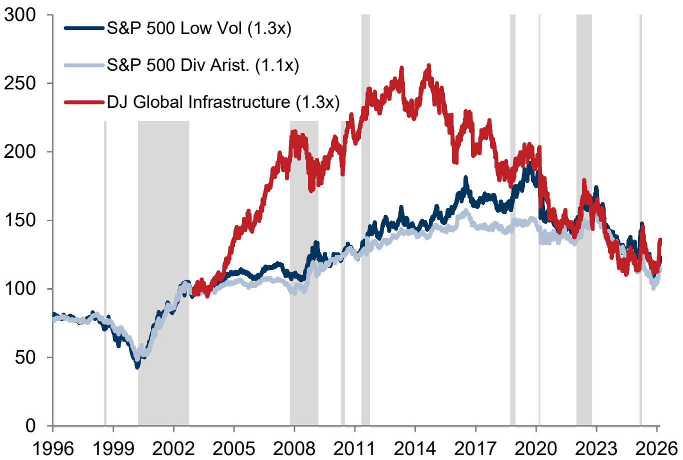
Source: Bloomberg, Datastream, Goldman Sachs GIR.

To manage innovation and inflation risks within portfolios, investors may need to continue rotating further from growth to value and capital-light to capital-heavy businesses, both in equity and credit. However, after the aggressive rotations since the beginning of the year, we favor being more selective. Within equities, we continue to see diversification benefits from defensive styles such as low volatility stocks, dividend aristocrats, and infrastructure, which materially outperformed on a risk-adjusted basis during the Tech bubble burst.

We also expect more dispersion as AI will likely continue to produce more winners and losers. This should support strategies that buy individual stock and sell index options and long/short hedge funds in both equity and credit. And we continue to view geographical diversification as an attractive strategy. TMT exposure is typically lower in non-US equity markets and tends to be more hardware-focused, with such markets also offering more capital-heavy sector opportunities.

Defensive styles have lagged in recent years but outperformed over the long run

Total return relative performance vs. S&P 500 (grey bars indicate periods of 15%+ market selloffs)*



Source: Datastream, Bloomberg, Goldman Sachs GIR.

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Software disruption & credit: no hard fall

Shamshad Ali argues that software loan losses owing to AI disruption fears are unlikely to catalyze a turn in the credit default cycle

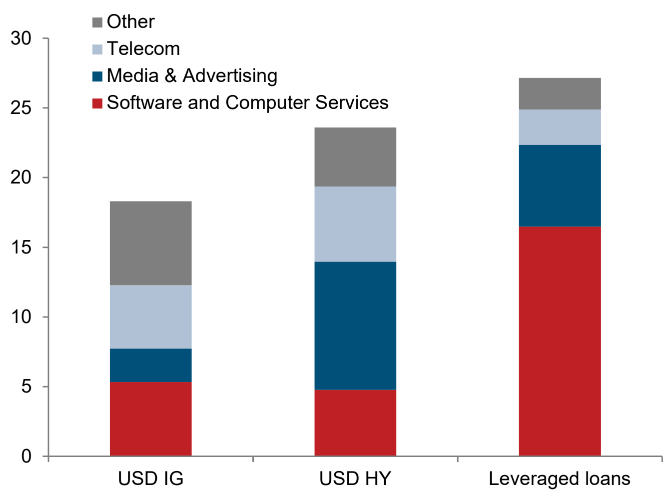
Recent headlines around AI-driven disruption—and the associated sharp drawdown in software stocks—have weighed on software-exposed credit, most notably in the broadly syndicated loan (BSL) market. As sectors like insurance, brokers, and other services exposed to AI disintermediation have also come under pressure as more new AI tools have emerged and investors worry about downstream implications, we believe the BSL market faces the greatest risk should disruption concerns translate into weaker earnings trajectories for traditional software companies, which would erode interest coverage ratios for these already levered borrowers. Given that software accounts for a meaningful share of the BSL market, some degree of index-level stress would be difficult to avoid if fundamentals continue to deteriorate. That said, the benign macro backdrop and lower funding costs should help limit the risk of a turn in the credit default cycle.

A more exposed BSL market

Software accounts for only a small slice of the USD IG and HY bond indices (roughly 5%) but a meaningfully larger share of the BSL market (16%). While many IG software issuers are large incumbents (including the AI hyperscalers), software-exposed issuers in the HY bond and leveraged loan markets have underperformed broader benchmarks as investors have repriced disruption risk. HY software has shifted from trading tighter than the average spread level of the overall HY index to roughly double index spreads, while secondary prices for software loans have fallen to the lowest levels since April 2025.

Software accounts for only a small slice of IG and HY but a meaningfully larger share of the BSL market

TMT subindex exposure in USD credit, %

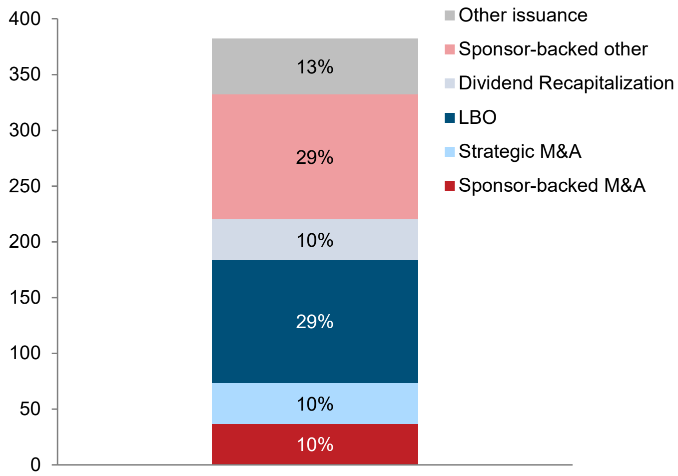


Source: Bloomberg, Pitchbook LCD, Goldman Sachs GIR.

A defining feature of software-related loans is the linkage to sponsor-backed (typically private equity) activity. Since 2021, nearly 40% of technology leveraged loan supply has been tied to sponsor-driven M&A and LBO financing. Another 10% of issuance stemmed from dividend recapitalization—firms taking on new debt to pay dividends—as sponsors sought liquidity amid a less favorable exit environment. The upshot is that underwriting is no longer just about borrower fundamentals; investors also need to carefully consider sponsor incentives, time horizon, and sponsors' willingness to support a firm's capital structure through difficult periods as the industry navigates a changing paradigm.

80% of tech loan issuance since 2021 has been sponsor-backed

Share of total gross Computer and Electronics loan issuance since 2021 by category, $bn



Source: Pitchbook LCD, Goldman Sachs GIR.

Some impairment likely, but not a turn in the default cycle

The potential persistence of AI disruption pressures would likely most clearly manifest via downgrades and lower secondary prices in the software-heavy portion of the loan index. Particularly important to watch would be the implications of such shifts for the CLO community, the main buyer of leveraged loans. CLOs have built-in rules or covenants designed to ensure that the overall diversification and quality of the underlying loans in their portfolios remain within acceptable limits (concentration, over-collateralization and collateral-quality tests). The downgrading of too many loans to a CCC rating (BSL CLOs typically have a limit of 7.5% for CCC exposure) could trigger various penalties. As a result, prices for these loans could move beyond fundamentals as the investor base potentially shifts from CCC-averse CLO buyers.

That said, bifurcation in loans isn't new. Even in periods when lower-quality loans cheapen, the broader index can remain resilient when higher-quality borrowers repeatedly reprice their loans at lower spreads and new loan supply is muted, as has been the case this year. Overall, while some impairment in software-exposed loans is hard to avoid, we don't think this impairment is sufficient to catalyze a credit default cycle, for two reasons. First, the macro backdrop remains benign, and history suggests that defaults typically coincide with rather than predict the broader cycle. Second, lower interest rates and improved funding conditions should continue to provide incremental relief to floating-rate borrowers.

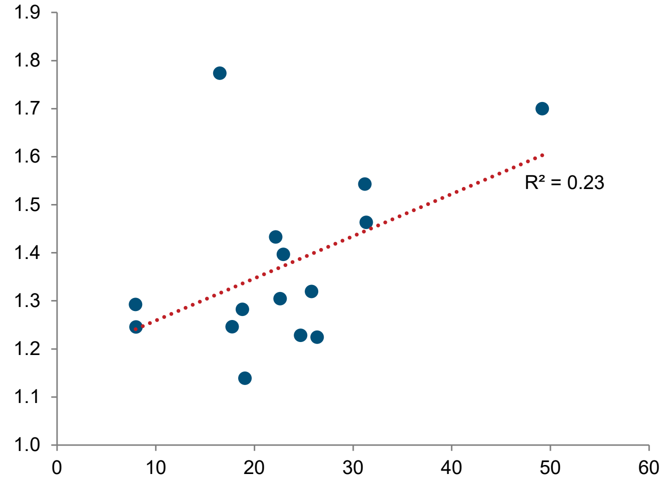
Pain in direct lending, but de-leveraging risk looks contained

Beyond the BSL market, direct lending—and by extension business development companies (BDCs) that invest in small and mid-sized, often private, firms—has been in the crosshairs given meaningful allocations to software. Indeed, our equity analysts estimate that software portfolio companies account for

22% of BDC loans (see pg. 20). Consistent with this, BDC equities have tended to trade at wider discounts to net asset value (NAV) while BDC credit spreads generally have widened in accordance with software exposure.

BDC credit spread widening correlates with software exposure

Software exposure as of 3Q25, % (x-axis) vs. on-the-run BDC senior unsecured bond spread change, ratio relative to Jan. 23 level (y-axis)



Source: Bloomberg, Goldman Sachs GIR.

We are closely monitoring signs of stress in BDC portfolios and have long argued that some writedown in asset valuations is likely. Recent liquidations of assets in BDC portfolios have also fueled speculation about asset health despite the loans selling at high, near-par, dollar prices. While full transparency isn't available on the sales, our equity analysts continue to think that credit trends in BDC portfolio data seem benign for now, but near-term portfolio data will remain under scrutiny.

While the rising rate of redemptions in BDCs is worth monitoring, we continue to think that the probability of a liquidity event or forced de-leveraging in direct lending is low owing to generally moderate leverage and substantial dry powder that can re-enter the complex and provide support, especially at more attractive valuations.

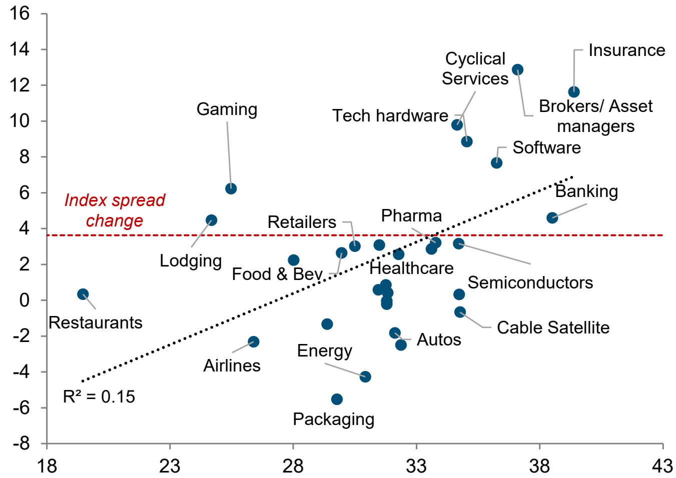
More dispersion from the next leg of the AI trade

While AI disruption fears remain centered on software, they have now spread to several other sectors, including insurance, brokers, and broader financial services (see pgs. 21-23), as the foundational model providers continue to roll out new domain-specific products. Amid this, IG and HY sector performance has become correlated with exposure to AI automation, mirroring the broader risk asset rotation toward "real economy" sectors that are less exposed at the expense of white-collar services that could face more disruption risk.

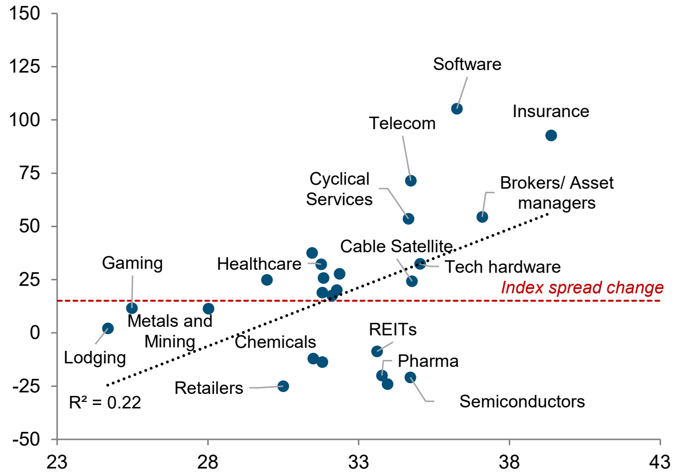
Looking ahead, we think this higher dispersion dynamic will persist, for two reasons. First, continued improvements in AI capabilities will likely impact more industries. Second, dispersion within industries should rise as company outcomes diverge based on the execution of AI adoption strategies. This dispersion will make it more difficult for investors to calibrate fair value for affected sectors, and issuer selection will become increasingly important. But we still see limited scope for serious impairment, especially for higher quality firms. Beyond their incumbent status and low starting points for leverage, we would note that the concerns around terminal values that have fueled much of the equity volatility are less salient for fixed duration credit assets, which are concerned with repayment of interest and principal.

Credit sector performance has become correlated with AI exposure

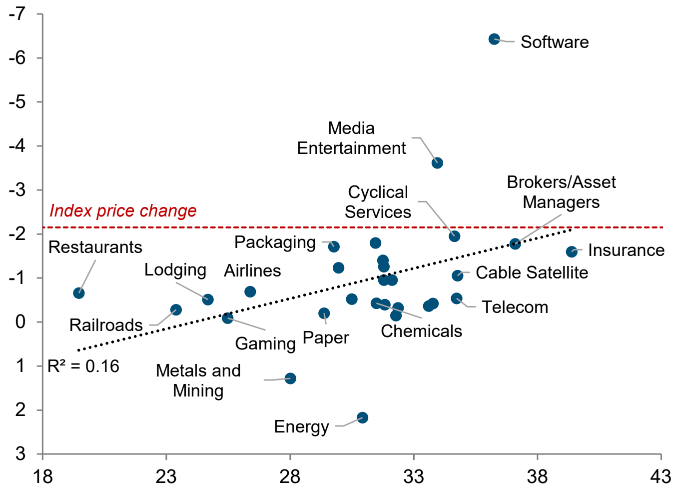
AI automation exposure, % (x-axis) vs. IG spread change YTD, bp (y-axis)



AI automation exposure, % (x-axis) vs. HY spread change YTD, bp (y-axis)



AI automation exposure, % (x-axis) vs. leveraged loan price change YTD, $ (y-axis, inverted)



Source: Revelio, Bloomberg, Goldman GIR.

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Alts managers: limited AI risk so far

Alexander Blostein and Michael Vinci argue that Alternative Asset Managers have limited exposure to AI-related software disruptions

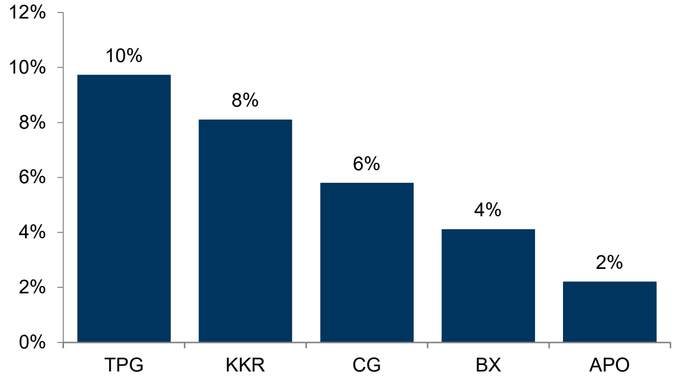
Concerns about AI-driven disruption risk have once again weighed on Alternative Asset Managers' stocks, which have declined ~27% YTD, bringing share prices and valuations below post-Liberation Day lows (NTM P/E net of stock-based compensation (SBC) at ~15X currently). This decline largely reflects concerns about software exposure across both private equity and private credit as well as fears that weaker investment performance could hamper growth. However, firmwide software exposure remains limited—averaging ~7% of total AUM—with software lending only modestly contributing to management fees, and managers have also been highly selective in their software investments, choosing those with more robust moats. While we think the recent selloff is overdone and continue to see significant upside across the sector, we remain cautious around names with outsized Retail Credit exposure given accelerating redemption requests.

Modest software exposure

Overall, we view Alternative Asset Managers' exposure to software investments as manageable. On the private equity (PE) front, while software has accounted for ~25% of industry-wide deployment volumes in recent years, exposure at Public Alternative Managers is more modest, with software activity among PE firms averaging only ~6% of total management fees. These estimates may even prove conservative as actual net exposure could be lower after managers monetized many investments in prior years.

PE software investments average ~6% of total management fees

2025 estimated firmwide management fees from Info Tech investments, %



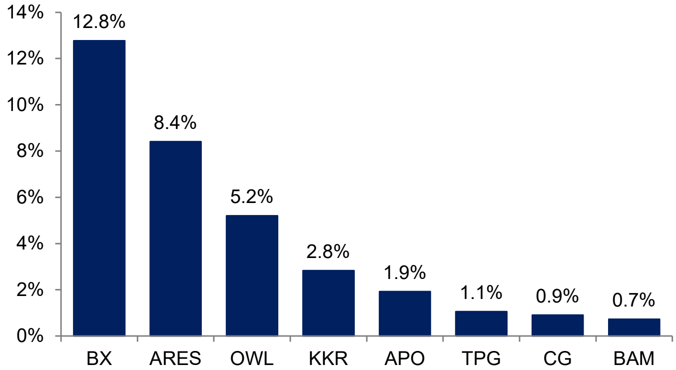
Source: Preqin, Goldman Sachs GIR.

On the credit front, across BDCs (where public data is available), we estimate that software loans represent ~22% of gross loans at an industry level, with a similar share among Public Alternative Managers—~20% of Direct Lending FPAUM on average. This translates to around 25% of Direct Lending base management fees.

But software lending is a relatively small portion of firmwide management fees when accounting for the broader spectrum of Alt asset classes that GPs manage. We estimate that Direct Lending to software-oriented portfolio companies accounts for 4% of firmwide base management fees on average. Even in the most adverse scenario of a significant markdown in software loans, our stress analysis (5%-15% mark-to-market) suggests the effect on firmwide base fees would remain in the low single-digits. And, importantly, these software loans are mostly senior secured and short duration, with the underlying companies generally showing strong EBITDA and cash flow.

Direct Lending to software-oriented portfolio companies accounts for 4% of firmwide base management fees on average

Estimated software direct lending firmwide base fees, %



Source: Company data, Goldman Sachs GIR.

Selective software investments

We also note that not all AI-related risk is created equal. Software holdings of most managers skew toward vertical, mission-critical businesses with proprietary data and platforms. These businesses are more likely to benefit from AI as opposed to the more disruption-prone horizontal software applications. AI-related risk has also become a central focus of investment processes, with one senior leader recently noting that AI is now among the first and most critical topics examined when bringing an investment before the investment committee. Evaluating AI-related risk has become a deeply embedded practice within GP investment committee deliberations and portfolio management frameworks, not just a reaction to recent market concerns.

We remain selective looking forward

While heightened market volatility and greater uncertainty related to AI disruption risks could dampen near-term PE exit activity, our conversations with management teams have given us more confidence around drawdown fund portfolio construction. That said, we continue to believe that sentiment/ headlines will weigh on retail/wealth flows. While the group's earnings streams remain highly durable, we see a slower pace of retail fundraising/accelerating Credit redemption trends potentially weighing on growth. Despite the disruption, deployment outlooks are constructive, and for Opportunistic Credit managers, most see opportunity in the current backdrop. We continue to think the group's earnings base and management fee growth are considerably more durable than current valuations suggest and see significant upside across the space.

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AI disruption: equity sector impacts

What types of companies in your coverage universe could be disrupted by and, conversely, benefit from new AI tools, and how much of this is already priced into stocks?

Consumer Internet

Eric Sheridan and team, GS Equity Research

- AI tools are currently having a democratizing effect on product development and innovation within the consumer internet sector, presenting disintermediation risk while also driving efficiency gains. Within our TMT coverage universe, digital advertising and gaming are the most exposed subsectors to AI disintermediation risk and have also seen the most advanced adoption of AI tools—a trend that the recent declines in several gaming and advertising stocks reflects.
- Digital advertising, in particular, has advanced the furthest in AI-driven product development and end-user adoption (outside of cloud computing). Companies within the subsector are currently integrating AI into four key workflows: campaign planning and strategy, creative development, media buying and optimization, and CRM and personalization. GOOGL's Performance Max and META's Advantage+ have been exceptionally successful in integrating multiple AI use cases into a single product. These AI-powered tools have the potential to both "abstract away"/automate many advertising services as well as improve the overall performance and efficiency of ad spend. This should result in shifting profit dynamics including: 1) accelerating the shift of ad spend from traditional to digital channels (an estimated $170bn addressable opportunity), 2) automating ad creative generation (~$114bn), 3) consolidating ad tech "intermediaries" (~$25bn), and 4) disrupting the traditional role of ad agencies (~$161bn).
- Beyond online advertising, the specter of AI disruption has caused outsized volatility across many other subsectors—including media & advertising, online marketplaces, online dating, travel, online education, and online gaming & gambling—which we expect to persist. Incumbent consumer internet platforms are playing a mix of "offense" (launching AI-driven consumer applications that enhance their existing offerings and revenue) and "defense" (developing AI capabilities to counter competitive applications that could be disruptive to existing legacy applications) to counter AI disruption risk. Several companies are also partnering directly with leading AI platforms to drive direct traffic and monetization. Examples include ChatGPT in-app integrations, agentic commerce, in-app chatbot placement, and data/content licensing for model training.

US Business and Information Services

George Tong, GS Equity Research

- Concerns over AI-driven disintermediation have sparked a sharp de-rating of the Information Services sector since mid-2025, with the sector experiencing a particularly notable valuation step-down in early Feb 2026. The median NTM P/E multiple in our Info Services coverage has fallen from 32x in June 2025 to 20x currently as investors worry that free or cheaper AI tools could diminish or eliminate demand for existing Info Services offerings. However, we believe some companies in the sector are better positioned to withstand AI disruption than others.
- Companies with a sustainable moat against AI are those with proprietary data, entrenched workflows with customers, scale and brand advantages, and/or a regulatory moat, and have made meaningful AI investments. These firms are not only insulated from AI risks but also are positioned to benefit from AI integration via product upgrades, accelerated pricing increases, increased cross-selling, higher retention rates, and new business growth. From a stock perspective, the multiples of many of these firms have recently contracted, making them attractive buying opportunities. The market may be overestimating the AI-related hit to these firms' organic revenue growth (with some estimates in the mid-to-high single digits) and we believe earnings at these firms remain stable and are not in a negative revision cycle. Upcoming earnings results that demonstrate beat-and-raise patterns and sustained momentum in organic revenue growth should act as a catalyst for valuation upside, in our view.
- By contrast, firms most vulnerable to AI disruption are those that have limited proprietary content, weak integration with client workflows, limited scale or brand advantages, no regulatory moat, and/or haven't invested enough in AI themselves. They also face potential headwinds from lower demand for white collar staffing roles due to AI automation.

North American Insurance

Robert Cox and team, GS Equity Research

- Recent reports that OpenAI has incorporated insurance services directly into ChatGPT sparked a significant selloff in insurance providers. We believe Private and Commercial (P&C) insurance companies will need to adapt to keep pace with these emerging AI tools. Certain personal lines & micro business insurance brokers are the most vulnerable to disruption as competitors increasingly leverage AI to simplify the purchase of lower complexity products like auto/home insurance and disrupt various consulting services. Shifts in distribution could also impact auto/home insurers' customer acquisition models, with auto insurers also facing long-term TAM concerns from increasing autonomous vehicle (AV) penetration. Commercial (re)insurers are well positioned as trusted, efficient, regulated capital allocators, especially if they quickly move to embrace AI efficiencies. Larger insurance brokers are also well positioned due to the data and structural moats associated with complex consulting and brokerage, however certain small account broking and less complex consulting could be exposed to AI disruption.
- We think both insurers and brokers can achieve significant efficiency gains through AI automation. Larger market cap firms with big technology budgets will likely benefit first. Insurers are using AI to automate claims processing and speed up policy handling, while brokers aim to automate routine processes/services. First movers are likely to see near- to medium- term margin/growth benefits, although whether these benefits will ultimately lead to lower premiums for customers remains an open question.
- The market has primarily focused on the risk of revenue disruptions from AI rather than on potential efficiency gains. While valuations for P&C insurers are under pressure from concerns about a softer pricing cycle, making it more difficult to isolate AI impacts, the market is clearly penalizing insurance brokers for potential revenue risks—a move which we think is overdone. Auto insurers have faced multiple contraction from AVs and could see revenue disruption from AI-based distribution platforms, but we think these headwinds will ultimately be manageable for underwriters with historically strong direct underwriting platforms. By contrast, commercial (re)insurers are in some cases benefiting from reported AI efficiency gains—CB has reported a ~20% headcount reduction over 3-4 years—and are seen as relatively safe from revenue disruption.

CRE Services

Julien Blouin, GS Equity Research

- We believe AI disruption concerns across CRE services businesses are significantly overstated. The physical nature of the CRE ecosystem provides meaningful insulation versus digitally-native industries. Historical precedents—listings platforms, online property management, virtual tours, automated valuation models—show that CRE technology adoption unfolds over years and decades, not quarters, and most importantly these innovations improved transparency, efficiency, and productivity but did not structurally disintermediate CRE brokerage and service providers.
- Transactional brokerage businesses face the most theoretical AI exposure, particularly for standardized mid-market mandates where AI can drive underwriting automation, OM generation, lease abstraction, and buyer matching. Yet wholesale disintermediation remains unlikely given the relationship-driven, localized, negotiated, and liability-bearing nature of CRE transactions (particularly for larger deals). Facilities, property, project, and investment management businesses are even more insulated as they are operationally complex, and often require on-premises workers/technicians. We believe AI will enhance functions like scheduling, cost control, and predictive maintenance rather than eliminate demand.
- The valuation opportunity is compelling. CRE services companies are benefiting from a transaction recovery that remains mid-cycle, with volumes still 44% below 2021–2022 peaks and 20% below 2017–2019 levels. Yet sector multiples have contracted sharply on AI fears, reverting to 2023 discounts versus the S&P Equal Weight—levels last seen when transactions were under extreme pressure. We view this selloff as an attractive entry point into a sector with powerful structural and cyclical tailwinds, supporting double-digit earnings CAGRs through 2028.

European Financials

Oliver Carruthers and team, GS Equity Research

- Within our European Financials coverage universe, AI disruption concerns have centered on market structure firms such as LSEG and DB1, though their AI exposure varies. LSEG's stock has declined since mid-2025 amid concerns that AI-powered analysis could disrupt its data business.
- More recently, focus has also turned to private markets and their exposure to software-based service business models. A de-rating of public software comparables off the back of AI-disruption concerns could lead to negative mark-to-market adjustments for portfolio companies and limit near-term monetization. That said, many private equity firms are benefitting from rising AI demand, including through digital infrastructure investments. And traditional asset managers have largely been shielded from AI disruptions with managers actively leveraging AI to boost efficiency and alleviate margin pressure as well as enhance efficiency across client servicing, risk management, and investment intelligence.
- AI disruption poses greater risk for private credit managers as fund performance is more sensitive to dispersion and, unlike buyout funds, private credit managers cannot offset losses with investments into "winners". We've seen significant underperformance among Alternative Asset Managers, with private credit-focused firms most impacted. Wealth managers and investment platforms also experienced an AI-related selloff after Altruist launched an AI-powered tax planning tool, which raised concerns that automated advice tools could increase competition and compress fees. However, we note that these tools also have the potential to enhance adviser productivity and increase efficiency for firms able to strategically integrate AI into their offerings.

European transport, infrastructure, and construction

Patrick Creuset and team, GS Equity Research

- When it comes to AI risk, the main debates in the European transport, infrastructure, and construction sector center around 1) whether the incumbents will be disrupted by AI technological progress, and 2) the extent to which AI productivity gains will be deflated away.
- We view the AI disruption risk as relatively low for the sector as incumbents benefit from large physical networks (e.g., sizable blue-collar workforces as well as numerous physical gateways and warehouses), have higher value-add proposition (e.g., end-to-end supply chain management), and benefit from network effects—all of which will be difficult for new AI-native entrants to match. Incumbents also stand to unlock significant advantages from integrating AI technology into their existing operations, further supporting their resilience.
- We are also less concerned about the prospect of AI-related productivity gains being deflated away. While some productivity gains will likely be passed on to customers—as has historically been the case within the industry—we expect the magnitude of productivity gains to be meaningfully larger than the potential yield erosion.
- Asset-light logistics companies are particularly well-positioned to benefit from integrating AI tools into their ongoing digitization push. The sector's SG&A operations have historically involved many repetitive, documentation-intensive tasks, including quoting, booking, carrier payment tracking, and customs clearance. We believe that further digitization and AI deployment will eventually enable industry players to automate some of these workflows and improve efficiency.

Glossary of GS proprietary indices

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GS CAIs measure the growth signal in a broad range of weekly and monthly indicators, offering an alternative to Gross Domestic Product (GDP). GDP is an imperfect guide to current activity: In most countries, it is only available quarterly and is released with a substantial delay, and its initial estimates are often heavily revised. GDP also ignores important measures of real activity, such as employment and the purchasing managers' indexes (PMIs). All of these problems reduce the effectiveness of GDP for investment and policy decisions. Our CAIs aim to address GDP's shortcomings and provide a timelier read on the pace of growth.

For more, see our CAI page and Global Economics Comment: Technical Updates to Our Global CAIs.

Dynamic Equilibrium Exchange Rates (DEER)

The GSDEER framework establishes an equilibrium (or "fair") value of the real exchange rate based on relative productivity and terms-of-trade differentials.

For more, see our GSDEER page, Global Economics Paper No. 227: Finding Fair Value in EM FX, 26 January 2016, and Global Markets Analyst: A Look at Valuation Across G10 FX, 29 June 2017.

Financial Conditions Index (FCI)

GS FCIs gauge the "looseness" or "tightness" of financial conditions across the world's major economies, incorporating variables that directly affect spending on domestically produced goods and services. FCIs can provide valuable information about the economic growth outlook and the direct and indirect effects of monetary policy on real economic activity.

FCIs for the G10 economies are calculated as a weighted average of a policy rate, a long-term risk-free bond yield, a corporate credit spread, an equity price variable, and a trade-weighted exchange rate; the Euro area FCI also includes a sovereign credit spread. The weights mirror the effects of the financial variables on real GDP growth in our models over a one-year horizon. FCIs for emerging markets are calculated as a weighted average of a short-term interest rate, a long-term swap rate, a CDS spread, an equity price variable, a trade-weighted exchange rate, and—in economies with large foreign-currency-denominated debt stocks—a debt-weighted exchange rate index.

For more, see our FCI page, Global Economics Analyst: Our New G10 Financial Conditions Indices, 20 April 2017, and Global Economics Analyst: Tracking EM Financial Conditions – Our New FCIs, 6 October 2017.

Goldman Sachs Analyst Index (GSAI)

The US GSAI is based on a monthly survey of GS equity analysts to obtain their assessments of business conditions in the industries they follow. The results provide timely "bottom-up" information about US economic activity to supplement and cross-check our analysis of "top-down" data. Based on analysts' responses, we create a diffusion index for economic activity comparable to the ISM's indexes for activity in the manufacturing and nonmanufacturing sectors.

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GS MAP scores facilitate rapid interpretation of new data releases for economic indicators worldwide. MAP summarizes the importance of a specific data release (i.e., its historical correlation with GDP) and the degree of surprise relative to the consensus forecast. The sign on the degree of surprise characterizes underperformance with a negative number and outperformance with a positive number. Each of these two components is ranked on a scale from 0 to 5, with the MAP score being the product of the two, i.e., from -25 to +25. For example, a MAP score of +20 (5;+4) would indicate that the data has a very high correlation to GDP (5) and that it came out well above consensus expectations (+4), for a total MAP value of +20.

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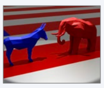
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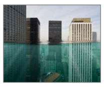
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TOP of MIND

AMERICAS FIRST

The Trump Administration kicked off 2026 with a new, more forceful foreign policy, driven by the so-called "Donroe Doctrine". So, what could "America First" evolving to "America*s* First" mean for economies, assets, and geopolitical risk? Trump's former Special Envoy for Latin America, Mauricio Claver-Carone, and Johns Hopkins SAIS' Hal Brands discuss the motives behind the Donroe Doctrine and agree it portends more action in the Western Hemisphere. GS' Alberto Ramos then explores the implications for regional economies, and GS' Neil Mehta and Daan Struyven dig into the implications for the asset at the center of recent events: oil. But with perhaps the most profound implication being the rise in geopolitical risk, we ask the Centre for Liberal Strategies' Ivan Krastev what's ahead for US-Europe relations (a rocky period, but not a divorce) and GS' Christian Mueller-Glissmann how to protect portfolios (diversify, don't time), with GS' Kamakshya Trivedi also seeing value in a surprisingly resilient corner of the market: EM assets.

"The prospect of a fundamental shift in US policy is more real now than at any point in the last 70 years.

- Hal Brands

President Trump has long believed that the US can't be the preeminent global power if it's not the preeminent regional power.

- Mauricio Claver-Carone

The [US-Europe] relationship looks to be headed for a rocky period, but not a divorce.

- Ivan Krastev"

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Macro news and views

We provide a brief snapshot on the most important economies for the global markets

US

Latest GS proprietary datapoints/major changes in views

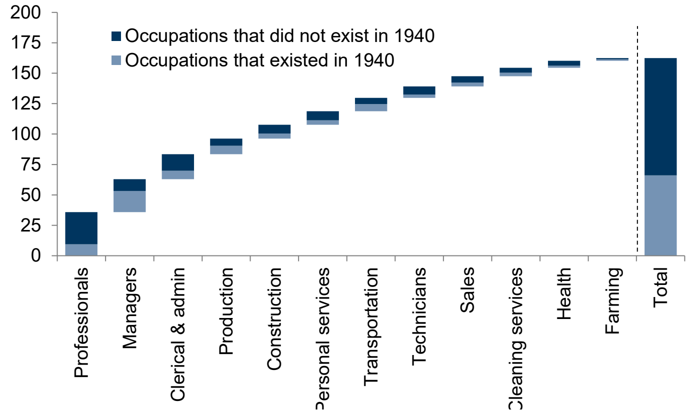
- We now expect two 25bp cuts in Jun and Sept (vs. Mar and Jun previously) to a terminal rate range of 3-3.25%.
- We recently lowered our 12m US recession probability to 20% (vs. 30%) partly owing to early signs of labor market stabilization.

Datapoints/trends we're focused on

- US growth; we expect somewhat stronger growth of 2.5% in 2026 (Q4/Q4) as the tariff drag gives way to a fiscal boost.
- US government shutdown, the odds of which have spiked.
- DoJ investigation into Fed Chair Powell, which we don't expect to impact Fed policy for now, but we find that a less independent Fed would put upward pressure on inflation.
- AI job apocalypse concerns, which we think are overblown.

Technological change: a key driver of job growth

Employment by new and pre-existing occupations, millions



Source: Goldman Sachs GIR.

Japan

Latest GS proprietary datapoints/major changes in views

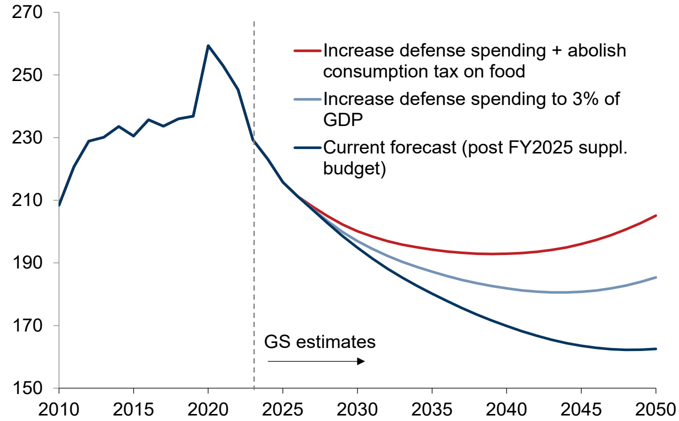
- No major changes in views.

Datapoints/trends we're focused on

- Japanese growth; we expect still-steady growth of 0.8% in 2026 (yoy) led by domestic demand, but see external demand slowing slightly amid higher Japan-China tensions.
- BoJ policy; we expect the BoJ to hike at a semiannual pace, with the next hike in July, to a policy rate of 1.5% by mid-2027, though the timing of the next hike is uncertain.
- Japanese fiscal fears, which have risen on the back of the looming Lower House election (February 8), with all political parties advocating for a consumption tax reduction.

Japan: a potential undermining of fiscal health ahead

Japanese government debt-to-GDP ratio by scenario, % of GDP



Source: Ministry of Finance, Cabinet Office, Goldman Sachs GIR.

Europe

Latest GS proprietary datapoints/major changes in views

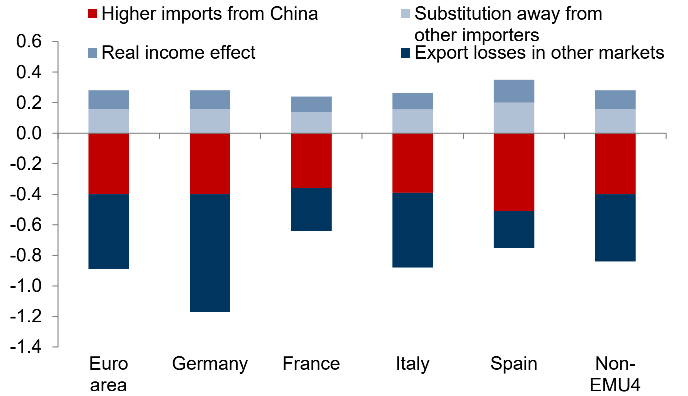
- No major changes in views.

Datapoints/trends we're focused on

- EA growth; we expect real GDP growth of 1.2% in 2026 (yoy) reflecting real income gains and a fading fiscal drag but growing headwinds from increased competition with China.
- UK economy, for which we expect another mixed year in 2026 characterized by trend-like growth and a further rise in unemployment but much lower inflation and more BoE cuts.
- Trade policy; we see only limited downside to European growth and limited upside to inflation if the US-EU trade deal implementation remains stalled.

Europe: a meaningful drag from China Shock 2.0

Estimated country-level real GDP impact of higher Chinese export growth by end-2029, %



Source: Haver Analytics, Goldman Sachs GIR.

Emerging Markets (EM)

Latest GS proprietary datapoints/major changes in views

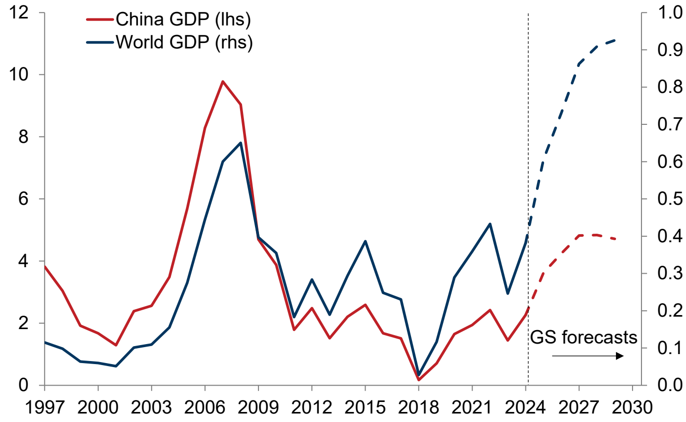
- No major changes in views.

Datapoints/trends we're focused on

- China's current account surplus, which we expect to rise further over the next several years to almost 1% of global GDP, the biggest of any country in recorded history, owing to China's manufacturing strength and weak domestic demand.
- China policy easing; we expect more easing this year, including two 10bp rate cuts, which should support our above-consensus growth and inflation forecasts.
- EM policy normalization, which we expect to continue amid further inflation normalization, including in Brazil and CEEMEA.

China current account surplus: rising to record levels

China current account surplus as a share of GDP, %



Source: IMF, Haver Analytics, Goldman Sachs GIR.

Americas First

The Trump Administration kicked off 2026 with a new, more forceful foreign policy, driven by the so-called "Donroe Doctrine"—a corollary to the Monroe Doctrine that aims to expand US influence and control in the Western Hemisphere (see pg. 8). What the evolution of "America First" to "America*s* First"—and potentially beyond—means for economies, assets, and geopolitical risk more broadly is Top of Mind.

We first turn to Mauricio Claver-Carone, President Trump's former Special Envoy for Latin America and Senior Director for Western Hemisphere Affairs at the NSC, and Johns Hopkins School of Advanced International Studies' Hal Brands for more insight into the Donroe Doctrine and the Trump Administration's potential next moves.

Claver-Carone explains that the Donroe Doctrine reflects Trump's long-held belief that "the US can't be the preeminent global power if it's not the preeminent regional power" (see pg. 9). This belief is driving Trump to reassert US dominance in Latin America with the overarching goal of strengthening US security—namely, national security and, importantly, energy security, which Claver-Carone says are closely linked. He sees no contradiction between this and Trump's "America First" policy, arguing that "Americas First" is its natural extension.

Brands, for his part, sees two motivating factors behind the Donroe Doctrine, which he characterizes as "a reversion to an earlier style of statecraft": the US' desire to consolidate its position in the Western Hemisphere amid a new era of great power rivalry, and Trump's desire to increase America's control of hemispheric resources (think oil and territory).

But Claver-Carone and Brands agree: the Donroe Doctrine portends more action in the Western Hemisphere ahead. Claver-Carone expects the Trump Administration to continue pursuing "practical partnerships" with Latin American countries. And Brands underscores the importance of the Greenland issue (see pg. 14), suggesting that it likely won't be easily resolved, though he also argues that "we shouldn't view the Donroe Doctrine as a constraint on action elsewhere" (e.g. Iran).

So, what could this all mean for regional economies? Alberto Ramos, GS Head of LatAm Economics Research, doesn't expect major economic spillovers from the developments in Venezuela to the rest of Latin America given their minimal linkages owing to Venezuela's strikingly prolonged and deep depression and hyperinflation. He sees similarly limited political implications as voters in several Latin American countries (Peru, Brazil, and Colombia) are set to head to the polls later this year.

And what about exposed assets? Ramos recounts the many uncertainties facing Venezuela's defaulted bonds amid potential debt restructuring. Neil Mehta, GS Head of North American Natural Resources Equity Research, and Daan Struyven, Co-head of Global Commodities Research, then dig into the global asset at the center of the recent developments—oil. Struyven expects the Administration's plan to revitalize Venezuela's beleaguered oil sector to result in a modest increase in Venezuelan oil production and a modestly negative impact on oil prices over the next several years, with more significant oil price impacts over the longer run as production ramps up further (see pg. 11 for a map of Venezuelan oil assets).

But the most profound and potentially lasting implication of the recent developments is the rise in geopolitical risk (see pg. 18). We speak with the Centre for Liberal Strategies' Ivan Krastev about the future of the US-Europe relationship, which he describes as at a "turning point but not a breaking point" given Europe's security, economic, and technological dependence on the US as well as American political realities that may constrain US policy. But he expects a rocky period in the relationship ahead given that trust has been broken, not just because of Greenland, but also because of US actions vis-à-vis Ukraine and the US' broader treatment of Europe.

And Brands sees even broader geopolitical implications, arguing that US designs on Greenland may test the global territorial status quo that has prevailed post WWII. He warns that the US deciding to play a fundamentally different role in the world than it has in the past 70 years could lead to significant changes in everything from nuclear proliferation, to the security of seaborne transit and democracy, to Dollar dominance.

So, what does this elevated geopolitical risk environment mean for global investors? Christian Mueller-Glissmann, GS Head of Asset Allocation Research, offers guidance on how to protect portfolios against this risk. He argues that rather than trying to time geopolitical shocks, investors' first line of defense should be diversification. He see value in several diversification strategies: turning to bonds and other safe havens like the Swiss Franc, increasing allocations to assets that could benefit from geopolitical shocks (like gold and the defense sector, which GS defense analysts Noah Poponak and Sam Burgess believe has not yet fully priced in elevated geopolitical risk), limiting direct and indirect exposures, and hedging with options.

Kamakshya Trivedi, GS Chief FX & EM Strategist, also sees value in a surprisingly resilient corner of the market: EM assets, which have held up despite elevated geopolitical risk, which typically weighs on EM relative performance. He expects this resilience to continue this year, with EM assets' role as diversifiers further increasing their value proposition. In this context, Trivedi believes frontier assets are also worth a look.

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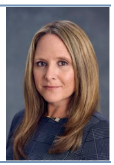


Interview with Hal Brands

Hal Brands is Henry A. Kissinger Distinguished Professor of Global Affairs at the Johns Hopkins School of Advanced International Studies and Senior Fellow at the American Enterprise Institute. Below, he discusses the US' renewed assertiveness in the Western Hemisphere, which he argues could have major repercussions for the global order.

The views stated herein are those of the interviewee and do not necessarily reflect those of Goldman Sachs.

Allison Nathan: In 2024, you argued that a second Trump presidency would feature a "revitalized Monroe Doctrine". What factors are motivating the US' more forceful posture in the Western Hemisphere?

Hal Brands: I see two motivating factors—one structural, one personal. The structural factor is the US' historic pattern of attempting to consolidate its position in the Western Hemisphere when the rest of the world is falling apart. During both World Wars and the Cold War, the US worked hard to try to ensure its dominance in the Western Hemisphere. So, it's not surprising that the US is now doing the same as the world enters a new era of great power rivalry.

The personal factor revolves around President Trump's long-held interest in renewing the Monroe Doctrine to increase the US' control of hemispheric resources, like oil, and territory, specifically from Panama, Canada, and Greenland. The President is also intensely focused on tangible threats to American sovereignty and security rooted in hemispheric issues like drug trafficking and migration. Putting all this together, it's not surprising that this Administration has focused on reconsolidating US primacy in the Western Hemisphere.

What is a bit more surprising is how energetic the Administration has been in doing so. The Venezuela intervention has received substantial attention, but it's just the capstone of a much bigger and broader campaign that has featured diplomatic pressure on Colombia and Panama, economic support to allies like the Milei government in Argentina, partnerships with El Salvador's Bukele government and other ideological fellow travelers, lethal strikes on drug boats, and an array of other military, diplomatic, and economic initiatives. In many ways, this has been the centerpiece of Trump's foreign policy so far.

Allison Nathan: So, it's right to think that oil was a key motivation for the Venezuela actions?

Hal Brands: Resource control is undoubtedly at the forefront of the President's calculus. He has long associated it with national power, often remarking that the primary mistake in the American invasion of Iraq was not "taking the oil". In describing US aims in Venezuela since the intervention, Trump has said very little about democracy and good governance, and even relatively little about drug trafficking. He talks *a lot* about the US exerting greater control over the disposition and sale of Venezuelan oil and the desire for greater openness of the Venezuelan oil industry to the US oil majors. Others in the Administration may value different things in this Trump Corollary to the Monroe Doctrine, but for Trump himself, the primary issues are essentially about economic gain and a neo-mercantilist approach to international economics.

Allison Nathan: Is that what differentiates the "Donroe Doctrine" from the Monroe Doctrine?

Hal Brands: The Monroe Doctrine's meaning has evolved over time. It was initially about preventing the reimposition of colonial control by European powers. During the 20th century, it became about keeping fascist and communist regimes out of the hemisphere. In some ways, the "Donroe Doctrine" is a throwback—it is a reversion to an earlier style of statecraft that is much less apologetic about the desire to control the resources of weaker countries and much more willing to talk about outright territorial acquisition and other ideas that went out of style with the end of the age of imperialism. It's sometimes said that Trump is trying to revive a 19th century style of foreign policy. I agree that he would likely feel quite at home if he were presiding over some of the US' interventions in Latin America in the late 19th and early 20th centuries. Indeed, the tactics he has used in Venezuela—overthrowing a government and exerting control of the country's trade revenues—aren't that different from what the US did in places like the Dominican Republic well over a century ago.

Allison Nathan: With that context, why do you think the Administration chose to leave the rest of the Maduro regime in place?

Hal Brands: Trump is seeking maximum leverage and gain with minimum long-term investment of resources and political capital. He burst onto the domestic political scene 10 years ago by arguing that the US had made terrible mistakes by undertaking nation-building missions in Afghanistan and Iraq. And flying Maria Corina Machado or Edmundo Gonzalez into Caracas in the hours after Maduro's apprehension probably would have incited a lot of instability by putting them in conflict with the remnants of the Maduro regime. So, the mechanism Trump has settled on is a 21st century version of gunboat diplomacy, where the US will work through the existing authorities while threatening them with punishment if they don't do the Administration's bidding.

Allison Nathan: What do these developments mean for other countries in the region, many of which are slated to hold elections this year?

Hal Brands: They serve notice to any vulnerable anti-American regimes playing strategic footsie with China and Russia that they will likely face greater US pressure. Cuba might be the best example. That said, I'm skeptical that the US would undertake a large-scale military intervention in Cuba because a collapse of that regime might produce a flood of refugees headed for the US. More broadly, countries from Colombia to Mexico will likely feel pressure to act with greater urgency on issues related to narcotrafficking, which we're already seeing evidence of in the discussions the US and Mexico appear to be having about a potential role for the US military in going after cartels operating in Mexico, as well as some of Trump's comments about Colombia.

The domestic political implications of these shifts will vary given the array of political cultures and levels of susceptibility to US interests throughout the region. In Brazil, the political culture to some degree has long been interwoven with resistance to US influence in Latin America, even in periods when conservative military governments in Brazil worked very closely with the US. So, I wouldn't be shocked if US efforts backfire to some degree in the next Brazilian election. But it could go the other way there and/or in other countries, where the US either has more influence or where a homegrown swing to the right or in the direction of neo-populist politics occurs.

Allison Nathan: What lessons do you think China and Russia are taking from the recent US actions, and how are they shifting their calculus in the region?

Hal Brands: Beijing and Moscow have run up hard against the reality of US hard power in the Western Hemisphere. Whether it's the US Navy and Coast Guard seizing shadow fleet tankers, the US forcibly deposing Maduro, or the Pentagon shooting up drug boats in the Caribbean, neither China nor Russia can do much to contest the application of American military power in the Western Hemisphere. That said, China's influence in the region in particular is multifaceted and deeply embedded in everything from trade relationships to the diffusion of technology to physical and digital infrastructure, all of which will persist to some degree.

It's important to note that China's influence is not simply imposed on Latin American countries, but a response to their needs. Big gaps in physical and digital infrastructure exist in the region, which China is well-placed to address. So, Beijing will play the long game and continue to attempt to win influence in the region through economic, political, and, in some cases, low profile security relationships that will focus more on internal security than traditional military functions. If anything, the recent developments likely mark the start of an even more intense phase of competition for influence in the region.

Allison Nathan: How does Greenland fit into Trump's plan, and what might that suggest about his next steps there?

Hal Brands: Greenland is both a hemispheric and global issue. It's a hemispheric issue in the sense that the Trump Administration views consolidating US control of Greenland as part of the larger effort to revive the Monroe Doctrine. And they can point to historic examples of the US increasing its influence in Greenland as part of a strategy of hemispheric defense, as President Roosevelt did in the run-up to US entry into WWII.

It's also a global issue, for two reasons. First, Greenland powerfully affects the fate of NATO and the transatlantic community; it's hard to overstate the degree of outright alarm that Canada and many European countries feel about US designs on the territory. Second, Greenland is the best test of whether the US will seek changes to the global territorial status quo. That's an explosive issue because China and Russia are attempting to do much the same thing in the South China Sea/Himalayas and Ukraine, respectively. So, a world in which the US also aims to redraw borders through coercion or even force is a world in which the three most powerful countries are all violently or coercively disrupting the territorial status quo. That's something we haven't seen since the 1930s and could be deeply corrosive to the post-1945 global order.

Allison Nathan: So, recent US actions have helped further the ambitions of China and Russia in some respects?

Hal Brands: Both countries were undoubtedly unhappy to see Maduro go, especially at the hands of ruthlessly efficient US military power. But recent developments aren't entirely bad news for Beijing and Moscow. A world in which international law and norms matter less, and great powers are free to do as they like, especially in their own backyards, is a world in which Chinese and Russian leaders would feel comfortable. In fact, that's exactly the world they aim to bring about. And if an intensification of US military activity in the Western Hemisphere pulls resources away from other regions, as was the case when Trump redeployed an aircraft carrier covering the Middle East and Europe from the Mediterranean to the Caribbean, that's even better news for them.

Allison Nathan: How does Iran fit in here?

Hal Brands: Iran is a key reason why we should be skeptical of the idea that the world is rapidly transitioning into a spheres of influence arrangement. Trump wants an American sphere of influence in the Western Hemisphere, but he also likes to maintain unhindered freedom of action globally when he views it as advantageous to him. This was clear in the recently-released national security strategy. This Administration has also conducted two major military interventions in the Middle East and threatened others. We shouldn't view the Donroe Doctrine as a constraint on action elsewhere.

Allison Nathan: Amid all this, what geopolitical risks should companies and investors be most focused on?

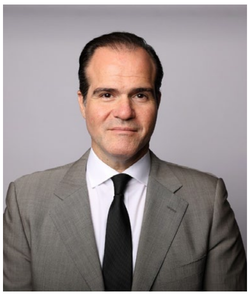
Hal Brands: The mega risk is the potential for the US to decide to play a fundamentally different role in the world. We have lived for so long in the post-1945 international order in which US power was wielded to secure strong prohibitions on forcible conquest, freedom of navigation to support and propel the global economy, and protection of human rights and the spread of democracy, that it's difficult to conceive of a world without that leadership. Such a world would likely see more countries pursuing nuclear weapons, greater disorder on the high seas that would impact global trade and finance, a challenge to Dollar dominance, and a global retreat of democratic values.

30 years ago, I would have said there was little chance of such a shift. 15 years ago, I would have said signs of retrenchment-minded inclinations in the US foreign policy debate had grown. We now have a US president who often expresses a desire for fundamental change in US foreign policy and the way the US engages in the global economy, even if the policies don't always match the rhetoric. So, the prospect of a fundamental shift in US policy is more real now than at any point in the last 70 years. The stakes are high, and the choices the US makes in the coming years will not only determine its own role on the world stage, but also the trajectory of the international order.

Interview with Mauricio Claver-Carone

Mauricio Claver-Carone is Managing Partner of the Latin America Real Assets Opportunity Fund. He served as US Special Envoy for Latin America and Senior Director for Western Hemisphere Affairs at the National Security Council in the Trump Administration. Below, he argues that President Trump views US dominance in the Western Hemisphere as essential for global preeminence, and that "Americas First" is a natural extension of "America First".

The views stated herein are those of the interviewee and do not necessarily reflect those of Goldman Sachs.



Jenny Grimberg: You played a key role in foreign policy in the Western Hemisphere during both Trump Administrations. What's behind President Trump's focus on the region, and do you sense a shift in that focus from his first term?

Mauricio Claver-Carone: President Trump has long believed that the US can't be the preeminent global power if it's not the preeminent regional power. That is the "Donroe Doctrine". And contrary to widespread perception, this reflects the President's own thinking, not his advisors' influence. President Trump is a developer from New York—he thinks in terms of building neighborhoods. He has traveled throughout the region and knows it well. And he believes that no region in the world impacts Americans' daily lives more than Latin America and the Caribbean from a security, economic, and social perspective.

I've seen firsthand many times how focused the President has been on this, and on the US' waning influence and positioning in the region over the last two decades, especially vis-à-vis China. During periods of conflict in Afghanistan and Syria in his first term, President Trump would incredulously remark on how eagerly and easily the US sent troops to those countries yet was too scared to have any type of presence in its own hemisphere. And I remember at one point during the intensifying crisis that gripped Venezuela in 2019, the President being disturbed to learn that the US Navy had no ships in the vicinity of the Caribbean because they were all deployed to the Persian Gulf, South Asia, and the Mediterranean.

That said, given his team was distracted by the Middle East and Asia at the time, President Trump's first term was largely about facilitating a pivot from China in the region. The Administration focused on setting frameworks to, mostly through soft power, execute this pivot, which led to the creation of the America Crece Initiative to catalyze private sector investment in energy and infrastructure projects across Latin America and the Caribbean.

The second term, by contrast, is about the implementation of President Trump's vision for the region. He made that vision extraordinarily clear during his second inauguration as he harkened back to President McKinley and the golden era of American influence, projected not only through policy—the Monroe Doctrine and, later, the Roosevelt Corollary to it—but also through brick-and-mortar initiatives like the Panama Canal. And this time around, his team is aligned with that vision. Secretary of State and National Security Advisor Marco Rubio understands and cares about the region and President Trump's vision for it. Deputy Chief of Staff for Policy and Homeland Security Advisor Stephen Miller is also focused on the region owing to his holistic approach on immigration and his view that immigration would be less of a problem for the US if Latin American countries were thriving. And, in contrast to the secretary of defense in the first Administration, who didn't believe that the Department of Defense should be involved in counternarcotics, Secretary of Defense Pete Hegseth is very involved.

Jenny Grimberg: What does the President hope to achieve by reasserting US influence in Latin America?

Mauricio Claver-Carone: The Trump Administration's actions in Latin America—from the Venezuela operation to the relationship it created with Mexico to the historic support it lent Argentina during its economic crisis last year—are based first and foremost on US security. This runs the gamut from national security in terms of halting narcotic and migrant flows and reducing terrorism risk to energy security. The Achilles heel of Latin America and the Caribbean in particular is energy. Fidel Castro always understood that, which is why he helped install Hugo Chavez and viewed Venezuela as his golden crown. And Caribbean countries' pressing need for energy led to the formation of PetroCaribe, an initiative launched by Chavez to supply Venezuelan oil to these countries on favorable financial terms, which allowed Venezuela to essentially control the Caribbean for many years.

Energy security is closely intertwined with national security, as high energy costs have been a key driver of migration throughout Central America. So, energy security is incredibly important, which is consistent with President Trump's desire to increase oil production in the Western Hemisphere. The US is an energy powerhouse in its own right, and many Latin American countries could be too with proper management. US influence can play a crucial role here. Guyana currently produces more oil than Venezuela, largely because the Guyanese government has worked hand-in-hand with US companies to develop the right policies from the get-go and leave behind the nationalist ideological strains that were embedded into the psyche of the regional population during the 20th century revolutions.

Jenny Grimberg: Does expanding US influence in Latin America conflict with the "America First" pillar of President Trump's Make America Great Again (MAGA) platform?

Mauricio Claver-Carone: Not at all. "Americas First" is the natural extension of America First as President Trump works to counter the notion of every DC think tank expert that the 21st century would be the Chinese century. The President was adamant during his inauguration that it would instead be another American century, and potentially the greatest American century. And contrary to what many believe, the President's Latin America strategy is not about boots on the ground or nation-building. It is about practical partnerships between nations as the US shifts from a policy of non-interventionism—which served neither the local population nor the US well by leading to the rise of Chavez and China—to non-imperial expansionism, a term coined to explain the strategy of deepening US influence in the region through partnerships.

These are not political partnerships—the US is currently working with the right-leaning governments of Argentina, El Salvador, Costa Rica, and Chile as well as the left-leaning governments of Mexico and Venezuela—but practical, business-oriented relationships that benefit not only the US but also these countries, whose populations desire security and economic opportunities. The best example of that is Argentina. Many people, including myself, worried that the Trump Administration's financial support ahead of last year's midterm elections could spark a backlash against President Milei's Conservative party given that strains of Peronism still exist in Argentina. But the election results showed that most of the party's support came from the poor, who are fed up with ideologies that neither pay the bills nor put food on the table. So, the region has undergone a dynamic social shift.

Jenny Grimberg: How does the Trump Administration's decision to depose Maduro fit into that?

Mauricio Claver-Carone: Let me first clarify that the Administration didn't "depose" Maduro. This was a law enforcement operation with military support to capture an individual indicted for narcoterrorism in the US. Now, this was a particularly complicated operation because Maduro was de facto running a nation-state with a military, which showed the world that nobody is outside of US reach and sent a message of US influence, ability, and capability. But the operation also underscored the Trump Administration's preference for partnership. President Trump gave Maduro plenty of opportunities to work together; he expressly rejected every offer, which led the Administration to execute a carefully thought-out operation to capture him. That sends an important message: when President Trump offers you a deal or an opportunity, you take it.

The Trump Administration's subsequent actions in Venezuela are also a testament to its partnership preference. The US undoubtedly had the capabilities for a broader operation, but assessed that the other people in Venezuela controlling the institutions of power would want to work with the US. And so far, that assessment has proven right. In the days following the operation, the US reached a deal to purchase 30-50 million barrels of Venezuelan oil and several million barrels moved shortly after. Venezuela is also increasingly and consistently releasing political prisoners. And the US is working with the Venezuelan authorities to put the country on a path toward becoming a more stable and secure presence in the region with an open, participative process. Much remains uncertain, but I'm optimistic.

Jenny Grimberg: The million-dollar question is what President Trump will do next in Latin America and the broader Western Hemisphere. What's your view?

Mauricio Claver-Carone: Further efforts at forming and consolidating partnerships with countries in the region are likely. Upcoming elections in Brazil and Colombia will be important. Conservative candidates recently won Honduras' and Chile's presidential elections, and if right-wing candidates also prevail in Colombia and Brazil, it would result in a historic ideological consolidation for the region. But again, this isn't about ideology but whether the new leaders will choose the US first and foremost as their partner. That would give the US a historic opportunity to modernize its business relationships and commercial partnerships across Latin America.

That could start with Mexico and the upcoming USMCA renegotiation, which seems to be the path the President is heading down. But President Trump's intense focus on trade is reactive. He feels that the trade deals of the 1990s/early 2000s were all flawed, forcing him to focus on fixing them. As those issues resolve, the focus of US partnerships with regional allies could shift to making deals that ensure a US strategic presence in critical sectors like energy and infrastructure. The hope is that by the end of President Trump's term, nobody will ever again talk about China's expanding influence in the region, because it will be unquestionably clear that the US is the partner of choice for Latin America and the Caribbean.

Jenny Grimberg: As an investor in the region, what advice do you have for investors attempting to navigate Latin America's evolving geopolitical landscape?

Mauricio Claver-Carone: Investors looking to monetize the "Donroe Doctrine trade" by focusing more on Latin America should look beyond the big-ticket deals in Mexico and Brazil, which have often been walloped by currency issues, to smaller countries. The deals are smaller, but the returns are bigger as a percentage, and these countries are among the region's fastest-growing economies, offering great opportunities for investors. The secret sauce is local partners.

I would also advise against relying on multilateral development banks. As a former president of the Inter-American Development Bank, I've seen firsthand that these institutions aren't focused on finding or creating the best deals. Investors should instead focus on US domestic finance institutions like the International Development Finance Corporation (DFC), which partners with the private sector to mobilize capital for strategic investments around the world and recently secured a six-year reauthorization, to gain exposure to long-term deals in strategic sectors.

I also see significant short and medium-term investment opportunities in the energy and infrastructure space. The timing is right for these investments because the US government is focused on and geopolitically invested in Latin America and wants to encourage private investment in the region, and that focus isn't going away given President Trump's "Americas First" philosophy.

The Monroe Doctrine through the years

INVOKING THE MONROE DOCTRINE

The Monroe Doctrine, first articulated by President James Monroe in 1823, declared that European powers should not interfere in the affairs of the newly independent nations of the Americas, a stance motivated by concerns over potential European re-colonization. Since its inception, US presidents have frequently invoked the Doctrine to justify various foreign policy actions across the Western Hemisphere.

1823 Origin of the Monroe Doctrine: In an address to Congress, President James Monroe declared that the Western Hemisphere was off-limits to European colonization and, in return, promised US non-interference in Europe in what became known as the Monroe Doctrine.

1865 support for Mexico against France: In 1864, France installed an Austrian Habsburg prince as the Emperor of Mexico, which the Lincoln Administration saw as a violation of the Monroe Doctrine. In 1865, the US exerted diplomatic and military pressure to help Mexican President Benito Juárez lead a successful revolt against the Emperor.

1895 Cleveland and the Venezuela boundary dispute: The Cleveland Administration intervened on Venezuela's behalf in the country's boundary dispute with British-controlled Guiana (now Guyana) after gold was found in the contested region, asserting that the US had the right to intervene in the dispute based on the Monroe Doctrine.

1902-1904 Roosevelt Corollary: In 1902, a fleet of British, German, and Italian gunboats set up a blockade of Venezuela's ports after the nation defaulted on its debts to European lenders. President Theodore Roosevelt, as part of his foreign policy to "speak softly, but carry a big stick," issued a corollary to the Monroe Doctrine that gave the US "international police power" over the Western Hemisphere, justifying US intervention in Latin American affairs to prevent European influence. Roosevelt used this to justify US intervention in Venezuela, as well as Panama in 1903 and Cuba in 1904.

1915 US intervention in Haiti: Following the assassination of the Haitian President, President Woodrow Wilson dispatched US Marines to Haiti with the goal of restoring order and maintaining political and economic stability in the Caribbean. The US maintained control over Haiti until 1934.

1933 FDR and the "Good Neighbor Policy": By the early 1930s, US relations with Latin America were strained due to decades of military interventions. In an effort to improve relations, President Franklin D. Roosevelt established the "Good Neighbor Policy," which renounced direct US intervention in Latin America in favor of economic cooperation with the region.

1954 US-backed coup in Guatemala: The US intervened in Guatemala in a CIA-backed coup to overthrow President Jacobo Árbenz amid fears that his government was being influenced by communist groups.

1962 JFK and the Cuban Missile Crisis: President Kennedy invoked the Monroe Doctrine during a televised address in October 1962, announcing the discovery that the Soviet Union was secretly installing offensive nuclear missile sites in Cuba. Kennedy ordered a naval blockade around the island, characterizing the Soviet action as a direct challenge to the long-standing principle of preventing hostile foreign powers from establishing military bases in the Western Hemisphere.

1983 Reagan and the invasion of Grenada: The US launched an operation in Grenada to oust its Marxist government after the execution of leader Maurice Bishop. President Reagan invoked the Monroe Doctrine in order to safeguard American lives in Grenada and extinguish communist influences. Reagan called the invasion of Grenada the first successful "rollback" of communist influence since the onset of the Cold War.

1989 capture of Manuel Noriega in Panama: President George H.W. Bush deployed US troops to Panama to arrest and extradite Manuel Noriega on drug trafficking and money laundering charges. The US cited the Monroe Doctrine to justify the invasion, claiming that Noriega's regime threatened US citizens living in Panama as well as the security of the Panama Canal.

2018 Trump 1.0 revitalization: During his first term, President Trump invoked the Monroe Doctrine in relation to US policy toward Venezuela and other nations in the Western Hemisphere, reaffirming his "enduring commitment" to the Doctrine.

2026 Trump Corollary and the Donroe Doctrine: President Trump laid out a Trump Corollary to the Monroe Doctrine that aims to restore US preeminence in the Western Hemisphere. Referring to it as the "Donroe Doctrine," Trump used it as justification for the arrest and extradition of Venezuelan leader Nicolás Maduro on drug trafficking charges.

Not an exhaustive list of all policy actions related to the Monroe Doctrine.
Source: US Department of State, National Archives, Office of the Historian, compiled by Goldman Sachs GIR.

The Donroe Doctrine, in Trump's words

Jan 20, 2017: "From this moment on, it's going to be **America First**. Every decision on trade, on taxes, on immigration, on foreign affairs, will be made to benefit American workers and American families… We will seek friendship and goodwill with the nations of the world–but we do so with the understanding that it is the right of all nations to put their own interests first."

June 16, 2017: "Last year, I promised to be a voice against repression in our region—remember, tremendous oppression—and a voice for the freedom of the Cuban people… America will expose the crimes of the Castro regime and stand with the Cuban people in their struggle for freedom. Because we know it is best for America to have freedom in our hemisphere, whether in Cuba or Venezuela."

August 11, 2017: "We have many options for Venezuela, including a possible military option, if necessary… We don't talk about it. But a military operation, a military option, is certainly something we could pursue."

September 25, 2018: "Here in the Western Hemisphere, we are committed to maintaining our independence from the encroachment of expansionist foreign powers. It has been the formal policy of our country since President Monroe that we reject the interference of foreign nations in this hemisphere and in our own affairs… In that spirit, we ask the nations gathered here to join us in calling for the restoration of democracy in Venezuela. Today, we are announcing additional sanctions against the repressive regime, targeting Maduro's inner circle and close advisors."

August 20, 2019: "Denmark is a very special country with incredible people, but based on Prime Minister Mette Frederiksen's comments, that she would have no interest in discussing the [US] purchase of Greenland, I will be postponing our meeting scheduled in two weeks for another time."

September 25, 2019: "For all of the countries of the Western Hemisphere, our goal is to help people invest in the bright futures of their own nation… We want every nation in our region to flourish and its people to thrive in freedom and peace. In that mission, we are also committed to supporting those people in the Western Hemisphere who live under brutal oppression, such as those in Cuba, Nicaragua, and Venezuela... We are watching the Venezuela situation very closely. We await the day when democracy will be restored, when Venezuela will be free, and when liberty will prevail throughout this hemisphere."

April 1, 2020: Trump deploys US Navy destroyers, surveillance aircraft, and additional military personnel to the Caribbean near Venezuela, saying "As governments and nations focus on the coronavirus, there is a growing threat that cartels, criminals, terrorists, and other malign actors will try to exploit the situation for their own gain. We must not let that happen."

May 8, 2020: "If I wanted to go into Venezuela, I wouldn't make a secret about it. I wouldn't send a small, little group, no, no, no. It would be called an army."

July 18, 2023: "When I'm back in the White House, on Day One, we are returning to a foreign policy that puts America's interests first… I will restore the proud culture and honor traditions of America's armed forces. And there will be no Marxism allowed, no communism allowed, and we'll get rid of the fascists."

May 2025: "For far too long, the communist Cuban regime has trampled on the rights of its people to lead lives of freedom—and has caused endless poverty, suffering, and hardship. America will continue to work alongside our allies and partners in the Western Hemisphere to bring stability, liberty, cooperation, and a free future to the people of Cuba."

December 2, 2025: "The United States will never waver in defense of our homeland, our interests, or the well-being of our citizens. Today, my Administration proudly reaffirms this promise under a new **'Trump Corollary'** to the Monroe Doctrine: That the American people—not foreign nations nor globalist institutions—will always control their own destiny in our hemisphere... Reinvigorated by my Trump Corollary, the Monroe Doctrine is alive and well—and American leadership is coming roaring back stronger than ever before."

January 3, 2026: "The Monroe Doctrine is a big deal, but we've superseded it by a lot, by a real lot. They now call it the **Donroe document**... [the] Monroe Doctrine, we sort of forgot about it. It was very important, but we forgot about it. We don't forget about it anymore. Under our new national security strategy, American dominance in the Western Hemisphere will never be questioned again. It won't happen… For decades other Administrations have neglected or even contributed to these growing security threats in the Western Hemisphere. Under the Trump Administration, we are reasserting American power in a very powerful way in our home region."

January 8, 2026: "It's a doctrine of 'don't send drugs into our country.' That's what the doctrine is. Don't send drugs into our country."

January 14, 2026: "The United States needs Greenland for the purpose of National Security. It is vital for the Golden Dome that we are building. NATO should be leading the way for us to get it. IF WE DON'T, RUSSIA OR CHINA WILL, AND THAT IS NOT GOING TO HAPPEN!"

January 14, 2026: "We are making tremendous progress, as we help Venezuela stabilize and recover… This partnership between the United States and Venezuela will be a spectacular one FOR ALL. Venezuela will soon be great and prosperous again, perhaps more so than ever before."

January 16, 2026: "I may put a tariff on countries if they don't go along with Greenland, because we need Greenland for national security."

January 21, 2026: "We probably won't get anything [on Greenland] unless I decide to use excessive strength and force where we would be, frankly, unstoppable. But I won't do that."

Source: White House, Truth Social, CBS, X, compiled by Goldman Sachs GIR.

Venezuela: what's next?

Alberto Ramos explores what developments in Venezuela could mean for the country's economy and the broader LatAm region

Following the US operation in Venezuela, the situation on the ground remains very fluid and the outlook complex and uncertain. The Trump Administration is now working with the remnants of the Maduro regime to stabilize the Venezuelan economy, focusing first and foremost on the battered oil sector, and eventually hold open elections.

An economic collapse of historic proportions

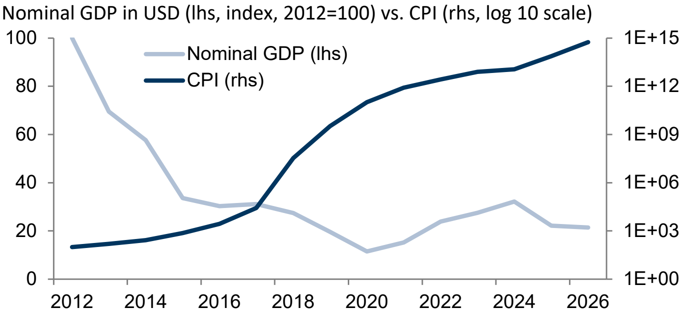
Venezuela has been trapped in a seemingly never-ending economic depression and hyperinflationary spiral. Large macro and financial imbalances have spawned an economic collapse of historic proportions. In fact, Venezuela has experienced one of the worst peacetime economic downfalls in modern history, with real GDP contracting far more than in the US during the Great Depression (-28%), Spain during the Spanish Civil War (-28%), and the 2009 Greece crisis (-27%). Per the IMF, in Dollar terms, Venezuelan GDP has shrunk by ~80% since 2012, with seven consecutive years of negative real GDP growth over 2014-2020 for an astonishing cumulative decline of 74.3%.

Quite telling and illustrative of the major economic collapse of the last 20 years is the dramatic shift within Venezuela's oil sector—the country's key source of foreign exchange. Oil production fell from 2.7mb/d in 2015 to 0.93mb/d in 2025, resulting in an oil-export hard-currency crunch that imposed a severe and debilitating constraint on imports. As a result, Venezuela's economy has contracted to ~$80bn today, roughly the size of Uruguay and less than a quarter the size of Chile.

Hyperbolic inflation immiserizes

Annual inflation over the decade leading up to 2025 was an astronomical 14,544% on average—reflecting years of hyperinflation (inflation was ~130,000% in 2018). Over the same period, Venezuela's currency (the Bolivar) lost ~98% of its value. The economic hardship has helped generate a mass exodus of people over the last 16 years. According to the UNHCR, nearly 8mn people fled the county, close to a quarter of the population and one of the world's largest displacements, with most settling in nearby countries, particularly Colombia.

Venezuela has experienced hyperinflation and depression
Nominal GDP in USD (lhs, index, 2012=100) vs. CPI (rhs, log 10 scale)



Source: IMF, Goldman Sachs GIR.

Likely limited economic and financial spillovers…

It therefore comes as no surprise that the economic and financial linkages between Venezuela and the rest of LatAm have become minor. Accordingly, we do not anticipate major economic spillovers from the situation in Venezuela to the rest of LatAm. However, if the situation on the ground were to deteriorate into widespread lawlessness and violence, we could see renewed migration flows, likely affecting neighboring countries first and foremost. Beyond migration, the situation in Venezuela could indirectly impact the region through their impact on key commodity prices, particularly oil and gold.

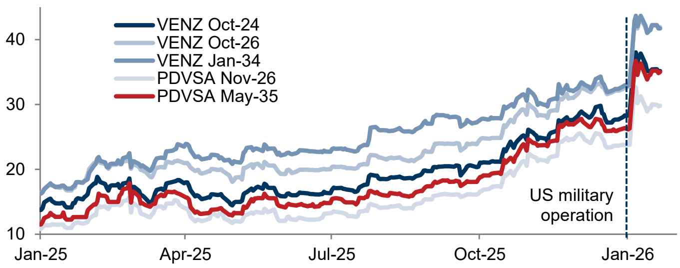
…and political implications

The political implications for the rest of LatAm also seem limited. The reactions of regional governments were mixed and clearly divided along political/ideological lines, but even those critical of the situation were cautious and contained in their responses. However, US actions in Venezuela could stress its relationship with Mexico, Colombia, and Brazil, especially if the US expands its focus to the regime in Cuba and/or conducts military strikes on drug cartels inside Mexico or Colombia. At this juncture, we do not expect the ongoing developments in Venezuela to impact upcoming elections in Brazil (general election in October), Colombia (congressional elections in March, presidential in May), or Peru (general elections in April).

Debt restructuring key to watch from here

Venezuela defaulted on international bonds in late 2017 due to the compounding effects of the deep economic crisis and sanctions. Defaulted debt is estimated at north of $150bn, including around $60bn (face value) in defaulted bonds (sovereign and state oil company PDVSA), other PDVSA obligations (e.g., suppliers' debt), bilateral loans (e.g., with China, Russia), and multilateral debt, with significant unpaid interest as well as legal claims from expropriation cases compounding the total since the 2017 default.

Venezuela bond prices rallied following the US military operation
Venezuela and PDVSA bond price, bp



Source: Bloomberg, Goldman Sachs GIR.

Venezuela bond prices rallied in 4Q25 on expectations that rising US pressure and military buildup would eventually lead to regime change. And the initial market rection to the January 3 US airstrikes was positive. Indeed, defaulted Venezuelan bond prices rallied 8-9pts (including PDVSA) on January 5 on expectations of a brighter outlook for Venezuela's economy and oil sector as well as the potential for debt restructuring.

However, bond prices have since given back some of these gains due to the complex and still highly uncertain economic and political picture, the perception that debt restructuring may not be a near-term priority, President Trump's executive order protecting Venezuelan oil revenue held in US Treasury accounts, and bond valuations commensurate with lower-bound recovery levels. The market debate has now shifted to "*what type of restructuring*" and *when* and with *whom* creditors will eventually be negotiating.

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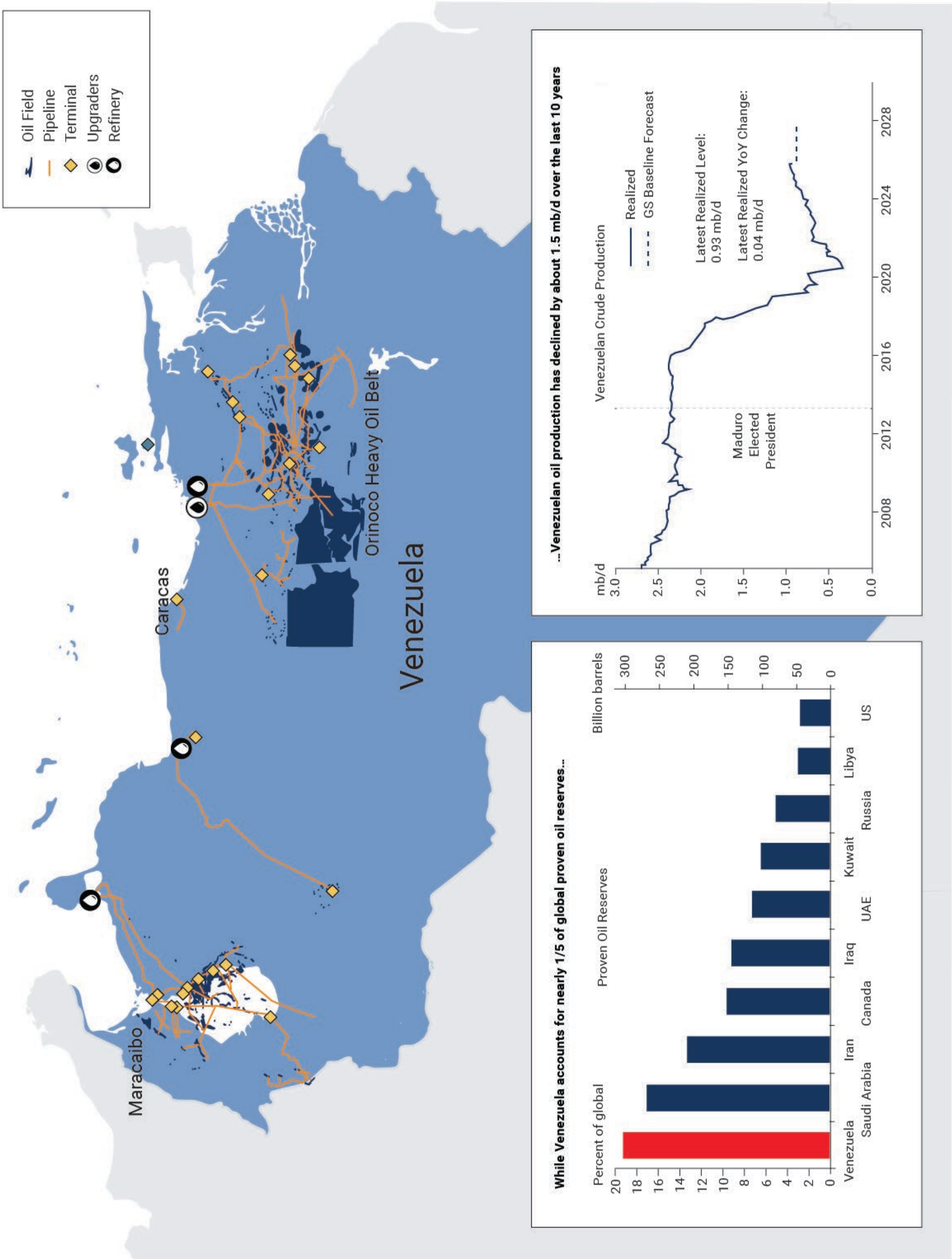
Tel: *212-357-5768*

Venezuela oil infrastructure, mapped out



**We use OPEC data on proven crude oil reserves as of 2024 for all the countries above except Canada, where we use Energy Institute proven reserves estimates as of 2020 (latest available).*

Sources: OPEC, Energy Institute, secondary sources, S&P Global Energy, Goldman Sachs GIR.

Special thanks to GS Commodities Research team for charts.

Q&A on implications for Venezuelan oil

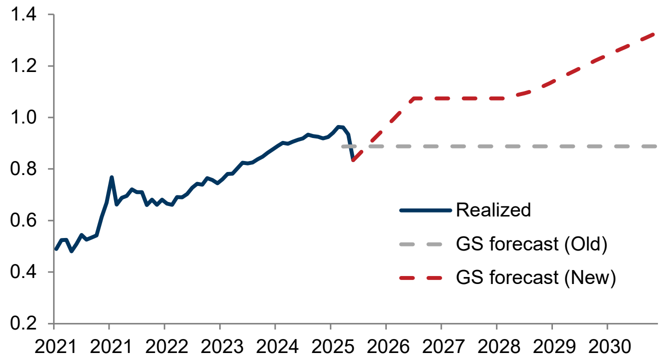
Neil Mehta and Daan Struyven answer key questions about the implications of US plans to revitalize Venezuela's oil sector[1]

Q: What's the size of the opportunity for the oil industry of the US' plan to revitalize Venezuela's oil sector?

A: Venezuela accounts for roughly one-fifth of the world's proven oil reserves[2], yet its production is constrained at ~800-900kb/d, which is less than ~1% of global output. The Trump Administration has called for ~$100bn of investment into the country over the next decade to rebuild infrastructure and bring production back to peak levels of ~3mb/d. We believe gains of a few hundred thousand barrels per day are achievable in the near term. Specifically, we expect Venezuelan crude production to rise by 0.2mb/d over the next 12m (from 0.8-0.9mb/d in Dec 2025) as an anticipated easing of sanctions and modest investment in existing assets help ease some operational bottlenecks. We expect production to rise by an additional 0.2mb/d by 2030 (to 1.3mb/d). But a complete restoration of the country's output to ~3mb/d would likely extend into the next decade and require significant capital investment. Some estimates suggest an initial ~$5-10bn of capital investment over the next 2-3 years could yield an additional ~500kb/d in Venezuela, but the subsequent ~1mb/d would require an estimated ~$70-80bn. Upstream production expansions would also require a reactivation of damaged upgraders that process heavy crude, extension of pipeline infrastructure, investments in the power grid, and continued political stability.

We expect a modest increase in Venezuelan crude supply

Venezuelan crude supply, mb/d



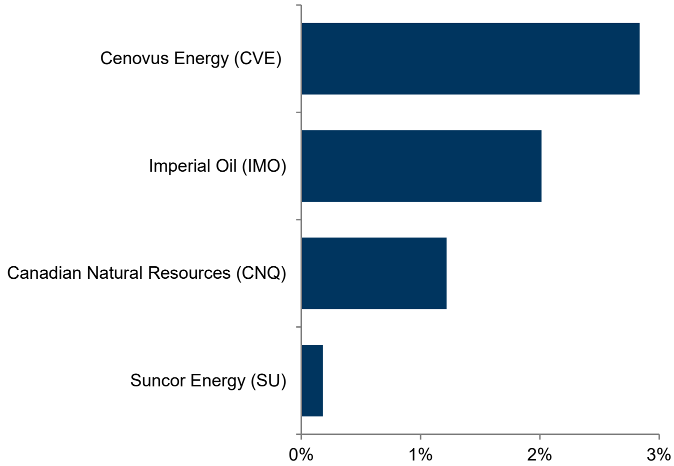
Source: Secondary sources, Goldman Sachs GIR.

Q: What types of companies could be harmed by the revitalization of the Venezuelan oil industry?

A: On a relative basis, Canadian oil companies could be negatively impacted by higher oil production from Venezuela. In particular, Venezuelan barrels could displace Canadian heavy crude on the Gulf Coast and result in wider WTI-WCS differentials, which would reduce realized prices for Canadian producers.

Venezuelan barrels could displace Canadian heavy crude on the Gulf Coast and result in wider WTI-WCS differentials

Estimated annual cash flow sensitivity to $1/b WCS price change, %



Source: Goldman Sachs GIR.

Q: How have US oil companies responded to these potential opportunities/risks so far?

A: Among the US Majors, Chevron is doubling its liftings from the country immediately and raising its production by 50% (or ~100kb/d) over the next 18-24 months. By contrast, ConocoPhillips and Exxon remain focused on recovering previous debt following the nationalization of their assets. Exxon has also publicly discussed its view that Venezuela is difficult to invest in without significant changes to the legal and commercial environment. Among oilfield services companies, SLB has discussed its ability to scale quickly as needed in the country, as it currently has operations on the ground. Halliburton and Weatherford have also commented on their ability and interest in allocating resources to Venezuela if activity and production ramp up.

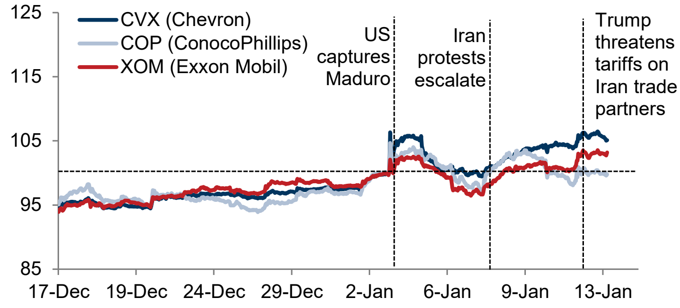
Q: How much of the upside/downside of these developments is already priced into stocks?

A: Energy equities reacted sharply on the back of the initial Venezuela headlines, including the stocks of US majors, US refiners, international service companies, and Canadian oil producers. While we believe the latest developments are now largely reflected in the equities, we continue to highlight strong fundamentals across the energy complex.

[1] The research views in this section are those of the analysts who cover the relevant areas. Equity research views are expressed by Neil Mehta, an Equity Research Analyst covering Americas Integrated Oil & Refiners. Commodities views are expressed by Daan Struyven, a Macro Research Analyst covering global commodities.
[2] According to OPEC data.

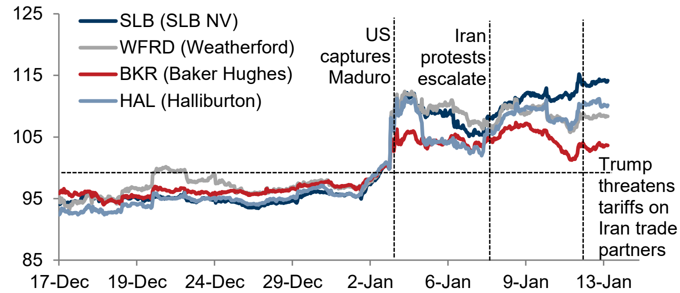
Energy equities reacted sharply on the back of the initial Venezuela headlines, including across US Majors…

Stocks of US Majors, index, Jan 2 close=100



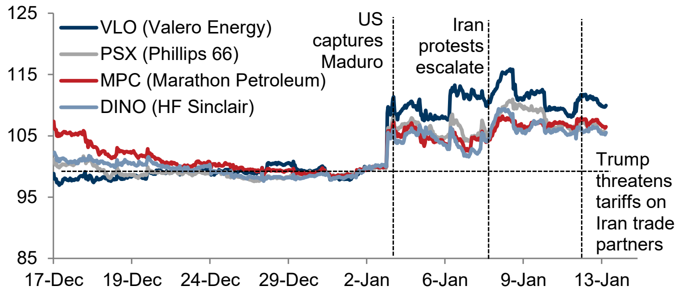
…international services…

Stocks of international services, index, Jan 2 close=100



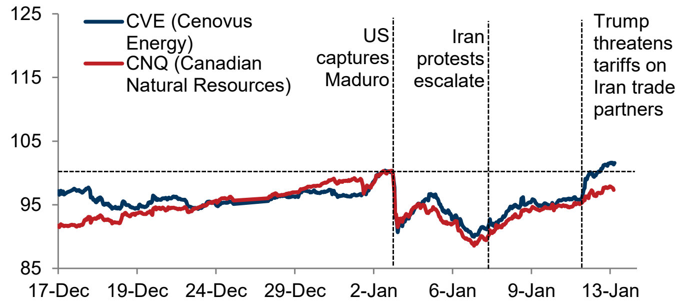
…US Gulf Coast refineries…

Stocks of US Gulf Coast refineries, index, Jan 2 close=100



…and Canadian producers

Stocks of Canadian producers, index, Jan 2 close=100



Source: Bloomberg, Goldman Sachs GIR.

Q: What, if anything, are investors underappreciating about the potential impacts of the Venezuela developments for US oil companies?

A: Given the state of Venezuela's energy infrastructure—which has deteriorated amid underinvestment—investors may be underestimating the challenges associated with sustainably ramping Venezuelan volumes. We also believe the scale of investment required to increase their Venezuelan presence as well as the political and legal risks of doing so could deter companies from making significant new capital commitments in the country, which is a major reason we expect an only modest increase in Venezuela's production by 2030.

Q: What are you watching most closely ahead to gauge how this could all play out for the US oil industry?

A: Despite the recent developments in Venezuela, the outlook for the US oil industry should remain largely unchanged from our base case scenario, with three primary forces continuing to drive the industry's trajectory. First, after rising from 5-6mb/d in early 2000s to 14mb/d of crude today, the industry is now entering a period of slowing growth and maturity. Accordingly, US producers will likely focus on harvesting cash flow domestically, investing in international exploration, and driving consolidation through M&A.

Second, the larger companies have structural advantages of scale and the ability to execute an international development plan. That leaves the US majors, as well as international services companies, well-positioned for growth. Third, we believe that refiners are structurally well-positioned on a multi-year basis as crack spreads are likely to remain above historical averages and global oil demand is expected to outpace new capacity additions between now and 2030.

Q: What could the Venezuela developments mean for oil prices?

A: We estimate that the 0.2mb/d increase in Venezuelan oil production we expect over the next 12m will have a $1 impact on Brent/WTI prices this year (baseline forecast of $56/$52/bbl average 2026 prices vs. $57/$53/bbl in a scenario of flat production). But the impact on long-run oil prices will likely be more significant. We estimate that the cumulative 0.4mb/d increase in Venezuelan production we expect by 2030 will have a $2 impact on long-term oil prices. This, together with higher expected US and Russian production, led us to lower our long-run Brent/WTI price forecasts to $75/$71/bbl earlier this month (2030-35, from $80/$76/bbl). We estimate that a larger recovery in Venezuelan production to 2mb/d by 2030 would pose $3 of downside to these revised forecasts.

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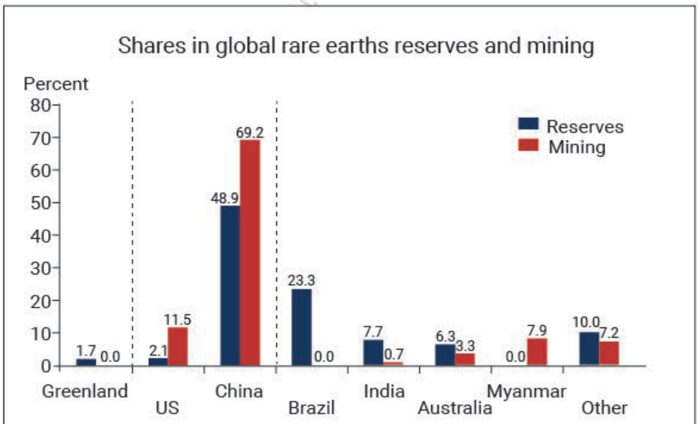
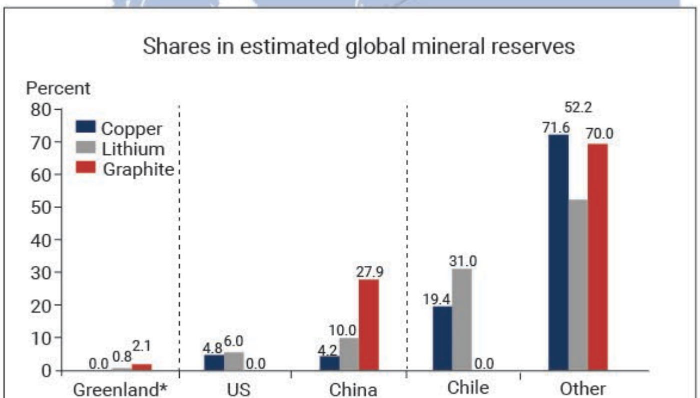
Greenland's strategic value, mapped out





*Greenland shares are unreported according to the latest USGS report; instead, data from the Geological Survey of Denmark and Greenland (GEUS) is shown.

Source: Bloomberg, USGS, GEUS, Goldman Sachs GIR.

Special thanks to GS Commodities Research team for charts.

Interview with Ivan Krastev

Ivan Krastev is Chair of the Centre for Liberal Strategies in Sofia, Bulgaria. Below, he argues that US-Europe relations have reached a turning point, but not a breaking point.

The views stated herein are those of the interviewee and do not necessarily reflect those of Goldman Sachs.

Allison Nathan: Do the Greenland developments mark a turning point in US-Europe relations?

Ivan Krastev: They mark a turning point but not a breaking point because alliances don't exist on paper, but in the minds of the public and policymakers. And the Greenland story, combined with the Trump Administration's recently-released national security strategy and Venezuela actions, have substantially shifted how Europe views transatlantic relations. I was part of a recent European Council on Foreign Relations group that conducted global polling revealing that only 16% of Europeans see the US as an ally, while 20% view it as a rival, with most others seeing it as only a necessary partner. This marks a big shift from the past.

The Greenland issue has been especially difficult for Europeans to digest because the Danes are ready to cooperate with the US on almost anything it wanted from a security and even economic perspective. And the Danish government is one of the most pro-Atlantic governments in the European Union. So, Europeans are shockingly no longer sure about three things: one, the extent to which the Trump Administration believes that its relationships with Russia and Europe would be more difficult in NATO's absence; two, that one of the objectives of US trade policies is not the European Union's destruction; and, three, that the EU would be able to count on the US in times of major crisis, which was a key part of the magic of NATO.

Allison Nathan: So, why isn't this a breaking point?

Ivan Krastev: First of all, it's one thing to understand that your marriage is not working in the same way as before, but it's completely different to imagine what divorced life would look like. Europe is too dependent on the US not only for its security, but also technologically and economically, to prefer the divorced life today. And the stakes are too high to find out what such a life would look like, with the very survival of the European project on the line. Remember that the European Union was largely an American project. In the moment of crisis, Americans played a critical role in securing European unity.

Second, some Europeans believe that Trump's actions don't reflect the consensus among American society or the elite, and point to the walking back of the potential use of military force in Greenland as evidence of the constraints Trump faces. So, Europeans believe that when they stand against Trump on this issue, they are not standing against America. Third, while Trump has strong allies within some European countries on issues like immigration, they have not sided with him on Greenland because no European leader can convince their voters that the US getting its way in Greenland is in Europe's best interest. It's noteworthy that the European Parliament stalled ratification of the US-EU trade deal, and all MPs, from the mainstream parties to the nationalistic right, applauded the decision to stand behind Denmark and Greenland.

Now, this is largely about territory for both sides. But while Trump views land from the lens of a real estate manager, European countries, especially small ones, view land as a critically important part of their national sovereignty, and the idea that small countries don't matter is not good news for anyone in Europe. So, European leaders agree that caving on Greenland would be disastrous for Europe and its credibility. In that light, this issue has strengthened European unity. And Europe's weakness paradoxically means that it cannot *not* retaliate when it comes to this red line, because its weakness would be exposed. My sense is that Trump initially miscalculated on this point but now seems to understand it.

Allison Nathan: How could US-Europe relations evolve?

Ivan Krastev: The relationship looks to be headed for a rocky period, but not a divorce. For Europe, while there is much talk about finding new allies, aligning with others won't be a quick or easy process. The European Parliament essentially blocking the EU-Mercosur trade deal shows that too many competing interests exist among EU countries. And I'm skeptical that Europe and China will suddenly have a smoother relationship just because the US-Europe marriage has hit hard times. Instead, Europe will be focused on showing the US that Europe matters, not only because it can be helpful to the US, but also because it can be harmful. One action Europe can take to show that it is strong and serious about consolidating its economic power is making major progress on capital market integration.

Europe will also look inward in terms of focusing on building up its defenses, but here too it will encounter difficulties. Europe is struggling against time, and the blame doesn't lie with Trump. 15 years ago, then US Secretary of Defense Robert Gates warned Europe that the US would not pay for Europe's defense if it wasn't willing to increase its defense budgets, which Europeans chose to ignore, and now time is not on its side. But Europe's biggest problem is that budgets don't fight wars—people do. And Europe culturally is not prepared for the more hostile world it now finds itself in. That said, some European countries are taking their defense more seriously, and I expect closer cooperation among those countries, especially those closest to the Ukrainian crisis, as well as Germany and France in its capacity as the only European nuclear power. The Greenland issue has also brought the UK closer to the EU on defense, which will be critically important in bolstering European defense capabilities.

For the US, midterm elections loom large, which I suspect will continue to serve as a constraint on the Administration's actions. So, both the US and Europe will likely look for ways to de-escalate tensions and compromise, as has already begun. But the trust between the US and Europe has been broken, not just because of Greenland but also because of Ukraine in terms of US policy related to the conflict, and the broader US treatment of Europe. So, I don't expect the relationship to go back to the way it was even if, as we've discussed, the current moment doesn't prove to be a breaking point.

When geopolitics meets portfolios

Christian Mueller-Glissmann discusses how to navigate and hedge against geopolitical risk

Geopolitical risk has returned to the forefront following the recent developments around Venezuela, Iran, and Greenland, and is likely to remain there for some time as the US continues to reshape its geopolitical and economic relationships with the world. This has raised questions about how investors can protect themselves from geopolitical shocks, which can have material impacts on global growth, inflation, and sentiment, especially as the linkages between economies and markets have become increasingly complex. Rather than try to time geopolitical shocks, which can be difficult, we think robust portfolio construction and diversification should be investors' first line of defense.

Geopolitical shocks: rhyming, but seldom repeating

While geopolitical shocks are a key risk for portfolios, they tend to differ every time and are particularly difficult to position for as the timing and potential economic and market impacts are hard to anticipate. Geopolitical tensions can also linger for a long time before they escalate. To further complicate matters, geopolitical shocks often start out as local shocks but can extend globally and have lasting knock-on effects.

The broader macro backdrop also affects how markets react to geopolitical shocks. We find an imperfect relationship between the news-based Geopolitical Risk Index and equity volatility, with an increase in geopolitical tensions often leading to higher—but not extreme—volatility. Geopolitical risks alone have seldom pushed the VIX above 40. But the combination of such risks and an already-weak macro backdrop has often sent volatility sharply higher. A more resilient US economy and a Fed able to buffer negative growth shocks may help markets digest geopolitical shocks better today, though low volatility, elevated equity valuations, and bullish investor sentiment to start the year increase the risk of a volatility spike.

The individual asset impacts of geopolitical shocks can also vary. Historically, large geopolitical shocks like the Middle East wars have driven oil prices sharply higher, often resulting in a worsening global growth/inflation mix. By contrast, oil prices have remained in a downtrend amid the recent geopolitical developments due to excess supply, with the Dollar also weakening as investors have become more concerned about US assets.

Finally, the inherent unpredictability of geopolitical events, including the possibility of a reversal of the shock, creates material market timing risk. Equities on average have recovered and risen after geopolitical shocks. In some cases, uncertainty can decline rapidly as more information becomes available or a policy pivot occurs, which can drive sharp equity recoveries, as happened around "Liberation Day" last April, which showcased the difficulty of timing equity moves.

Protecting portfolios: diversify, don't time

Given these challenges, investors' first line of defense against geopolitical shocks should be robust portfolio construction and diversification. In a balanced multi-asset portfolio, bonds can provide a buffer for risky assets against unanticipated growth shocks. However, bonds become a less effective hedge if the geopolitical shock increases inflation risk. US policy uncertainty and global fiscal risks have also weighed on bonds, with recent concerns about potential fiscal expansion in Japan increasing the upward pressure on long-dated bond yields globally. Long-dated bonds may provide a smaller buffer for risky assets going forward if fiscal risks persist.

Investors can also turn to other global safe haven assets like the Swiss Franc, which has become a more effective "risk-off" hedge vs. the Euro in periods of rising geopolitical risk. But it's important to keep in mind that while safe haven assets tend to outperform during risk-off episodes, they usually have lower returns through the cycle. So, as with bonds, investors need to consider the tradeoffs between risk reduction and returns.

Increased allocations to assets that could benefit from an escalating shock can also buffer geopolitical risk. Commodities like oil and gold have often been useful diversifiers in periods of geopolitical uncertainty, though their hedge value can vary depending on the drivers (if there are oil supply disruptions, for example) and again they tend to have relatively low risk-adjusted returns over longer horizons. That said, gold and precious metals more broadly have recently helped buffer both geopolitical and fiscal risks, with higher central bank demand further boosting their risk-adjusted returns. Defense stocks are another potential beneficiary of geopolitical shocks (see pg. 20).

Investors can also manage geopolitical risk by avoiding or reducing direct and indirect exposures. This could include avoiding the country/region facing elevated geopolitical risk or large energy consumers (in the event of sharply higher oil prices) as well as reducing indirect exposure from trade linkages. With global investor portfolios currently dominated by US assets due to their large weights in equity and bond benchmarks, investors could also increase home bias in portfolios to reduce FX risk, though the potential impacts of large shifts out of US assets on portfolio returns and risk should be carefully considered.

Finally, the fact that geopolitical shocks can reverse quickly suggests that options can be a valuable hedge. Hedging tail risk with options can undoubtedly be expensive, making prolonged exposure to risk management strategies that protect portfolios systematically unfeasible. That said, realized and implied volatility has been very low coming into the year, increasing the value of tactical portfolio overlay hedges.

So, for investors looking for ways to protect their portfolios from geopolitical shocks, the message is: diversify, don't time. Specifically, we recommend more regional, sector, and style diversification relative to global benchmarks and like safe assets such as gold and selectively hedging Dollar risk. With the volatility reset, we also like shorter-dated rates receivers as a hedge for major growth shocks, credit payer spreads/VIX call spreads to hedge corrections, and longer-dated calls to maintain overweight equity allocations despite the potential for more geopolitical shocks.

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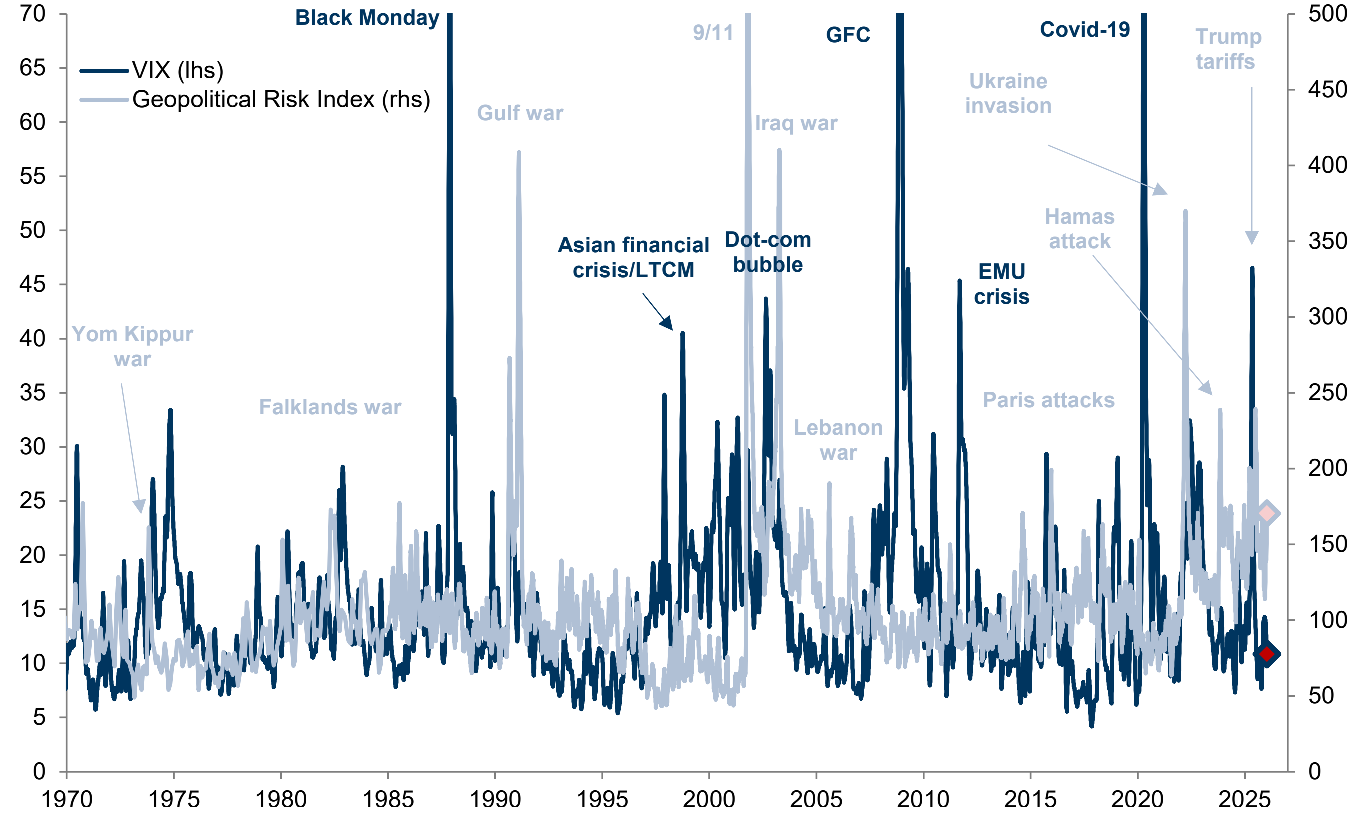
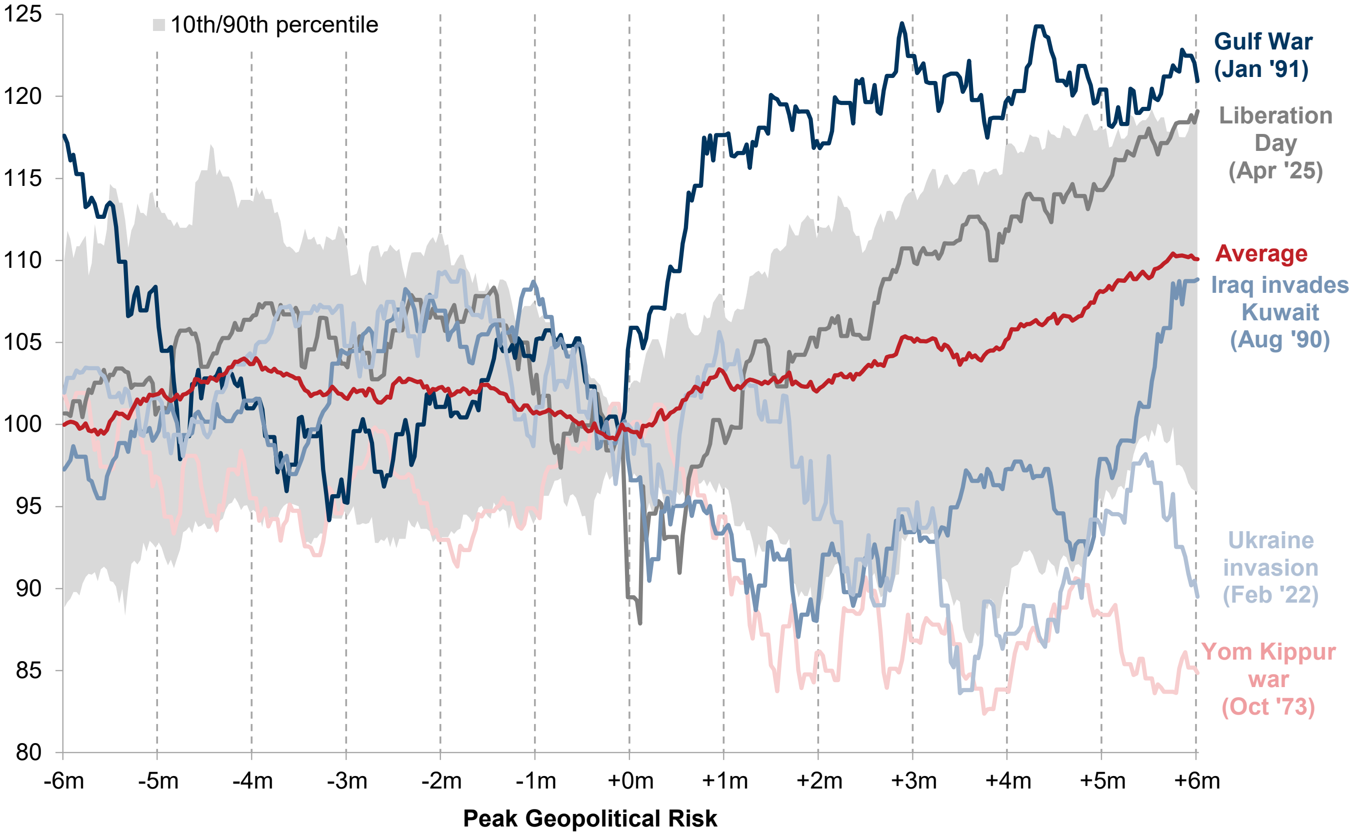
The relationship between geopolitical risk and equity volatility is mixed

VIX Index (lhs) vs. Geopolitical Risk Index (rhs), 1-month average



Timing equities around geopolitical shocks has been very difficult

Equity response to geopolitical shocks since 1970 (15 geopolitical shocks included)



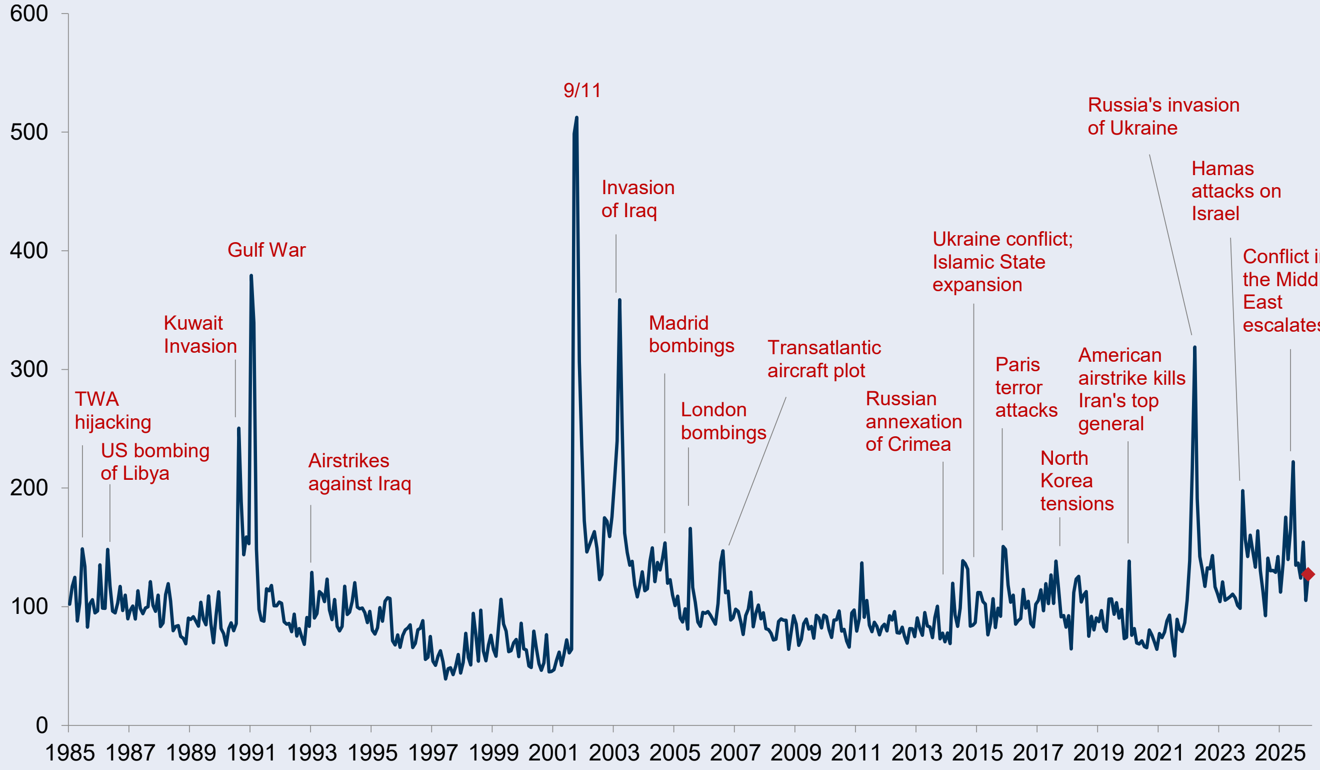
Source: Datastream, Bloomberg, Dario Caldara and Matteo Iacoviello, Goldman Sachs GIR.

Geopolitical risk through the years

Geopolitical tensions, which take many different forms, are difficult to measure. One proxy for assessing the geopolitical environment is the news-based Geopolitical Risk Index developed by economists from the Federal Reserve Board.

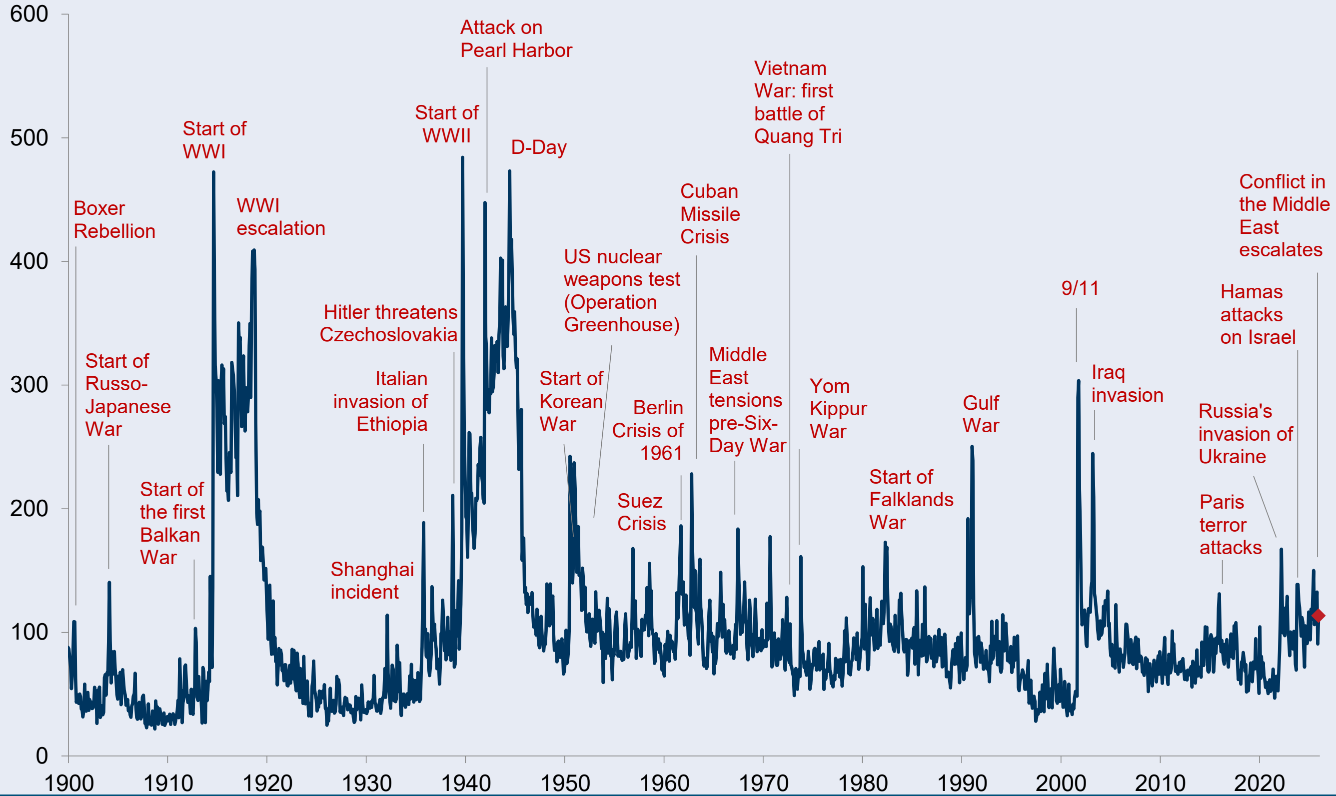
1985-Present

Geopolitical Risk Index, 1900-2019=100



1900-Present

Geopolitical Risk Index, 1985-2019=100



Note: The index from 1985 on counts the number of articles in 11 US, UK, and Canadian newspapers mentioning phrases related to geopolitical tensions. The index from 1900 on performs the same analysis using the archives of three newspapers: the New York Times, the Wall Street Journal, and the Financial Times. The choice of newspapers for both indices implies a measure of geopolitical risk as covered by the Anglo-Saxon press. See here for more information.

Source: Dario Caldara and Matteo Iacoviello, Federal Reserve Board, Goldman Sachs GIR.

EM assets: continued resilience

Kamakshya Trivedi discusses the outlook for EM assets amid elevated geopolitical risk

Geopolitical risks have dominated headlines in 2026, from developments in Venezuela and Iran to Greenland. Despite that, EM equities have started the year strong and have comfortably outpaced DM equity markets, EM currencies are among the top performers year-to-date, and EM rates and spreads have held on. We think this is indicative of the resilient upside that we expect out of EM this year. Indeed, despite heightened geopolitical risks and several upcoming elections, our global macro outlook of sturdy growth, low inflation, and further Fed cuts is a friendly one for EM assets, even if less intensely so than in 2025. Better local EM fundamentals and still-low global investor allocations should also support returns in 2026.

Limited spillovers from Venezuela

Venezuelan developments—while highly pertinent from a geopolitical standpoint—have had limited spillovers in part because of limited exposure in EM investor portfolios. Political shifts can create significant volatility in local markets—as seen in Argentina last year and the political ructions in Türkiye before that—but their relatively small share of aggregate EM portfolios typically limits any impact. Equally, the improvement in EM fundamentals in recent years—solid growth, further inflation normalization, and scope for rate cuts as well as contained fiscal and current account positions—has helped insulate the broader EM complex from political and geopolitical shocks.

A positive returns story for EM assets

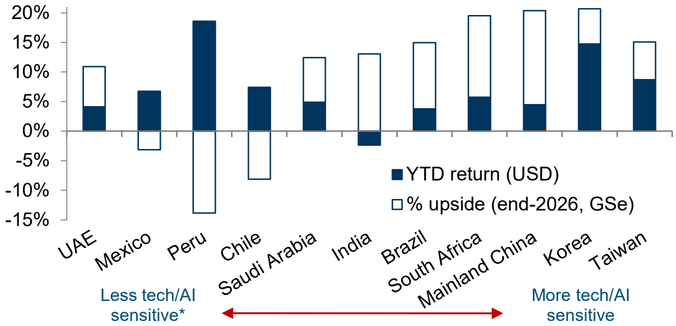
We continue to expect positive returns for EM assets this year amid a friendly global macro backdrop. In particular, we expect EM equities to outperform DM equities again, though we look for more moderate returns of ~15% (vs. last year's record 30% gains) driven primarily by solid earnings growth (19%/12% in 2026/2027) spread across the EM geographies. In EM FX, after the year of carry in 2025, we think that cyclical exposure—via both equity and commodity price linkages—will be as important as carry levels in 2026 for relative spot returns. This includes ZAR, CLP, and KRW, which have high betas to cyclical pricing and to the appreciation we expect in the CNY anchor and remain among the more undervalued currencies vs. USD. In EM fixed income, the scope for deeper cutting cycles in EM high-yielders (like Hungary and Brazil) should provide the best opportunities in local rates. We remain defensive on EM sovereign credit, with total returns likely to be largely a function of yield given our expectation of a modest widening in very tight spreads.

EM assets as diversifiers

EM assets can also provide an attractive opportunity for investors given their role as diversifiers. In particular, EM assets can help balance AI exposure amid growing investor concerns about valuations. For example, within EM, while we continue to favor the tech-sensitive equity markets of Korea, Taiwan (strong semiconductor earnings), and Mainland China (AI-driven growth and external market share gains), we prefer rotating some portfolio weight into the more domestic-oriented equity markets of South Africa (where a cyclical growth recovery and easing rates should help laggard domestic sectors catch up to the miners), India (with its mass-consumption revival), and Brazil (where rate cuts should drive further gains).

EM has seen a broad-based rally this year

Returns, %, USD



*MSCI weight in tech/hardware, semis, internet, software, media/entertainment.
Source: FactSet, Goldman Sachs GIR.

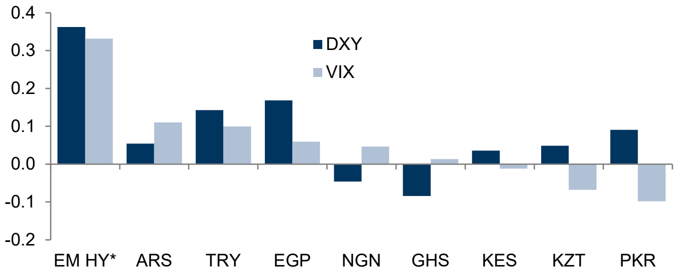
Uncorrelated risks and returns on the EM Frontier

Frontier markets also provide a source of returns that are relatively uncorrelated to the broader macro backdrop (although they come with their own risks, as Venezuela shows). Amid renewed geopolitical risks and the potential for higher volatility this year, we think that Frontier currencies continue to represent an attractive investment proposition in a more range-bound Dollar environment owing to elevated carry levels together with low correlation to Dollar strength and global macro volatility relative to other EM counterparts.

In particular, given still-healthy GSDEER model valuations and our economists' constructive external outlook, we favor carry longs in CEEMEA, including EGP, NGN, and KES, where risks are skewed toward appreciation given historically high FX reserves. Specifically, a still-cautious central bank in Egypt, which could keep real rates elevated for longer than we previously expected, led us to recently extend the target on our short USD/EGP trade recommendation (opened last July). And in Türkiye, we expect TRY monthly total returns to remain around 1-1.5% as a slight reduction in carry is offset by a decline in the pace of nominal depreciation amid slowing year-on-year inflation. The main risk to this constructive view is elevated positioning in still-relatively illiquid FX markets that could reverse sharply on a negative shock. Finally, we expect ARS to weaken along wider FX flotation bands given Argentine policymakers' objective to rapidly accumulate international reserves this year from critically low levels.

Frontier currencies are less correlated with USD and global vol

Last six-month correlation of 12m forwards



Note: We take log changes of all series before computing correlations.
**Median of BRL, COP, MXN, and ZAR.*
Source: Goldman Sachs GIR.

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Q&A on defense sector geopolitical risk

We ask our equity analysts about the potential implications of elevated geopolitical risk for the defense sector.

US Defense

Noah Poponak, GS Equity Research

Q: Is the upside for defense stocks on the back of elevated geopolitical risk already fully priced in?

A: A lot of upside is priced in some stocks, but it is not fully priced across the sector. The US Defense sector has outperformed year-to-date on the back of geopolitical developments and statements from the White House regarding potentially much higher defense budgets. The Administration currently appears to want to further grow the US defense budget, but efforts are also underway to shift more financial and technology development risk onto the industrial base, creating a somewhat mixed picture for defense firms' financials. Product category prioritizations are also driving different growth profiles across verticals and individual companies in the sector. And with the US defense budget already at an all-time high and questions remaining about the appropriate level of US government spending, the annual level of defense spending will continue to be debated notwithstanding the clear focus on national security. Valuations at the sector level are on average above their historical midpoint but below their highs, with enough dispersion at the company level to still find value.

Q: Where does value remain in the sector?

A: We see value in stocks leveraged to product and vertical prioritizations that have growth and margin upside potential (rather than facing more financial risk as the Administration and Pentagon seek to shift more of the investment burden onto firms) and trade at relatively attractive valuations. We also see value in select Defense Tech companies.

Q: What key catalysts are you watching to realize the upside?

A: The White House recently discussed the possibility of a much higher US defense budget. The government typically makes its official budget request for the next fiscal year in the spring, which then goes through the congressional armed services committees through the summer, and should be finalized by the fall. Where that budget ultimately lands will go a long way toward informing future growth. We're also watching the moving pieces under the hood to continue to understand the verticals and products in higher demand. New Administration frameworks have been incrementally informing major changes in the Pentagon's defense procurement process, capital requirements, and risk sharing processes. And Defense budgets have been rising in other places, especially Europe, which will impact defense companies in those regions as well as US companies given that other countries procure military technologies from the US.

European Defense

Sam Burgess, GS Equity Research

Q: Is the upside for defense stocks on the back of elevated geopolitical risk already fully priced in?

A: No. While geopolitical tensions, alongside an increasingly fractious NATO alliance, have driven a sharp re-rating in European defense equities—with the GS EU Defense basket (GSSBDEFE) up over 20% year-to-date—we still see value in the sector. On growth-adjusted metrics, European defense valuations remain materially cheaper than US peers, with the discount currently at one of the highest points over the past three years. As a result, we would push back on the idea of a European defense bubble—the sector has clearly re-rated, but valuations are not uniformly stretched.

Q: What key catalysts are you watching to realize the upside?

A: As global geopolitical developments continue to emerge, we are watching for catalysts that reinforce defense spending as a necessary "insurance" outlay rather than a discretionary budget item for European policymakers. We are also keeping an eye on European defense budgets as well as disbursements from the ~€150bn European Security Action for Europe (SAFE) fund, a financial instrument providing loans to EU member states for defense procurement, which should drive meaningful orders for European primes. And we will be monitoring European primes' ability to expand capacity and meet demand in the new environment of increased military spending.

Summary of our key forecasts

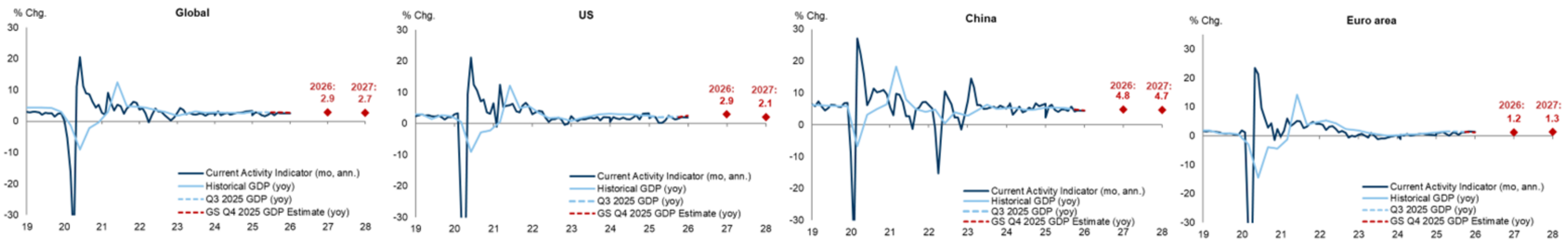
GS GIR: Macro at a glance

Watching

- **Globally,** we expect sturdy and above-consensus real GDP growth of 2.9% yoy in 2026 amid fading headwinds from US tariffs and rising real income growth. We expect global core inflation to decline to 2.1% by the end of 2026 as the tariff-driven boost fades and shelter and wage inflation fall further.
- **In the US,** we expect real GDP growth to rise to an above-consensus 2.5% on a Q4/Q4 basis in 2026 as tax cuts provide a tailwind to disposable incomes and financial conditions ease, but uncertainty around trade policy presents a potential risk. We expect core PCE inflation to decline to 2.1% yoy by the end of 2026 as tariff pass-through fades while underlying wage and shelter inflation trends improve. We expect the unemployment rate to end 2026 at 4.5%.
- **We expect the Fed** to deliver two 25bp cuts this year in June and September for a terminal rate range of 3-3.25%.
- **In the Euro area,** we expect real GDP growth of 1.2% yoy in 2026 reflecting real income gains and a fading fiscal policy drag but growing headwinds from more intense export competition with China. We expect core inflation to decline to 1.8% by end-2026 amid lower energy prices, a stronger Euro, and the potential for a negative inflation effect from Chinese trade rerouting.
- **We expect the ECB** to remain on hold for the foreseeable future given our and the ECB's forecast of better growth ahead and target-consistent inflation over the medium term.
- **In China,** we expect above-consensus real GDP growth of 4.8% yoy in 2026 given resilient export growth, continued government policy easing, and a decreasing drag from the ongoing property market downturn, although sluggish domestic demand should remain a headwind. On the inflation front, China's reflation process will likely remain gradual, but CPI/PPI inflation should rise to 0.6%/-0.7% yoy this year owing to government policy efforts and a low base.
- **WATCH US POLICY, GEOPOLITICAL DEVELOPMENTS, AND FISCAL DYNAMICS**. Uncertainty about US policy remains elevated, especially around Greenland, tariffs, and a possible US government shutdown. Geopolitical developments more broadly pose risks worth watching, especially as the situation in Venezuela develops, tension in the Middle East continues, a potential Russia-Ukraine peace deal remains uncertain, and US-China relations remain fragile. And fiscal pressures—including in the US, UK, France, and Japan—remain important to watch.

Goldman Sachs Global Investment Research.

Growth



Source: Haver Analytics, Goldman Sachs Global Investment Research.
Note: GS CAI is a measure of current growth. For more information on the methodology of the CAI please see "Technical Updates to Our Global CAIs," Global Economics Comment, Sep. 01, 2025.

Forecasts

Economics

GDP growth (%)	2026 GS (Q4/Q4)	2026 Cons. (Q4/Q4)	2026 GS (CY)	2026 Cons. (CY)	2027 GS (CY)	2027 Cons. (CY)
Global	2.9	--	2.9	2.6	2.7	2.6
US	2.5	2.1	2.9	2.0	2.1	2.3
China	4.7	4.7	4.8	4.9	4.7	4.5
Euro area	1.4	1.4	1.2	1.4	1.3	1.2

Policy rates (%)	2026 GS	2026 Mkt.	2027 GS	2027 Mkt.
US	3.13	3.23	3.13	3.33
Euro area	2.00	1.93	2.00	2.20
China	1.20	1.34	1.10	--
Japan	1.00	1.22	1.50	1.60

Markets

Interest rates 10Yr (%)	Last	E2026	E2027
US	4.24	4.20	4.25
Germany	2.86	3.25	3.00
Japan	2.26	2.00	2.00
UK	4.42	4.00	4.00

Commodities	Last	3m	12m
Crude Oil, Brent ($/bbl)	66	57	55
Nat Gas, NYMEX ($/mmBtu)	5.28	3.75	4.00
Nat Gas, TTF (EUR/MWh)	39.47	38	30
Copper ($/mt)	13,049	12,665	10,870
Gold ($/troy oz)	4,946	4,805	5,445

FX	Last	3m	12m
EUR/$	1.18	1.18	1.25
GBP/$	1.36	1.33	1.36
$/JPY	158	157	152
$/CNY	6.94	6.95	6.85

Credit (bp)		Last	2Q26	4Q26
USD	IG	72	85	90
	HY	256	295	315
EUR	IG	85	95	99
	HY	257	298	315

Equities

S&P 500	E2026 GS	E2026 Cons.	E2027 GS	E2027 Cons.
Price	7,600			
EPS	$305	$311	$336	$356
Growth	12%	15%	10%	15%

Returns (%)	12m	YTD	E2026 P/E
S&P 500	9.9	1.0	22.6x
MXAPJ	12.0	5.7	15.2x
Topix	7.4	6.5	18.1x
STOXX 600	2.7	2.8	15.4x

Consumer	2026 CPI (%, yoy)	2026 Unemp. Rate	2027 CPI (%, yoy)	2027 Unemp. Rate	Wage Tracker 2025 (%) Q1	Q2	Q3	Q4
US	2.4	4.5	1.9	4.5	4.1	3.9	3.7	3.4
Euro area	1.8	6.4	1.8	6.3	--	--	--	--
China	0.6	--	0.9	--	--	--	--	--

Source: Bloomberg, Goldman Sachs Global Investment Research. For important disclosures, see the Disclosure Appendix or go to www.gs.com/research/hedge.html.

Market pricing as of January 23, 2026

Glossary of GS proprietary indices

Current Activity Indicator (CAI)

GS CAIs measure the growth signal in a broad range of weekly and monthly indicators, offering an alternative to Gross Domestic Product (GDP). GDP is an imperfect guide to current activity: In most countries, it is only available quarterly and is released with a substantial delay, and its initial estimates are often heavily revised. GDP also ignores important measures of real activity, such as employment and the purchasing managers' indexes (PMIs). All of these problems reduce the effectiveness of GDP for investment and policy decisions. Our CAIs aim to address GDP's shortcomings and provide a timelier read on the pace of growth.

For more, see our CAI page and Global Economics Comment: Technical Updates to Our Global CAIs.

Dynamic Equilibrium Exchange Rates (DEER)

The GSDEER framework establishes an equilibrium (or "fair") value of the real exchange rate based on relative productivity and terms-of-trade differentials.

For more, see our GSDEER page, Global Economics Paper No. 227: Finding Fair Value in EM FX, 26 January 2016, and Global Markets Analyst: A Look at Valuation Across G10 FX, 29 June 2017.

Financial Conditions Index (FCI)

GS FCIs gauge the "looseness" or "tightness" of financial conditions across the world's major economies, incorporating variables that directly affect spending on domestically produced goods and services. FCIs can provide valuable information about the economic growth outlook and the direct and indirect effects of monetary policy on real economic activity.

FCIs for the G10 economies are calculated as a weighted average of a policy rate, a long-term risk-free bond yield, a corporate credit spread, an equity price variable, and a trade-weighted exchange rate; the Euro area FCI also includes a sovereign credit spread. The weights mirror the effects of the financial variables on real GDP growth in our models over a one-year horizon. FCIs for emerging markets are calculated as a weighted average of a short-term interest rate, a long-term swap rate, a CDS spread, an equity price variable, a trade-weighted exchange rate, and—in economies with large foreign-currency-denominated debt stocks—a debt-weighted exchange rate index.

For more, see our FCI page, Global Economics Analyst: Our New G10 Financial Conditions Indices, 20 April 2017, and Global Economics Analyst: Tracking EM Financial Conditions – Our New FCIs, 6 October 2017.

Goldman Sachs Analyst Index (GSAI)

The US GSAI is based on a monthly survey of GS equity analysts to obtain their assessments of business conditions in the industries they follow. The results provide timely "bottom-up" information about US economic activity to supplement and cross-check our analysis of "top-down" data. Based on analysts' responses, we create a diffusion index for economic activity comparable to the ISM's indexes for activity in the manufacturing and nonmanufacturing sectors.

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GS MAP scores facilitate rapid interpretation of new data releases for economic indicators worldwide. MAP summarizes the importance of a specific data release (i.e., its historical correlation with GDP) and the degree of surprise relative to the consensus forecast. The sign on the degree of surprise characterizes underperformance with a negative number and outperformance with a positive number. Each of these two components is ranked on a scale from 0 to 5, with the MAP score being the product of the two, i.e., from -25 to +25. For example, a MAP score of +20 (5;+4) would indicate that the data has a very high correlation to GDP (5) and that it came out well above consensus expectations (+4), for a total MAP value of +20.

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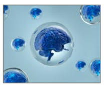

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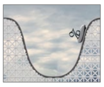

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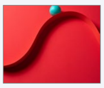

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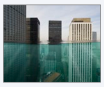

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Reg AC

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We, Neil Mehta, Noah Poponak, CFA and Sam Burgess, hereby certify that all of the views expressed in this report accurately reflect our personal views about the subject company or companies and its or their securities. We also certify that no part of our compensation was, is or will be, directly or indirectly, related to the specific recommendations or views expressed in this report.

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